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Heterodimeric proteins

Abstract

In one aspect, the present invention provides heterodimeric antibodies comprising a first monomer comprising a first heavy chain constant domain comprising a first variant Fc domain and a first antigen binding domain and a second monomer comprising a second heavy chain constant domain comprising a second variant Fc domain and a second antigen binding domain. In an additional aspect the heterodimeric antibody comprises a first monomer comprising a heavy chain comprising a first Fc domain and a single chain Fv region (scFv) that binds a first antigen, wherein the scFv comprises a charged scFv linker. The heterodimeric antibody further comprises a second monomer comprising a first heavy chain comprising a second Fc domain and a first variable heavy chain and a first light chain.

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Background/Summary

PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/087,467, filed Nov. 2, 2020, now U.S. Pat. No. 11,814,423, which is a continuation of U.S. patent application Ser. No. 14/216,705, filed Mar. 17, 2014, now U.S. Pat. No. 10,858,417, which is a continuation-in-part of International Patent Application No. PCT/US14/11549, filed Jan. 14, 2014, U.S. patent application Ser. No. 14/155,334, filed Jan. 14, 2014, now U.S. Pat. No. 10,738,132, U.S. patent application Ser. No. 14/205,248, filed Mar. 11, 2014, now U.S. Pat. No. 9,650,446 and U.S. patent application Ser. No. 14/207,489, filed Mar. 12, 2014, now U.S. Pat. No. 10,131,710. Further, U.S. patent application Ser. No. 14/216,705, filed Mar. 17, 2014, now U.S. Pat. No. 10,858,417, claims the benefit of U.S. Provisional Patent Application No. 61/818,153, filed May 1, 2013, U.S. Provisional Patent Application No. 61/818,344, filed May 1, 2013, U.S. Provisional

Patent Application No. 61/794,896, filed Mar. 15, 2013, U.S. Provisional Patent Application No. 61/818,401, filed May 1, 2013, U.S. Provisional Patent Application No. 61/913,879, filed Dec. 9, 2013, U.S. Provisional Patent Application No. 61/913,832, filed Dec. 9, 2013, U.S. Provisional Patent Application No. 61/938,095, filed Feb. 10, 2014 and U.S. Provisional Patent Application No. 61/913,870, filed Dec. 9, 2013, all of which are expressly incorporated by reference in their entirety, and particularly for all Figures and associated Legends, and for the amino acid variants disclosed therein.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML file, created on Sep. 7, 2023, is named 067461-5167-US06.xml and is 1,056,651 bytes in size.

BACKGROUND OF THE INVENTION

(2) Antibody-based therapeutics have been used successfully to treat a variety of diseases, including cancer and autoimmune/inflammatory disorders. Yet improvements to this class of drugs are still needed, particularly with respect to enhancing their clinical efficacy. One avenue being explored is the engineering of additional and novel antigen binding sites into antibody-based drugs such that a single immunoglobulin molecule co-engages two different antigens. Such non-native or alternate antibody formats that engage two different antigens are often referred to as bispecifics. Because the considerable diversity of the antibody variable region (Fv) makes it possible to produce an Fv that recognizes virtually any molecule, the typical approach to bispecific generation is the introduction of new variable regions into the antibody.

(3) A number of alternate antibody formats have been explored for bispecific targeting (Chames & Baty, 2009, mAbs 1[6]:1-9; Holliger & Hudson, 2005, Nature Biotechnology 23[9]:1126-1136; Kontermann, mAbs 4(2):182 (2012), all of which are expressly incorporated herein by reference). Initially, bispecific antibodies were made by fusing two cell lines that each produced a single monoclonal antibody (Milstein et al., 1983, Nature 305:537-540). Although the resulting hybrid hybridoma or quadroma did produce bispecific antibodies, they were only a minor population, and extensive purification was required to isolate the desired antibody. An engineering solution to this was the use of antibody fragments to make bispecifics. Because such fragments lack the complex quaternary structure of a full length antibody, variable light and heavy chains can be linked in single genetic constructs. Antibody fragments of many different forms have been generated, including diabodies, single chain diabodies, tandem scFv's, and Fab.sub.2 bispecifics (Chames & Baty, 2009, mAbs 1[6]:1-9; Holliger & Hudson, 2005, Nature Biotechnology 23[9]:1126-1136; expressly incorporated herein by reference). While these formats can be expressed at high levels in bacteria and may have favorable penetration benefits due to their small size, they clear rapidly in vivo and can present manufacturing obstacles related to their production and stability. A principal cause of these drawbacks is that antibody fragments typically lack the constant region of the antibody with its associated functional properties, including larger size, high stability, and binding to various Fc receptors and ligands that maintain long half-life in serum (i.e. the neonatal Fc receptor FcRn) or serve as binding sites for purification (i.e. protein A and protein G).

(4) More recent work has attempted to address the shortcomings of fragment-based bispecifics by engineering dual binding into full length antibody-like formats (Wu et al., 2007, Nature Biotechnology 25[11]:1290-1297; U.S. Ser. No. 12/477,711; Michaelson et al., 2009, mAbs 1[2]:128-141; PCT/US2008/074693; Zuo et al., 2000, Protein Engineering 13[5]:361-367; U.S. Ser. No. 09/865,198; Shen et al., 2006, J Biol Chem 281[16]:10706-10714; Lu et al., 2005, J Biol Chem 280[20]:19665-19672; PCT/US2005/025472; expressly incorporated herein by reference). These formats overcome some of the obstacles of the antibody fragment bispecifics, principally because they contain an Fc region. One significant drawback of these formats is that, because they build new antigen binding sites on top of the homodimeric constant chains, binding to the new

antigen is always bivalent.

(5) For many antigens that are attractive as co-targets in a therapeutic bispecific format, the desired binding is monovalent rather than bivalent. For many immune receptors, cellular activation is accomplished by cross-linking of a monovalent binding interaction. The mechanism of cross-linking is typically mediated by antibody/antigen immune complexes, or via effector cell to target cell engagement. For example, the low affinity Fc gamma receptors (FcγRs) such as FcγRIIa, FcγRIIb, and FcγRIIIa bind monovalently to the antibody Fc region. Monovalent binding does not activate cells expressing these FcγRs; however, upon immune complexation or cell-to-cell contact, receptors are cross-linked and clustered on the cell surface, leading to activation. For receptors responsible for mediating cellular killing, for example FcγRIIIa on natural killer (NK) cells, receptor cross-linking and cellular activation occurs when the effector cell engages the target cell in a highly avid format (Bowles & Weiner, 2005, *J Immunol Methods* 304:88-99, expressly incorporated by reference). Similarly, on B cells the inhibitory receptor FcγRIIb downregulates B cell activation only when it engages into an immune complex with the cell surface B-cell receptor (BCR), a mechanism that is mediated by immune complexation of soluble IgG's with the same antigen that is recognized by the BCR (Heyman 2003, *Immunol Lett* 88[2]:157-161; Smith and Clatworthy, 2010, *Nature Reviews Immunology* 10:328-343; expressly incorporated by reference). As another example, CD3 activation of T-cells occurs only when its associated T-cell receptor (TCR) engages antigen-loaded MHC on antigen presenting cells in a highly avid cell-to-cell synapse (Kuhns et al., 2006, *Immunity* 24:133-139). Indeed nonspecific bivalent cross-linking of CD3 using an anti-CD3 antibody elicits a cytokine storm and toxicity (Perruche et al., 2009, *J Immunol* 183[2]:953-61; Chatenoud & Bluestone, 2007, *Nature Reviews Immunology* 7:622-632; expressly incorporated by reference). Thus for practical clinical use, the preferred mode of CD3 co-engagement for redirected killing of targets cells is monovalent binding that results in activation only upon engagement with the co-engaged target.

(6) Thus while bispecifics generated from antibody fragments suffer biophysical and pharmacokinetic hurdles, a drawback of those built with full length antibody-like formats is that they engage co-target antigens multivalently in the absence of the primary target antigen, leading to nonspecific activation and potentially toxicity. The present invention solves this problem by introducing a novel set of bispecific formats that enable the multivalent co-engagement of distinct target antigens. In addition, the present invention provides novel heterodimerization variants that allow for better formation and purification of heterodimeric proteins, including antibodies.

BRIEF SUMMARY OF THE INVENTION

(7) In one aspect, the present invention provides heterodimeric antibodies comprising a first monomer comprising a first heavy chain constant domain comprising a first variant Fc domain and a first antigen binding domain and a second monomer comprising a second heavy chain constant domain comprising a second variant Fc domain and a second antigen binding domain.

(8) In an additional aspect the heterodimeric antibody comprises a first monomer comprising a heavy chain comprising a first Fc domain and a single chain Fv region (scFv) that binds a first antigen, wherein the scFv comprises a charged scFv linker. The heterodimeric antibody further comprises a second monomer comprising a first heavy chain comprising a second Fc domain and a first variable heavy chain and a first light chain. In an additional aspect this charged linker has either a positive charge from 3 to 8 or a negative charge from 3 to 8 and is selected from the group consisting of those linkers depicted in FIG. 9.

(9) In a further aspect, the invention provides heterodimeric antibody compositions comprising a first monomer comprising a first heavy chain sequence comprising a first variant Fc domain as compared to a human Fc domain; and a first antigen-binding domain that binds to a first antigen; and a second heavy chain sequence comprising: a second variant Fc domain as compared to a human Fc domain; and a second antigen binding domain that binds to a second antigen; wherein the first and second variant Fc domains comprise a set of amino acid substitutions selected from the

group consisting of the amino acid sets depicted in FIG. 3A-3C.

(10) In an additional aspect, the invention provides heterodimeric antibody compositions comprising: a first monomer comprising a first heavy chain sequence comprising a first variant Fc domain as compared to a human Fc domain; and a first antigen-binding domain that binds to a first antigen; and a second heavy chain sequence comprising a second variant Fc domain as compared to a human Fc domain; and a second antigen binding domain that binds to CD19. The second antigen binding domain comprises a variable heavy chain domain comprising the amino acid sequence of H1.227 (SEQ ID NO:464) and a variable light chain selected from the group consisting of the amino acid sequence of L1.198 (SEQ ID NO:465) and the amino acid sequence of 1.199 (SEQ ID NO:466) as depicted in FIG. 21.

(11) In a further aspect, the invention provides a heterodimeric antibody composition comprising a first monomer comprising a first heavy chain sequence comprising a first variant Fc domain as compared to a human Fc domain a first antigen binding domain comprising an anti-CD3 variable region having a sequence comprising a vhCDR1 having the sequence T-Y-A-M-Xaa1, wherein Xaa1 is N, S or H (SEQ ID NO:435), a vhCDR2 having the sequence R-I-R-S-K-Xaa1-N-Xaa2-Y-A-T-Xaa3-Y-Y-A-Xaa4-S-V-K-G, wherein Xaa1 is Y or A, Xaa2 is N or S, Xaa3 is Y or A and Xaa4 is D or A (SEQ ID NO:436), a vhCDR3 having the sequence H-G-N-F-G-Xaa1-S-Y-V-S-W-F-Xaa2-Y, wherein Xaa1 is N, D or Q and Xaa2 is A or D (SEQ ID NO:437), a v1CDR1 having the sequence Xaa1-S-S-T-G-A-V-T-Xaa2-Xaa3-Xaa4-Y-A-N, wherein Xaa1 is G, R or K, Xaa2 is T or S, Xaa3 is S or G and Xaa4 is N or H, (SEQ ID NO:438), a v1CDR2 having the sequence Xaa1-T-N-Xaa2-R-A-Xaa3, wherein Xaa1 is G or D, Xaa2 is K or N, and Xaa3 is P or S (SEQ ID NO:439) and a v1CDR3 having the sequence Xaa1-L-W-Y-S-N-Xaa2-W-V, wherein Xaa1 is A or L and Xaa2 is L or H (SEQ ID NO:440). The heterodimeric antibody further comprises a second monomer comprising a second heavy chain sequence comprising a second variant Fc domain as compared to a human Fc domain; and an anti-CD19 antigen binding domain comprising a variable heavy chain domain comprising the amino acid sequence of H1.227 (SEQ ID NO:464) and a variable light chain selected from the group consisting of the amino acid sequence of L1.198 (SEQ ID NO:465) and the amino acid sequence of 1.199 (SEQ ID NO:466) as depicted in FIG. 21.

(12) In an additional aspect, the invention provides a heterodimeric antibody comprising a first monomer comprising a heavy chain comprising a first variant Fc domain; and a single chain Fv region (scFv) that binds a first antigen, wherein the scFv comprises a charged scFv linker; and a second monomer comprising a first heavy chain comprising a second variant Fc domain and a first variable heavy chain and the second monomer also comprises a first light chain, wherein the first and second variant Fc domains comprise amino acid substitution(s) selected from the group consisting of those depicted in FIG. 7.

(13) In a further aspect, the invention provides a heterodimeric antibody composition comprising a first monomer comprising a first antigen binding domain comprising an anti-CD3 variable region having a sequence comprising a vhCDR1 having the sequence T-Y-A-M-Xaa1, wherein Xaa1 is N, S or H (SEQ ID NO:435), a vhCDR2 having the sequence R-I-R-S-K-Xaa1-N-Xaa2-Y-A-T-Xaa3-Y-Y-A-Xaa4-S-V-K-G, wherein Xaa1 is Y or A, Xaa2 is N or S, Xaa3 is Y or A and Xaa4 is D or A (SEQ ID NO:436), a vhCDR3 having the sequence H-G-N-F-G-Xaa1-S-Y-V-S-W-F-Xaa2-Y, wherein Xaa1 is N, D or Q and Xaa2 is A or D (SEQ ID NO:437), a v1CDR1 having the sequence Xaa1-S-S-T-G-A-V-T-Xaa2-Xaa3-Xaa4-Y-A-N, wherein Xaa1 is G, R or K, Xaa2 is T or S, Xaa3 is S or G and Xaa4 is N or H, (SEQ ID NO:438), a v1CDR2 having the sequence Xaa1-T-N-Xaa2-R-A-Xaa3, wherein Xaa1 is G or D, Xaa2 is K or N, and Xaa3 is P or S (SEQ ID NO:439) and a v1CDR3 having the sequence Xaa1-L-W-Y-S-N-Xaa2-W-V, wherein Xaa1 is A or L and Xaa2 is L or H (SEQ ID NO:440). The first monomer also comprises a first heavy chain sequence comprising a first variant Fc domain as compared to a human Fc domain. The heterodimeric antibody also comprises a second monomer comprising a second antigen-binding domain; and a second heavy chain sequence comprising a second variant Fc domain as compared to a human Fc domain and

wherein the first and second variant Fc domains have different amino acid sequences. In some embodiments, the anti-CD3 variable region comprises a vhCDR1 having the sequence T-Y-A-M-Xaa1, wherein Xaa1 is N, S or H (SEQ ID NO:435), a vhCDR2 having the sequence R-I-R-S-K-Xaa1-N-Xaa2-Y-A-T-Xaa3-Y-Y-A-Xaa4-S-V-K-G, wherein Xaa1 is Y or A, Xaa2 is N or S, Xaa3 is Y or A and Xaa4 is D or A (SEQ ID NO:436), a vhCDR3 having the sequence H-G-N-F-G-Xaa1-S-Y-V-S-W-F-Xaa2-Y, wherein Xaa1 is N, D or Q and Xaa2 is A or D (SEQ ID NO:437), a v1CDR1 having the sequence Xaa1-S-S-T-G-A-V-T-Xaa2-Xaa3-Xaa4-Y-A-N, wherein Xaa1 is G, R or K, Xaa2 is T or S, Xaa3 is S or G and Xaa4 is N or H, (SEQ ID NO:438), a v1CDR2 having the sequence Xaa1-T-N-Xaa2-R-A-Xaa3, wherein Xaa1 is G or D, Xaa2 is K or N, and Xaa3 is P or S (SEQ ID NO:439) and a v1CDR3 having the sequence Xaa1-L-W-Y-S-N-Xaa2-W-V, wherein Xaa1 is A or L and Xaa2 is L or H (SEQ ID NO:440).

(14) In a further aspect the invention provides heterodimer proteins comprising a first monomer comprising a first variant heavy chain constant region and a first fusion partner; and a second monomer comprising a second variant heavy chain constant region and a second fusion partner, wherein the Fc region of the first and second constant regions comprise a set of amino acid substitutions from FIGS. 3A-3C and 12A-12J. IN some cases, the first monomer comprises a third fusion partner and optionally the second monomer comprises a fourth fusion partner. The fusion partners are independently selected from the group consisting of an immunoglobulin component, a peptide, a cytokine, a chemokine, an immune receptor and a blood factor. In some cases, the immunoglobulin component is selected from the group consisting of Fab, VH, VL, scFv, scFv2, dAb.

(15) In many aspects, one of the first and second variant Fc domains comprise amino acid substitution(s) selected from the group consisting of those depicted in FIGS. 6, 7 and/or 12A-12J. In some aspects, the first antigen binding domain is a scFv covalent attached to the first heavy chain constant domain. In additional aspects, the heterodimeric antibody has a structure selected from the structures of FIGS. 1B to 1L and 2A to 2M. In further aspects, the first and/or second Fc domain of the heterodimeric antibody further comprises amino acid substitution(s) selected from the group consisting of 434A, 434S, 428L, 308F, 259I, 428L/434S, 259I/308F, 436I/428L, 436I or V/434S, 436V/428L, 252Y, 252Y/254T/256E, 259I/308F/428L, 236A, 239D, 239E, 332E, 332D, 239D/332E, 267D, 267E, 328F, 267E/328F, 236A/332E, 239D/332E/330Y, 239D, 332E/330L, 236R, 328R, 236R/328R, 236N/267E, 243L, 298A and 299T. In some aspects, one of the first and the second variant Fc domains comprises the amino acid substitutions 364K/E357Q and the other of comprises the amino acid substitutions 368D/370S. These antibodies can further comprise amino acid substitution(s) selected from the group consisting of those listed in FIG. 7.

(16) In additional aspects the present invention provides nucleic acids, expression vectors and host cells that will produce the heterodimeric proteins and antibodies of the invention.

(17) In further aspects the invention provides methods of making the heterodimeric proteins of the invention by culturing host cells comprising the nucleic acids encoding the heterodimeric proteins and antibodies of the invention under conditions wherein the heterodimer is produced and recovering the heterodimer.

(18) In a further aspect the invention provides methods of making a heterodimeric antibody of the invention comprising providing a first nucleic acid encoding a first heavy chain comprising a first heavy chain comprising a first Fc domain and a single chain Fv region (scFv) that binds a first antigen, wherein said scFv comprises a charged linker; and providing a second nucleic acid encoding a second heavy chain comprising a second Fc domain a first variable heavy chain; and providing a third nucleic acid comprising a light chain. The method additionally comprises expressing the first, second and third nucleic acids in a host cell to produce a first, second and third amino acid sequence, respectively, loading the first, second and third amino acid sequences onto an ion exchange column; and collecting the heterodimeric fraction.

(19) In additional aspects the invention provides methods of treating an individual in need thereof by administering a heterodimeric antibody or protein herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Heterodimerization Formats and Variants

(2) FIGS. 1A-1M depict a number of heterodimeric protein formats, including heterodimeric Fc fusion proteins as well as heterodimeric antibodies. FIG. 1A shows the basic concept of a dimeric Fc region with four possible fusion partners A, B, C and D. A, B, C and D are optionally and independently selected from immunoglobulin domain(s) (e.g. Fab, vH, vL, scFv, scFv2, scFab, dAb, etc.), peptide(s), cytokines (e.g. IL-2, IL-10, IL-12, GCSF, GM-CSF, etc.), chemokine(s) (e.g. RANTES, CXCL9, CXCL10, CXCL12, etc.), hormone(s) (e.g. FSH, growth hormone), immune receptor(s) (e.g. CTLA-4, TNFR1, TNFR2, other TNFRSF, other TNFRSF, etc.) and blood factor(s) (e.g. Factor VII, Factor VIII, Factor IX, etc.). Domains filled with solid white or solid black are engineered with heterodimerization variants as outlined herein. FIG. 1B depicts the “triple F” format (sometimes also referred to as the “bottle-opener” configuration as discussed below). FIG. 1C shows a “triple F” configuration with another scFv attached to the Fab monomer (this one, along with FIG. 1F, has a greater molecular weight differential as well). FIG. 1D depicts a “triple F” with another scFv attached to the scFv monomer. FIG. 1E depicts a “three scFv” format. FIG. 1F depicts an additional Fab attached to the Fab monomer. FIG. 1G depicts a Fab hooked to one of the scFv monomers. FIGS. 1H-1L show additional varieties of “higher multispecificity” embodiments of the “triple F” format, all with one monomer comprising an scFv (and all of which have molecular weight differentials which can be exploited for purification of the heterodimers). FIG. 1H shows a “Fab-Fv” format with binding to two different antigens, with FIG. 1I depicting the “Fab-Fv” format with binding to a single antigen (e.g. bivalent binding to antigen 1). FIGS. 1J and 1K depicts a “Fv-Fab” format with similar bivalent or monovalent additional antigen binding. FIG. 1L depicts one monomer with a CH1-CL attached to the second scFv. FIG. 1M depicts a dual scFv format. In some embodiments the triple F format is not preferred.

(3) FIGS. 2A to 2U depicts a wide variety of the multispecific (e.g. heterodimerization) formats and the combinations of different types of heterodimerization variants that can be used in the present invention (these are sometimes referred to herein as “heterodimeric scaffolds”). Note in addition that all of these formats can include addition variants in the Fc region, as more fully discussed below, including “ablation” or “knock out” variants (FIG. 7), Fc variants to alter FcγR binding (FcγRIIb, FcγRIIIa, etc.), Fc variants to alter binding to FcRn receptor, etc. FIG. 2A shows a dual scFv-Fc format, that, as for all heterodimerization formats herein can include heterodimerization variants such as pI variants, knobs in holes (KIH, also referred to herein as steric variants or “skew” variants), charge pairs (a subset of steric variants), isosteric variants, and SEED body (“strand-exchange engineered domain”; see Klein et al., mAbs 4:6 653-663 (2012) and Davis et al, Protein Eng Des Sel 2010 23:195-202) which rely on the fact that the CH3 domains of human IgG and IgA do not bind to each other. FIG. 2B depicts a bispecific IgG, again with the option of a variety of heterodimerization variants. FIG. 2C depicts the “one armed” version of DVD-Ig which utilizes two different variable heavy and variable light domains. FIG. 2D is similar, except that rather than an “empty arm”, the variable heavy and light chains are on opposite heavy chains. FIG. 2E is generally referred to as “mAb-Fv”. FIG. 2F depicts a multi-scFv format; as will be appreciated by those in the art, similar to the “A, B, C, D” formats discussed herein, there may be any number of associated scFvs (or, for that matter, any other binding ligands or functionalities). Thus, FIG. 2F could have 1, 2, 3 or 4 scFvs (e.g. for bispecifics, the scFv could be “cis” or “trans”, or both on one “end” of the molecule). FIG. 2G depicts a heterodimeric FabFc with the Fab being

formed by two different heavy chains one containing heavy chain Fab sequences and the other containing light chain Fab sequences. FIG. 2H depicts the “one armed Fab-Fc”, where one heavy chain comprises the Fab. FIG. 2I depicts a “one armed scFv-Fc”, wherein one heavy chain Fc comprises an scFv and the other heavy chain is “empty”. FIG. 2J shows a scFv-CH3, wherein only heavy chain CH3 regions are used, each with their own scFv. FIG. 2K depicts a mAb-scFv, wherein one end of the molecule engages an antigen bivalently with a monovalent engagement using an scFv on one of the heavy chains. FIG. 2L depicts the same structure except that both heavy chains comprise an additional scFv, which can either bind the same antigen or different antigens. FIG. 2M shows the “CrossMab” structure, where the problem of multiplex formation due to two different light chains is addressed by switching sequences in the Fab portion. FIG. 2N depicts an scFv, FIG. 2O is a “BiTE” or scFv-scFv linked by a linker as outlined herein, FIG. 2P depicts a DART, FIG. 2Q depicts a TandAb, and FIG. 2R shows a diabody. FIGS. 2S, 2T and 2U depict additional alternative scaffold formats that find use in the present invention.

(4) FIG. 3A-3C depicts a number of suitable heterodimerization variants, including skew/steric variants, isosteric variants, pI variants, KIH variants, etc. for use in the heterodimeric proteins of the invention. As for all the heterodimeric structures herein, each set of these heterodimerization variants can be combined, optionally and independently and in any combination in any heterodimeric scaffold. The variants at the end of the monomer 1 list are isosteric pI variants, which are generally not use in pairs or sets. In this case, one monomer is engineered to increase or decrease the pI without altering the other monomer. Thus, although depicted in the “monomer 1” list, these can be incorporated in the appropriate monomer, preserving “strandedness”. That is, what is important is that the “strandedness” of the monomer pairs remains intact although variants listed as “monomer 1” variants in the steric list can be crossed with “monomer 2” variants in the pI list. That is, any set can be combined with any other, regardless of which “monomer” list to which they are associated (as is more fully discussed below, in the case where changes in pI are to be used to purify the heterodimeric proteins, the “pI strandedness” is also preserved; for example, if there are skew variants that happen to alter charge, they are paired with pI variants on the correct strand; skew variants that result in increases in pI are added to the monomer that has increased pI variants, etc. This is similar to the addition of charged scFv linkers; in that case, as more fully described herein, the correctly charged scFv linker is added to the correct monomer to preserve the pI difference. In addition, each pair of amino acid variants (or where there is a single monomer being engineered) can be optionally and independently included or excluded from any heterodimeric protein, as well as can be optionally and independently combined.

(5) FIGS. 4A, 4B and 4C depicts a subset of heterodimerization variants of FIG. 3A-3C finding particular use in the invention.

(6) FIG. 5 depicts a subset of heterodimerization variants of FIG. 3A-3C.

(7) FIG. 6 depicts a list of isotypic and isosteric variant antibody constant regions and their respective substitutions. pI₍₋₎ indicates lower pI variants, while pI₍₊₎ indicates higher pI variants. These can be optionally and independently combined with other heterodimerization variants of the invention.

(8) FIG. 7 depicts a number of suitable “knock out” (“KO”) variants to reduce binding to some or all of the FcγR receptors. As is true for many if not all variants herein, these KO variants can be independently and optionally combined, both within the set described in FIG. 35 and with any heterodimerization variants outlined herein, including steric and pI variants. For example, E233P/L234V/L235A/G236del can be combined with any other single or double variant from the list. In addition, while it is preferred in some embodiments that both monomers contain the same KO variants, it is possible to combine different KO variants on different monomers, as well as have only one monomer comprise the KO variant(s). Reference is also made to the Figures and Legends of U.S. Ser. No. 61/913,870, all of which is expressly incorporated by reference in its entirety as it relates to “knock out” or “ablation” variants.

(9) FIG. 8 depicts a number of anti-CD3 scFv engineered disulfides.

(10) FIG. 9 depicts a number of charged scFv linkers that find use in increasing or decreasing the pI of heterodimeric proteins that utilize one or more scFv as a component. A single prior art scFv linker with a single charge is referenced as “Whitlow”, from Whitlow et al., Protein Engineering 6(8):989-995 (1993). It should be noted that this linker was used for reducing aggregation and enhancing proteolytic stability in scFvs.

(11) FIGS. 10A and 10B is an additional list of potential heterodimerization variants for use in the present invention, including isotypic variants.

(12) FIG. 11 depicts a matrix of possible combinations of heterodimerization formats, heterodimerization variants (separated into pI variants and steric variants (which includes charge pair variants), Fc variants, FcRn variants and combinations. Legend A are suitable FcRn variants: 434A, 434S, 428L, 308F, 259I, 428L/434S, 259I/308F, 436I1428L, 436I or V/434S, 436V/428L, 252Y, 252Y/254T/256E and 259I/308F/428L. That is, the Triple F format of FIG. 1B can have any of these FcRn variants on either or both monomer sequences. For clarity, as each heavy chain is different, FcRn variants (as well as the Fc variants) can reside on one or both monomers. Legend B are suitable Fc variants: 236A, 239D, 239E, 332E, 332D, 239D/332E, 267D, 267E, 328F, 267E/328F, 236A/332E, 239D/332E/330Y, 239D, 332E/330L, 236R, 328R, 236R/328R, 236N/267E, 243L, 298A and 299T. (Note, additional suitable Fc variants are found in FIG. 41 of US 2006/0024298, the figure and legend of which are hereby incorporated by reference in their entirety). In some cases as described herein, “knock out” or “ablation” variants are used such as depicted in FIG. 7, and they are included in the definition of Fc variants. As for FcRn variants, the Fc variants can reside on either strand. Legend C are suitable pI variants, and these, for brevity are imported from FIGS. 3A-3C and 12A-12J, again with the understanding that there is a “strandedness” to pI variants. Legend D are suitable steric variants (including charge pair variants); again, for brevity are imported from FIGS. 3A-3C and 12A-12J, again with the understanding that there is a “strandedness” to steric variants. Legend E reflects the following possible combinations, again, with each variant being independently and optionally combined from the appropriate source Legend: 1) pI variants plus FcRn variants; 2) pI variants plus Fc variants; 3) pI variants plus FcRn variants plus Fc variants; 4) steric variants plus FcRn variants; 5) steric variants plus Fc variants; 6) steric variants plus FcRn variants plus Fc variants; 7) pI variants plus steric variants plus FcRn variants; 8) pI variants plus steric variants plus Fc variants; 9) pI variants plus steric variants plus FcRn variants plus Fc variants; and 10) pI variants plus steric variants. Note any or all of these combinations can optionally include or exclude the knock out/ablation variants in either or both monomers.

(13) FIGS. 12A to 12J depicts additional heterodimerization variant pairs.

(14) Specific Sequences of the Inventions

(15) FIG. 13 depicts the amino acid sequences of wild-type constant regions used in the invention and the IgG1/G2 fusion.

(16) FIGS. 14A to 14YY depict the amino acid sequences of stability-optimized, humanized anti-CD3 variant scFvs, variable heavy and variable light sequences. (Note also that the first sequence is the histidine tagged version for ease of purification). CDRs are underlined. It should be understood that the increased stability of the optimized variable and optimized light chains (as well as the scFv chains) can be attributed to framework regions as well as the CDRs. Thus, it should be understood that the disclosure of the entire variable region includes the disclosure of the framework regions, although they are not separately numbered. In addition, the scFv linkers are shown in grey. Each scFv linker can be replaced with a charged scFv linker as depicted in FIG. 5. That is, any charged scFv linker, whether positive or negative, including those depicted in FIG. 5, can be substituted for the highlighted region in FIGS. 14A to 14YY.

(17) FIGS. 15A to 15I depict a collation of all the CD3 vhCDR1-3 and v1CDR1-3 sequences useful in the present invention. The sequences of the consensus CDRs are shown at the end of the

Figure.

(18) FIG. 16 shows the sequence of XENP13790, which is XENP12912 (CD3 scFv+disulfide) with the addition of a charged linker.

(19) FIGS. 17A, 17B and 17C. FIG. 17A depicts two different Triple F embodiments.

(20) FIGS. 17B and 17C show the sequences of the Triple F embodiment of FIG. 17A.

(21) FIG. 18 depicts the sequences of a preferred embodiment of the invention. The variable regions are underlined, and the charged scFv linker is in grey.

(22) FIGS. 19A and 19B. The T_m and change in T_m for stability-optimized, humanized anti-CD19 variant scFvs. Amino acid numbering is Kabat numbering. FIG. 19A as determined by DSF (Differential Scanning Fluorimetry) of stability-optimized, humanized anti-CD19 variant scFvs. done at a concentration of 0.2 mg/ml and FIG. 19B was done at 0.4 mg/ml.

(23) FIGS. 20A-20K. Amino acid sequences of stability-optimized, humanized anti-CD19 variant scFvs, variable heavy and variable light sequences. (Note also that the first sequence is the histidine tagged version for ease of purification). It should be understood that the increased stability of the optimized variable and optimized light chains (as well as the scFv chains) can be attributed to framework regions as well as the CDRs. Thus, it should be understood that the disclosure of the entire variable region includes the disclosure of the framework regions, although they are not separately numbered.

(24) FIG. 21. Depicts stabilized anti-CD19 Fv regions.

(25) FIGS. 22A and 22B depicts dual-scFv constructs (e.g. as shown in FIG. 1M).

(26) FIGS. 23A and 23B depict “bottle opener” constructs (e.g. as shown in FIG. 1B).

(27) FIGS. 24A-24K shows additional sequences of the invention including isosteric heterodimerization variants.

(28) Data Materials

(29) FIG. 25. Stabilized anti-CD19 variable regions—competition binding with labeled anti-CD19 IgG1 @ 1 μ g/mL.

(30) FIG. 26 shows the characterization and comparison of a dual scFv-Fc format, an anti-CD3/anti-CD19 pair, with the “BiTE” format, using the same scFvs but no Fc region. As shown, the dual scFv-Fc is less potent than the BiTE format, but the addition of the Fc region increases the half-life in mice by 10 fold.

(31) FIG. 27 depicts that the scFv portions each crossreact with cynomolgus monkey antigens in an RTCC test. That is, the potency difference between the formats (dual scFv-Fc versus BiTE) translates into cyno monkeys.

(32) FIG. 28 shows that the half life difference also translates into cyno monkeys as between the two formats. The dual scFv-Fc antibody was run at three different concentrations as shown.

(33) FIG. 29 depicts the projection of pharmacodynamics in monkeys, with the duration of the serum concentration when greater than EC50 is longer for the dual scFv-Fc format than for BiTE, at 2-3 weeks versus 2-3 days.

(34) FIG. 30 shows the extensive and prolonged B cell killing with the dual scFv-Fc bispecific format. The longer PK of this format enables prolonged B cell depletion out to 14 days.

(35) FIG. 31 depicts the stability engineering of the anti-CD3/anti-CD19 scFv-Fc scFv portions. By identifying and replacing rare amino acids, identifying and replacing amino acids with unusual contacting residues, linker engineering and conversion to VL-VH orientation, substantially increased stability was achieved.

(36) FIG. 32 depicts the improved PK in mice as a result of the stabilization, which resulted in a doubling of the half life in mice for the anti-CD19 stabilization.

(37) FIG. 33 shows the production and purification of the “triple F” or “bottle opener” (or as referred to in some of the figures, Fab-scFv-Fc).

(38) FIG. 34 shows the characterization of the anti-CD19/anti-CD3 triple F format, which exhibits picomolar cytotoxicity with only monovalent binding to the target antigens.

(39) FIG. 35 shows the improvement in PK in mice that results from replacing one scFv of a dual scFv-Fc with a Fab. Replacing the anti-CD19 scFv with a Fab doubles the half-life in BL/6 mice from 3 to 6 days.

(40) FIG. 36 depicts the scheme for the “plug and play platform” for the triple F format. A Fab from any existing mAb can be combined with the anti-CD3 scFv-Fc bispecific format.

(41) FIGS. 37A and 37B depict the characterization of a “plug and play” combination of existing antibodies with the triple F format. FIG. 37A shows the an anti-CD38 Fab with the anti-CD3 scFv into the triple F format, and FIG. 37B shows the Her2/CD3 combination.

(42) FIG. 38 depicts the remarkable “skew” towards heterodimerization using variants of the invention. Heterodimerization of over 95% was accomplished using one monomer with L368E/K370T and the other with S364K as compared to the same molecule without the Fc variants.

(43) FIG. 39 shows the B cell depletion in cyno monkey lymph nodes and spleen using the dual scFv format as compared to the BiTE format.

(44) FIG. 40. List of bevacizumab, Fc-only, and anti-CD19×CD3 heterodimers containing isosteric pI substitutions. pI values of each expected protein species are indicated.

(45) FIG. 41. Cation exchange chromatography showing purification of the heterodimer species of bevacizumab containing isosteric engineered constant regions.

(46) FIG. 42. Cation exchange chromatography showing purification of the heterodimer species of Fc-only variants containing isosteric engineered constant regions.

(47) FIG. 43. Cation exchange chromatography showing purification of the heterodimer species of an anti-CD19×CD3 bispecific antibody containing isosteric engineered constant regions. Also shown is an IEF gel of protein A purified material as well as the isolated heterodimer bispecific.

(48) FIG. 44. List of bevacizumab and Fc-only variants containing isosteric pI substitutions as well as T_m values obtained from DSF.

(49) FIG. 45. List of anti-CD3 and anti-CD19 scFvs containing positively and negatively charged linkers. Also shown are DSF T_m values.

(50) FIG. 46. Example SEC chromatograms from purified scFvs with positively charged linkers.

(51) FIG. 47. Direct binding of anti-CD3 scFvs containing positively charged linkers binding to CD4⁺ T cells (left) or CD20⁺ cells from PBMCs (to check for non-specific binding; right).

(52) FIG. 48. Direct binding of anti-CD3 scFvs containing positively charged linkers binding to CD20⁺ cells from PBMCs (left) or 293E cells (right).

(53) FIG. 49. Example cation exchange purification of XENP13124, which is a Fab-scFv-Fc format bispecific antibody targeting CD19 and CD3. The anti-CD3 scFv contains the positively charged linker (GKPGS)₄ to enable purification.

(54) FIG. 50. Example SEC chromatograms of purified Fab-scFv-Fc format bispecific antibodies targeting CD19 and CD3 incubated at various concentrations. XENP13121 (left) contains the standard (GGGGS)₄ linker while XENP13124 (right) contains the (GKPGS)₄ charged linker. The charged linker has the unexpected property of decreasing the amount of high molecular aggregates present.

(55) FIG. 51. RTCC assay with PBMCs and Fab-scFv-Fc format bispecific anti-CD19×CD3 antibodies containing different scFv linkers. Linkers have little impact on RTCC activity, except for the highly charged linker (GKGKS)₃ which has lower activity.

(56) FIG. 52A to 52D show sequences of the invention that include charged scFv linkers as well as corresponding controls.

(57) Miscellaneous Other Material

(58) FIG. 53. Literature pIs of the 20 amino acids. It should be noted that the listed pIs are calculated as free amino acids; the actual pI of any side chain in the context of a protein is different, and thus this list is used to show pI trends and not absolute numbers for the purposes of the invention.

(59) FIGS. 54A, 54B and 54C. List of all possible reduced pI variants created from isotypic substitutions of IgG1-4. Shown are the pI values for the three expected species as well as the average delta pI between the heterodimer and the two homodimer species present when the variant heavy chain is transfected with IgG1-WT heavy chain.

(60) FIG. 55. List of all possible increased pI variants created from isotypic substitutions of IgG1-4. Shown are the pI values for the three expected species as well as the average delta pI between the heterodimer and the two homodimer species present when the variant heavy chain is transfected with IgG1-WT heavy chain.

(61) FIG. 56 shows the amino acid sequence of the CK and CX, light constant chains. Residues which contribute to a higher pI (K, R and H) or lower pI (D and E) are highlighted in bold. Preferred positions for modification to lower the pI are shown in gray. For scaffolds that contain one or more light chains, these changes can be used to alter the pI of one or both of the monomers, and can be independently and optionally combined with all heavy chain variants.

(62) FIGS. 57A to 57E depict the sequences of a number of disulfide constructs; the first sequence is the scFv construct including the His(6) tag for convenience of purification, the second sequence is the scFv construct without the tag, the third sequence is the variable heavy chain alone and the fourth sequence is the variable light sequence alone. The CDRs are underlined.

DETAILED DESCRIPTION OF THE INVENTION

(63) FIGS. 66 to 80 of WO/2013/055809, the sequences and the accompanying legends are incorporated specifically by reference herein. FIGS. 2 to 111 and their accompanying legend from U.S. Ser. No. 13/648,951 are specifically incorporated herein by reference.

I. Overview of Heterodimerization Proteins

(64) The present invention is directed to novel constructs to provide heterodimeric proteins that allow binding to more than one antigen or ligand, e.g. to allow for multispecific binding. The heterodimeric protein constructs are based on the self-assembling nature of the two Fc domains of the heavy chains of antibodies, e.g. two “monomers” that assemble into a “dimer”. Heterodimeric proteins are made by altering the amino acid sequence of each monomer as more fully discussed below. Thus, the present invention is generally directed to the creation of heterodimeric proteins including antibodies, which can co-engage antigens in several ways, relying on amino acid variants in the constant regions that are different on each chain to promote heterodimeric formation and/or allow for ease of purification of heterodimers over the homodimers. As discussed more fully below, the heterodimeric proteins can be antibody variants or based on Fc fusion proteins. In general, heterodimeric antibodies are the focus of the discussion, but as will be appreciated by those in the art and more fully described below, the discussion applies equally to heterodimeric proteins that

(65) Thus, the present invention provides bispecific antibodies (or, as discussed below, trispecific or tetraspecific antibodies can also be made). An ongoing problem in antibody technologies is the desire for “bispecific” (and/or multispecific) antibodies that bind to two (or more) different antigens simultaneously, in general thus allowing the different antigens to be brought into proximity and resulting in new functionalities and new therapies. In general, these antibodies are made by including genes for each heavy and light chain into the host cells. This generally results in the formation of the desired heterodimer (A-B), as well as the two homodimers (A-A and B-B). However, a major obstacle in the formation of multispecific antibodies is the difficulty in purifying the heterodimeric antibodies away from the homodimeric antibodies and/or biasing the formation of the heterodimer over the formation of the homodimers.

(66) There are a number of mechanisms that can be used to generate the heterodimers of the present invention. In addition, as will be appreciated by those in the art, these mechanisms can be combined to ensure high heterodimerization. Thus, amino acid variants that lead to the production of heterodimers are referred to as “heterodimerization variants”. As discussed below, heterodimerization variants can include steric variants (e.g. the “knobs and holes” or “skew” variants described below and the “charge pairs” variants described below) as well as “pI variants”,

which allows purification of homodimers away from heterodimers.

(67) One mechanism is generally referred to in the art as “knobs and holes” (“KIH”) or sometimes herein as “skew” variants, referring to amino acid engineering that creates steric influences to favor heterodimeric formation and disfavor homodimeric formation can also optionally be used; this is sometimes referred to as “knobs and holes”; as described in U.S. Ser. No. 61/596,846 and U.S. Ser. No. 12/875,0015, Ridgway et al., Protein Engineering 9(7):617 (1996); Atwell et al., J. Mol. Biol. 1997 270:26; U.S. Pat. No. 8,216,805, US 2012/0149876, all of which are hereby incorporated by reference in their entirety. The Figures identify a number of “monomer A-monomer B” pairs that include “knobs and holes” amino acid substitutions. In addition, as described in Merchant et al., Nature Biotech. 16:677 (1998), these “knobs and hole” mutations can be combined with disulfide bonds to skew formation to heterodimerization.

(68) An additional mechanism that finds use in the generation of heterodimers is sometimes referred to as “electrostatic steering” or “charge pairs” as described in Gunasekaran et al., J. Biol. Chem. 285(25):19637 (2010), hereby incorporated by reference in its entirety. This is sometimes referred to herein as “charge pairs”. In this embodiment, electrostatics are used to skew the formation towards heterodimerization. As those in the art will appreciate, these may also have have an effect on pI, and thus on purification, and thus could in some cases also be considered pI variants. However, as these were generated to force heterodimerization and were not used as purification tools, they are classified as “steric variants”. These include, but are not limited to, D221E/P228E/L368E paired with D221R/P228R/K409R (e.g. these are “monomer corresponding sets) and C220E/P228E/368E paired with C220R/E224R/P228R/K409R and others shown in the Figures.

(69) In the present invention, in some embodiments, pI variants are used to alter the pI of one or both of the monomers and thus allowing the isoelectric purification of A-A, A-B and B-B dimeric proteins.

(70) In the present invention, there are several basic mechanisms that can lead to ease of purifying heterodimeric proteins; one relies on the use of pI variants, such that each monomer has a different pI, thus allowing the isoelectric purification of A-A, A-B and B-B dimeric proteins. Alternatively, some scaffold formats, such as the “triple F” format, also allows separation on the basis of size. As is further outlined below, it is also possible to “skew” the formation of heterodimers over homodimers. Thus, a combination of steric heterodimerization variants and pI or charge pair variants find particular use in the invention. Additionally, as more fully outlined below, scaffolds that utilize scFv(s) such as the Triple F format can include charged scFv linkers (either positive or negative), that give a further pI boost for purification purposes. As will be appreciated by those in the art, some Triple F formats are useful with just charged scFv linkers and no additional pI adjustments, although the invention does provide the use of skew variants with charged scFv linkers as well (and combinations of Fc, FcRn and KO variants).

(71) In the present invention that utilizes pI as a separation mechanism to allow the purification of heterodimeric proteins, amino acid variants can be introduced into one or both of the monomer polypeptides; that is, the pI of one of the monomers (referred to herein for simplicity as “monomer A”) can be engineered away from monomer B, or both monomer A and B change be changed, with the pI of monomer A increasing and the pI of monomer B decreasing. As is outlined more fully below, the pI changes of either or both monomers can be done by removing or adding a charged residue (e.g. a neutral amino acid is replaced by a positively or negatively charged amino acid residue, e.g. glycine to glutamic acid), changing a charged residue from positive or negative to the opposite charge (aspartic acid to lysine) or changing a charged residue to a neutral residue (e.g. loss of a charge; lysine to serine.). A number of these variants are shown in the Figures.

(72) Accordingly, in this embodiment of the present invention provides for creating a sufficient change in pI in at least one of the monomers such that heterodimers can be separated from homodimers. As will be appreciated by those in the art, and as discussed further below, this can be

done by using a “wildtype” heavy chain constant region and a variant region that has been engineered to either increase or decrease its pI (wt A++B or wt A--B), or by increasing one region and decreasing the other region (A+-B- or A-B+).

(73) Thus, in general, a component of some embodiments of the present invention are amino acid variants in the constant regions of antibodies that are directed to altering the isoelectric point (pI) of at least one, if not both, of the monomers of a dimeric protein to form “pI heterodimers” (when the protein is an antibody, these are referred to as “pI antibodies”) by incorporating amino acid substitutions (“pI variants” or “pI substitutions”) into one or both of the monomers. As shown herein, the separation of the heterodimers from the two homodimers can be accomplished if the pIs of the two monomers differ by as little as 0.1 pH unit, with 0.2, 0.3, 0.4 and 0.5 or greater all finding use in the present invention.

(74) As will be appreciated by those in the art, the number of pI variants to be included on each or both monomer(s) to get good separation will depend in part on the starting pI of the scFv and Fab of interest. That is, to determine which monomer to engineer or in which “direction” (e.g. more positive or more negative), the Fv sequences of the two target antigens are calculated and a decision is made from there. As is known in the art, different Fvs will have different starting pIs which are exploited in the present invention. In general, as outlined herein, the pIs are engineered to result in a total pI difference of each monomer of at least about 0.1 logs, with 0.2 to 0.5 being preferred as outlined herein.

(75) Furthermore, as will be appreciated by those in the art and outlined herein, heterodimers can be separated from homodimers on the basis of size. For example, as shown in FIGS. 1A-1M and 2A-2U, heterodimers with two scFvs can be separated by those of the “triple F” format and a bispecific mAb. This can be further exploited in higher valency with additional antigen binding sites being utilized. For example, as additionally shown, one monomer will have two Fab fragments and the other will have one scFv, resulting in a differential in size and thus molecular weight.

(76) In addition, as will be appreciated by those in the art and outlined herein, the format outlined herein can be expanded to provide trispecific and tetraspecific antibodies as well. In this embodiment, some variations of which are depicted in the FIG. 1A, it will be recognized that it is possible that some antigens are bound divalently (e.g. two antigen binding sites to a single antigen; for example, A and B could be part of a typical bivalent association and C and D can be optionally present and optionally the same or different). As will be appreciated, any combination of Fab and scFvs can be utilized to achieve the desired result and combinations.

(77) In the case where pI variants are used to achieve heterodimerization, by using the constant region(s) of the heavy chain(s), a more modular approach to designing and purifying multispecific proteins, including antibodies, is provided. Thus, in some embodiments, heterodimerization variants (including skew and purification heterodimerization variants) are not included in the variable regions, such that each individual antibody must be engineered. In addition, in some embodiments, the possibility of immunogenicity resulting from the pI variants is significantly reduced by importing pI variants from different IgG isotypes such that pI is changed without introducing significant immunogenicity. Thus, an additional problem to be solved is the elucidation of low pI constant domains with high human sequence content, e.g. the minimization or avoidance of non-human residues at any particular position.

(78) A side benefit that can occur with this pI engineering is also the extension of serum half-life and increased FcRn binding. That is, as described in U.S. Ser. No. 13/194,904 (incorporated by reference in its entirety), lowering the pI of antibody constant domains (including those found in antibodies and Fc fusions) can lead to longer serum retention in vivo. These pI variants for increased serum half life also facilitate pI changes for purification.

(79) In addition, it should be noted that the pI variants of the heterodimerization variants give an additional benefit for the analytics and quality control process of bispecific antibodies, as,

particularly in the case of CD3 antibodies, the ability to either eliminate, minimize and distinguish when homodimers are present is significant. Similarly, the ability to reliably test the reproducibility of the heterodimeric protein production is important.

(80) In addition to all or part of a variant heavy constant domain, one or both of the monomers may contain one or two fusion partners, such that the heterodimers form multivalent proteins. As is generally depicted in the Figures, and specifically FIG. 1A, the fusion partners are depicted as A, B, C and D, with all combinations possible. In general, A, B, C and D are selected such that the heterodimer is at least bispecific or bivalent in its ability to interact with additional proteins.

(81) As will be appreciated by those in the art and discussed more fully below, the heterodimeric fusion proteins of the present invention can take on a wide variety of configurations, as are generally depicted in FIGS. 1A-1M and 2A-2U. Some figures depict “single ended” configurations, where there is one type of specificity on one “arm” of the molecule and a different specificity on the other “arm”. Other figures depict “dual ended” configurations, where there is at least one type of specificity at the “top” of the molecule and one or more different specificities at the “bottom” of the molecule. Furthermore as is shown, these two configurations can be combined, where there can be triple or quadruple specificities based on the particular combination. Thus, the present invention provides “multispecific” binding proteins, including multispecific antibodies. Thus, the present invention is directed to novel immunoglobulin compositions that co-engage at least a first and a second antigen. First and second antigens of the invention are herein referred to as antigen-1 and antigen-2 respectively.

(82) One heterodimeric scaffold that finds particular use in the present invention is the “triple F” or “bottle opener” scaffold format. In this embodiment, one heavy chain of the antibody contains an single chain Fv (“scFv”, as defined below) and the other heavy chain is a “regular” FAb format, comprising a variable heavy chain and a light chain. This structure is sometimes referred to herein as “triple F” format (scFv-FAb-Fc) or the “bottle-opener” format, due to a rough visual similarity to a bottle-opener (see FIG. 1B). The two chains are brought together by the use of amino acid variants in the constant regions (e.g. the Fc domain and/or the hinge region) that promote the formation of heterodimeric antibodies as is described more fully below.

(83) There are several distinct advantages to the present “triple F” format. As is known in the art, antibody analogs relying on two scFv constructs often have stability and aggregation problems, which can be alleviated in the present invention by the addition of a “regular” heavy and light chain pairing. In addition, as opposed to formats that rely on two heavy chains and two light chains, there is no issue with the incorrect pairing of heavy and light chains (e.g. heavy 1 pairing with light 2, etc.)

(84) In addition to all or part of a variant heavy constant domain, one or both of the monomers may contain one or two fusion partners, such that the heterodimers form multivalent proteins. As is generally depicted in the FIG. 64 of U.S. Ser. No. 13/648,951, hereby incorporated by reference with its accompanying legend, the fusion partners are depicted as A, B, C and D, with all combinations possible. In general, A, B, C and D are selected such that the heterodimer is at least bispecific or bivalent in its ability to interact with additional proteins. In the context of the present “triple F” format, generally A and B are an scFv and a Fv (as will be appreciated, either monomer can contain the scFv and the other the Fv/Fab) and then optionally one or two additional fusion partners.

(85) Furthermore, as outlined herein, additional amino acid variants may be introduced into the bispecific antibodies of the invention, to add additional functionalities. For example, amino acid changes within the Fc region can be added (either to one monomer or both) to facilitate increased ADCC or CDC (e.g. altered binding to Fcγ receptors); to allow or increase yield of the addition of toxins and drugs (e.g. for ADC), as well as to increase binding to FcRn and/or increase serum half-life of the resulting molecules. As is further described herein and as will be appreciated by those in the art, any and all of the variants outlined herein can be optionally and independently combined

with other variants.

(86) Similarly, another category of functional variants are “Fcγ ablation variants” or “Fc knock out (FcKO or KO) variants. In these embodiments, for some therapeutic applications, it is desirable to reduce or remove the normal binding of the Fc domain to one or more or all of the Fcγ receptors (e.g. FcγR1, FcγRIIa, FcγRIIb, FcγRIIIa, etc.) to avoid additional mechanisms of action. That is, for example, in many embodiments, particularly in the use of bispecific antibodies that bind CD3 monovalently and a tumor antigen on the other (e.g. CD19, her2/neu, etc.), it is generally desirable to ablate FcγRIIIa binding to eliminate or significantly reduce ADCC activity.

Definitions

(87) In order that the application may be more completely understood, several definitions are set forth below. Such definitions are meant to encompass grammatical equivalents.

(88) By “ablation” herein is meant a decrease or removal of activity. Thus for example, “ablating FcγR binding” means the Fc region amino acid variant has less than 50% starting binding as compared to an Fc region not containing the specific variant, with less than 70-80-90-95-98% loss of activity being preferred, and in general, with the activity being below the level of detectable binding in a Biacore assay. Of particular use in the ablation of FcγR binding are those shown in FIG. 7.

(89) By “ADCC” or “antibody dependent cell-mediated cytotoxicity” as used herein is meant the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell. ADCC is correlated with binding to FcγRIIIa; increased binding to FcγRIIIa leads to an increase in ADCC activity.

(90) By “ADCP” or antibody dependent cell-mediated phagocytosis as used herein is meant the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause phagocytosis of the target cell.

(91) By “modification” herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence or an alteration to a moiety chemically linked to a protein. For example, a modification may be an altered carbohydrate or PEG structure attached to a protein. By “amino acid modification” herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence. For clarity, unless otherwise noted, the amino acid modification is always to an amino acid coded for by DNA, e.g. the 20 amino acids that have codons in DNA and RNA.

(92) By “amino acid substitution” or “substitution” herein is meant the replacement of an amino acid at a particular position in a parent polypeptide sequence with a different amino acid. In particular, in some embodiments, the substitution is to an amino acid that is not naturally occurring at the particular position, either not naturally occurring within the organism or in any organism. For example, the substitution E272Y refers to a variant polypeptide, in this case an Fc variant, in which the glutamic acid at position 272 is replaced with tyrosine. For clarity, a protein which has been engineered to change the nucleic acid coding sequence but not change the starting amino acid (for example exchanging CGG (encoding arginine) to CGA (still encoding arginine) to increase host organism expression levels) is not an “amino acid substitution”; that is, despite the creation of a new gene encoding the same protein, if the protein has the same amino acid at the particular position that it started with, it is not an amino acid substitution.

(93) By “amino acid insertion” or “insertion” as used herein is meant the addition of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, –233E or 233E designates an insertion of glutamic acid after position 233 and before position 234. Additionally, –233ADE or A233ADE designates an insertion of AlaAspGlu after position 233 and before position 234.

(94) By “amino acid deletion” or “deletion” as used herein is meant the removal of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, E233– or E233 # or E233() designates a deletion of glutamic acid at position 233. Additionally, EDA233– or EDA233 # designates a deletion of the sequence GluAspAla that begins at position 233.

(95) By “variant protein” or “protein variant”, or “variant” as used herein is meant a protein that differs from that of a parent protein by virtue of at least one amino acid modification. Protein variant may refer to the protein itself, a composition comprising the protein, or the amino sequence that encodes it. Preferably, the protein variant has at least one amino acid modification compared to the parent protein, e.g. from about one to about seventy amino acid modifications, and preferably from about one to about five amino acid modifications compared to the parent. As described below, in some embodiments the parent polypeptide, for example an Fc parent polypeptide, is a human wild type sequence, such as the Fc region from IgG1, IgG2, IgG3 or IgG4, although human sequences with variants can also serve as “parent polypeptides”, for example the IgG1/2 hybrid of FIG. 13. The protein variant sequence herein will preferably possess at least about 80% identity with a parent protein sequence, and most preferably at least about 90% identity, more preferably at least about 95-98-99% identity. Variant protein can refer to the variant protein itself, compositions comprising the protein variant, or the DNA sequence that encodes it. Accordingly, by “antibody variant” or “variant antibody” as used herein is meant an antibody that differs from a parent antibody by virtue of at least one amino acid modification, “IgG variant” or “variant IgG” as used herein is meant an antibody that differs from a parent IgG (again, in many cases, from a human IgG sequence) by virtue of at least one amino acid modification, and “immunoglobulin variant” or “variant immunoglobulin” as used herein is meant an immunoglobulin sequence that differs from that of a parent immunoglobulin sequence by virtue of at least one amino acid modification. “Fc variant” or “variant Fc” as used herein is meant a protein comprising an amino acid modification in an Fc domain. The Fc variants of the present invention are defined according to the amino acid modifications that compose them. Thus, for example, N434S or 434S is an Fc variant with the substitution serine at position 434 relative to the parent Fc polypeptide, wherein the numbering is according to the EU index. Likewise, M428L/N434S defines an Fc variant with the substitutions M428L and N434S relative to the parent Fc polypeptide. The identity of the WT amino acid may be unspecified, in which case the aforementioned variant is referred to as 428L/434S. It is noted that the order in which substitutions are provided is arbitrary, that is to say that, for example, 428L/434S is the same Fc variant as M428L/N434S, and so on. For all positions discussed in the present invention that relate to antibodies, unless otherwise noted, amino acid position numbering is according to the EU index. The EU index or EU index as in Kabat or EU numbering scheme refers to the numbering of the EU antibody (Edelman et al., 1969, Proc Natl Acad Sci USA 63:78-85, hereby entirely incorporated by reference.) The modification can be an addition, deletion, or substitution. Substitutions can include naturally occurring amino acids and, in some cases, synthetic amino acids. Examples include U.S. Pat. No. 6,586,207; WO 98/48032; WO 03/073238; US2004-0214988A1; WO 05/35727A2; WO 05/74524A2; J. W. Chin et al., (2002), Journal of the American Chemical Society 124:9026-9027; J. W. Chin, & P. G. Schultz, (2002), ChemBioChem 11:1135-1137; J. W. Chin, et al., (2002), PICAS United States of America 99:11020-11024; and, L. Wang, & P. G. Schultz, (2002), Chem. 1-10, all entirely incorporated by reference.

(96) As used herein, “protein” herein is meant at least two covalently attached amino acids, which includes proteins, polypeptides, oligopeptides and peptides. The peptidyl group may comprise naturally occurring amino acids and peptide bonds, or synthetic peptidomimetic structures, i.e. “analogs”, such as peptoids (see Simon et al., PNAS USA 89(20):9367 (1992), entirely incorporated by reference). The amino acids may either be naturally occurring or synthetic (e.g. not an amino acid that is coded for by DNA); as will be appreciated by those in the art. For example, homo-phenylalanine, citrulline, ornithine and noreleucine are considered synthetic amino acids for the purposes of the invention, and both D- and L-(R or S) configured amino acids may be utilized. The variants of the present invention may comprise modifications that include the use of synthetic amino acids incorporated using, for example, the technologies developed by Schultz and colleagues, including but not limited to methods described by Cropp & Shultz, 2004, Trends Genet. 20(12):625-30, Anderson et al., 2004, Proc Natl Acad Sci USA 101 (2):7566-71, Zhang et al., 2003,

303(5656):371-3, and Chin et al., 2003, Science 301(5635):964-7, all entirely incorporated by reference. In addition, polypeptides may include synthetic derivatization of one or more side chains or termini, glycosylation, PEGylation, circular permutation, cyclization, linkers to other molecules, fusion to proteins or protein domains, and addition of peptide tags or labels.

(97) By “residue” as used herein is meant a position in a protein and its associated amino acid identity. For example, Asparagine 297 (also referred to as Asn297 or N297) is a residue at position 297 in the human antibody IgG1.

(98) By “Fab” or “Fab region” as used herein is meant the polypeptide that comprises the VH, CHL VL, and CL immunoglobulin domains. Fab may refer to this region in isolation, or this region in the context of a full length antibody, antibody fragment or Fab fusion protein. By “Fv” or “Fv fragment” or “Fv region” as used herein is meant a polypeptide that comprises the VL and VH domains of a single antibody.

(99) By “IgG subclass modification” or “isotype modification” as used herein is meant an amino acid modification that converts one amino acid of one IgG isotype to the corresponding amino acid in a different, aligned IgG isotype. For example, because IgG1 comprises a tyrosine and IgG2 a phenylalanine at EU position 296, a F296Y substitution in IgG2 is considered an IgG subclass modification.

(100) By “non-naturally occurring modification” as used herein is meant an amino acid modification that is not isotypic. For example, because none of the IgGs comprise a serine at position 434, the substitution 434S in IgG1, IgG2, IgG3, or IgG4 (or hybrids thereof) is considered a non-naturally occurring modification.

(101) By “amino acid” and “amino acid identity” as used herein is meant one of the 20 naturally occurring amino acids that are coded for by DNA and RNA.

(102) By “effector function” as used herein is meant a biochemical event that results from the interaction of an antibody Fc region with an Fc receptor or ligand. Effector functions include but are not limited to ADCC, ADCP, and CDC.

(103) By “IgG Fc ligand” as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an IgG antibody to form an Fc/Fc ligand complex. Fc ligands include but are not limited to FcγR1s, FcγR2s, FcγR3s, FcRn, C1q, C3, mannan binding lectin, mannose receptor, staphylococcal protein A, streptococcal protein G, and viral FcγR. Fc ligands also include Fc receptor homologs (FcRH), which are a family of Fc receptors that are homologous to the FcγRs (Davis et al., 2002, Immunological Reviews 190:123-136, entirely incorporated by reference). Fc ligands may include undiscovered molecules that bind Fc. Particular IgG Fc ligands are FcRn and Fc gamma receptors. By “Fc ligand” as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an antibody to form an Fc/Fc ligand complex.

(104) By “Fc gamma receptor”, “FcγR” or “FcγR” as used herein is meant any member of the family of proteins that bind the IgG antibody Fc region and is encoded by an FcγR gene. In humans this family includes but is not limited to FcγR1 (CD64), including isoforms FcγR1a, FcγR1b, and FcγR1c; FcγR2 (CD32), including isoforms FcγR2a (including allotypes H131 and R131), FcγR2b (including FcγR2b-1 and FcγR2b-2), and FcγR2c; and FcγR3 (CD16), including isoforms FcγR3a (including allotypes V158 and F158) and FcγR3b (including allotypes FcγR3b-NA1 and FcγR3b-NA2) (Jefferis et al., 2002, Immunol Lett 82:57-65, entirely incorporated by reference), as well as any undiscovered human FcγRs or FcγR isoforms or allotypes. An FcγR may be from any organism, including but not limited to humans, mice, rats, rabbits, and monkeys. Mouse FcγRs include but are not limited to FcγR1 (CD64), FcγR2 (CD32), FcγR3 (CD16), and FcγR3-2 (CD16-2), as well as any undiscovered mouse FcγRs or FcγR isoforms or allotypes.

(105) By “FcRn” or “neonatal Fc Receptor” as used herein is meant a protein that binds the IgG antibody Fc region and is encoded at least in part by an FcRn gene. The FcRn may be from any

organism, including but not limited to humans, mice, rats, rabbits, and monkeys. As is known in the art, the functional FcRn protein comprises two polypeptides, often referred to as the heavy chain and light chain. The light chain is beta-2-microglobulin and the heavy chain is encoded by the FcRn gene. Unless otherwise noted herein, FcRn or an FcRn protein refers to the complex of FcRn heavy chain with beta-2-microglobulin. A variety of FcRn variants used to increase binding to the FcRn receptor, and in some cases, to increase serum half-life, are shown in the Figure Legend of FIG. 11.

(106) By “parent polypeptide” as used herein is meant a starting polypeptide that is subsequently modified to generate a variant. The parent polypeptide may be a naturally occurring polypeptide, or a variant or engineered version of a naturally occurring polypeptide. Parent polypeptide may refer to the polypeptide itself, compositions that comprise the parent polypeptide, or the amino acid sequence that encodes it. Accordingly, by “parent immunoglobulin” as used herein is meant an unmodified immunoglobulin polypeptide that is modified to generate a variant, and by “parent antibody” as used herein is meant an unmodified antibody that is modified to generate a variant antibody. It should be noted that “parent antibody” includes known commercial, recombinantly produced antibodies as outlined below.

(107) By “Fc fusion protein” or “immunoadhesin” herein is meant a protein comprising an Fc region, generally linked (optionally through a linker moiety, as described herein) to a different protein, such as a binding moiety to a target protein, as described herein. In some cases, one monomer of the heterodimeric protein comprises an antibody heavy chain (either including an scFv or further including a light chain) and the other monomer is a Fc fusion, comprising a variant Fc domain and a ligand. In some embodiments, these “half antibody-half fusion proteins” are referred to as “Fusionbodies”.

(108) By “position” as used herein is meant a location in the sequence of a protein. Positions may be numbered sequentially, or according to an established format, for example the EU index for antibody numbering.

(109) By “target antigen” as used herein is meant the molecule that is bound specifically by the variable region of a given antibody. A target antigen may be a protein, carbohydrate, lipid, or other chemical compound. A wide number of suitable target antigens are described below.

(110) By “strandedness” in the context of the monomers of the heterodimeric proteins of the invention herein is meant that, similar to the two strands of DNA that “match”, heterodimerization variants are incorporated into each monomer so as to preserve the ability to “match” to form heterodimers. For example, if some pI variants are engineered into monomer A (e.g. making the pI higher) then steric variants that are “charge pairs” that can be utilized as well do not interfere with the pI variants, e.g. the charge variants that make a pI higher are put on the same “strand” or “monomer” to preserve both functionalities.

(111) By “target cell” as used herein is meant a cell that expresses a target antigen.

(112) By “variable region” as used herein is meant the region of an immunoglobulin that comprises one or more Ig domains substantially encoded by any of the V.kappa., V.lamda., and/or VH genes that make up the kappa, lambda, and heavy chain immunoglobulin genetic loci respectively.

(113) By “wild type or WT” herein is meant an amino acid sequence or a nucleotide sequence that is found in nature, including allelic variations. A WT protein has an amino acid sequence or a nucleotide sequence that has not been intentionally modified.

(114) The antibodies of the present invention are generally isolated or recombinant. “Isolated,” when used to describe the various polypeptides disclosed herein, means a polypeptide that has been identified and separated and/or recovered from a cell or cell culture from which it was expressed. Ordinarily, an isolated polypeptide will be prepared by at least one purification step. An “isolated antibody,” refers to an antibody which is substantially free of other antibodies having different antigenic specificities.

(115) “Specific binding” or “specifically binds to” or is “specific for” a particular antigen or an

epitope means binding that is measurably different from a non-specific interaction. Specific binding can be measured, for example, by determining binding of a molecule compared to binding of a control molecule, which generally is a molecule of similar structure that does not have binding activity. For example, specific binding can be determined by competition with a control molecule that is similar to the target.

(116) Specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a KD for an antigen or epitope of at least about 10^{-4} M, at least about 10^{-5} M, at least about 10^{-6} M, at least about 10^{-7} M, at least about 10^{-8} M, at least about 10^{-9} M, alternatively at least about 10^{-10} M, at least about 10^{-11} M, at least about 10^{-12} M, or greater, where KD refers to a dissociation rate of a particular antibody-antigen interaction. Typically, an antibody that specifically binds an antigen will have a KD that is 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for a control molecule relative to the antigen or epitope.

(117) Also, specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a KA or Ka for an antigen or epitope of at least 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for the epitope relative to a control, where KA or Ka refers to an association rate of a particular antibody-antigen interaction.

Heterodimeric Proteins

(118) The present invention is directed to the generation of multispecific, particularly bispecific binding proteins, and in particular, multispecific antibodies. The present invention generally relies on the use of engineered or variant Fc domains that can self-assemble in production cells to produce heterodimeric proteins, and methods to generate and purify such heterodimeric proteins.

Antibodies

(119) The present invention relates to the generation of multispecific antibodies, generally therapeutic antibodies. As is discussed below, the term “antibody” is used generally. Antibodies that find use in the present invention can take on a number of formats as described herein, including traditional antibodies as well as antibody derivatives, fragments and mimetics, described below. In general, the term “antibody” includes any polypeptide that includes at least one constant domain, including, but not limited to, CH1, CH2, CH3 and CL.

(120) Traditional antibody structural units typically comprise a tetramer. Each tetramer is typically composed of two identical pairs of polypeptide chains, each pair having one “light” (typically having a molecular weight of about 25 kDa) and one “heavy” chain (typically having a molecular weight of about 50-70 kDa). Human light chains are classified as kappa and lambda light chains. The present invention is directed to the IgG class, which has several subclasses, including, but not limited to IgG1, IgG2, IgG3, and IgG4. Thus, “isotype” as used herein is meant any of the subclasses of immunoglobulins defined by the chemical and antigenic characteristics of their constant regions. It should be understood that therapeutic antibodies can also comprise hybrids of isotypes and/or subclasses. For example, as shown in US Publication 2009/0163699, incorporated by reference, the present invention covers pI engineering of IgG1/G2 hybrids.

(121) The amino-terminal portion of each chain includes a variable region of about 100 to 110 or more amino acids primarily responsible for antigen recognition, generally referred to in the art and herein as the “Fv domain” or “Fv region”. In the variable region, three loops are gathered for each of the V domains of the heavy chain and light chain to form an antigen-binding site. Each of the loops is referred to as a complementarity-determining region (hereinafter referred to as a “CDR”), in which the variation in the amino acid sequence is most significant. “Variable” refers to the fact that certain segments of the variable region differ extensively in sequence among antibodies. Variability within the variable region is not evenly distributed. Instead, the V regions consist of relatively invariant stretches called framework regions (FRs) of 15-30 amino acids separated by shorter regions of extreme variability called “hypervariable regions” that are each 9-15 amino acids long or longer.

(122) Each VH and VL is composed of three hypervariable regions (“complementary determining

regions,” “CDRs”) and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1-CDR1-FR2-CDR2-FR3-CDR3-FR4.

(123) The hypervariable region generally encompasses amino acid residues from about amino acid residues 24-34 (LCDR1; “L” denotes light chain), 50-56 (LCDR2) and 89-97 (LCDR3) in the light chain variable region and around about 31-35B (HCDR1; “H” denotes heavy chain), 50-65 (HCDR2), and 95-102 (HCDR3) in the heavy chain variable region; Kabat et al., SEQUENCES OF PROTEINS OF IMMUNOLOGICAL INTEREST, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991) and/or those residues forming a hypervariable loop (e.g. residues 26-32 (LCDR1), 50-52 (LCDR2) and 91-96 (LCDR3) in the light chain variable region and 26-32 (HCDR1), 53-55 (HCDR2) and 96-101 (HCDR3) in the heavy chain variable region; Chothia and Lesk (1987) J. Mol. Biol. 196:901-917. Specific CDRs of the invention are described below.

(124) Throughout the present specification, the Kabat numbering system is generally used when referring to a residue in the variable domain (approximately, residues 1-107 of the light chain variable region and residues 1-113 of the heavy chain variable region) and the EU numbering system for Fc regions (e.g., Kabat et al., supra (1991)).

(125) The CDRs contribute to the formation of the antigen-binding, or more specifically, epitope binding site of antibodies. “Epitope” refers to a determinant that interacts with a specific antigen binding site in the variable region of an antibody molecule known as a paratope. Epitopes are groupings of molecules such as amino acids or sugar side chains and usually have specific structural characteristics, as well as specific charge characteristics. A single antigen may have more than one epitope.

(126) The epitope may comprise amino acid residues directly involved in the binding (also called immunodominant component of the epitope) and other amino acid residues, which are not directly involved in the binding, such as amino acid residues which are effectively blocked by the specifically antigen binding peptide; in other words, the amino acid residue is within the footprint of the specifically antigen binding peptide.

(127) Epitopes may be either conformational or linear. A conformational epitope is produced by spatially juxtaposed amino acids from different segments of the linear polypeptide chain. A linear epitope is one produced by adjacent amino acid residues in a polypeptide chain. Conformational and nonconformational epitopes may be distinguished in that the binding to the former but not the latter is lost in the presence of denaturing solvents.

(128) An epitope typically includes at least 3, and more usually, at least 5 or 8-10 amino acids in a unique spatial conformation. Antibodies that recognize the same epitope can be verified in a simple immunoassay showing the ability of one antibody to block the binding of another antibody to a target antigen, for example “binning.”

(129) The carboxy-terminal portion of each chain defines a constant region primarily responsible for effector function. Kabat et al. collected numerous primary sequences of the variable regions of heavy chains and light chains. Based on the degree of conservation of the sequences, they classified individual primary sequences into the CDR and the framework and made a list thereof (see SEQUENCES OF IMMUNOLOGICAL INTEREST, 5th edition, NIH publication, No. 91-3242, E. A. Kabat et al., entirely incorporated by reference).

(130) In the IgG subclass of immunoglobulins, there are several immunoglobulin domains in the heavy chain. By “immunoglobulin (Ig) domain” herein is meant a region of an immunoglobulin having a distinct tertiary structure. Of interest in the present invention are the heavy chain domains, including, the constant heavy (CH) domains and the hinge domains. In the context of IgG antibodies, the IgG isotypes each have three CH regions. Accordingly, “CH” domains in the context of IgG are as follows: “CH1” refers to positions 118-220 according to the EU index as in Kabat. “CH2” refers to positions 237-340 according to the EU index as in Kabat, and “CH3” refers to positions 341-447 according to the EU index as in Kabat. As shown herein and described below,

the pI variants can be in one or more of the CH regions, as well as the hinge region, discussed below.

(131) It should be noted that the sequences depicted herein start at the CH1 region, position 118; the variable regions are not included except as noted. For example, the first amino acid of SEQ ID NO: 2, while designated as position “1” in the sequence listing, corresponds to position 118 of the CH1 region, according to EU numbering.

(132) Another type of Ig domain of the heavy chain is the hinge region. By “hinge” or “hinge region” or “antibody hinge region” or “immunoglobulin hinge region” herein is meant the flexible polypeptide comprising the amino acids between the first and second constant domains of an antibody. Structurally, the IgG CH1 domain ends at EU position 220, and the IgG CH2 domain begins at residue EU position 237. Thus for IgG the antibody hinge is herein defined to include positions 221 (D221 in IgG1) to 236 (G236 in IgG1), wherein the numbering is according to the EU index as in Kabat. In some embodiments, for example in the context of an Fc region, the lower hinge is included, with the “lower hinge” generally referring to positions 226 or 230. As noted herein, pI variants can be made in the hinge region as well.

(133) The light chain generally comprises two domains, the variable light domain (containing the light chain CDRs and together with the variable heavy domains forming the Fv region), and a constant light chain region (often referred to as CL or C_κ).

(134) Another region of interest for additional substitutions, outlined below, is the Fc region. By “Fc” or “Fc region” or “Fc domain” as used herein is meant the polypeptide comprising the constant region of an antibody excluding the first constant region immunoglobulin domain and in some cases, part of the hinge. Thus Fc refers to the last two constant region immunoglobulin domains of IgA, IgD, and IgG, the last three constant region immunoglobulin domains of IgE and IgM, and the flexible hinge N-terminal to these domains. For IgA and IgM, Fc may include the J chain. For IgG, the Fc domain comprises immunoglobulin domains C_γ2 and C_γ3 (C_γ2 and C_γ3) and the lower hinge region between C_γ1 (C_γ1) and C_γ2 (C_γ2). Although the boundaries of the Fc region may vary, the human IgG heavy chain Fc region is usually defined to include residues C226 or P230 to its carboxyl-terminus, wherein the numbering is according to the EU index as in Kabat. In some embodiments, as is more fully described below, amino acid modifications are made to the Fc region, for example to alter binding to one or more Fc_γR receptors or to the FcRn receptor.

(135) Accordingly, in some embodiments the present invention provides heterodimeric antibodies that rely on the use of two different heavy chain variant Fc domains that will self-assemble to form heterodimeric antibodies.

(136) In some embodiments, the antibodies are full length. By “full length antibody” herein is meant the structure that constitutes the natural biological form of an antibody, including variable and constant regions, including one or more modifications as outlined herein, particularly in the Fc domains to allow either heterodimerization formation or the purification of heterodimers away from homodimers. A full length heterodimeric antibody is two heavy chains with different Fc domains and either two light chains or a common light chain.

(137) Alternatively, the antibodies can include a variety of structures as are generally shown in the Figures, including, but not limited to, antibody fragments, monoclonal antibodies, bispecific antibodies, minibodies, domain antibodies, synthetic antibodies (sometimes referred to herein as “antibody mimetics”), chimeric antibodies, humanized antibodies, antibody fusions (sometimes referred to as “antibody conjugates”), and fragments of each, respectively.

(138) In one embodiment, the antibody is an antibody fragment, as long as it contains at least one constant domain which can be engineered to produce heterodimers, such as pI engineering. Other antibody fragments that can be used include fragments that contain one or more of the CH1, CH2, CH3, hinge and CL domains of the invention that have been pI engineered. For example, Fc fusions are fusions of the Fc region (CH2 and CH3, optionally with the hinge region) fused to another protein. A number of Fc fusions are known the art and can be improved by the addition of the

heterodimerization variants of the invention. In the present case, antibody fusions can be made comprising CH1; CH1, CH2 and CH3; CH2; CH3; CH2 and CH3; CH1 and CH3, any or all of which can be made optionally with the hinge region, utilizing any combination of heterodimerization variants described herein.

scFv Embodiments

(139) In some embodiments of the present invention, one monomer comprises a heavy chain comprises a scFV linked to an Fc domain, and the other monomer comprises a heavy chain comprising a Fab linked to an Fc domain, e.g. a “typical” heavy chain, and a light chain. By “Fab” or “Fab region” as used herein is meant the polypeptide that comprises the VH, CHL VL, and CL immunoglobulin domains. Fab may refer to this region in isolation, or this region in the context of a full length antibody, antibody fragment or Fab fusion protein. By “Fv” or “Fv fragment” or “Fv region” as used herein is meant a polypeptide that comprises the VL and VH domains of a single antibody.

(140) Several of the heterodimeric antibody embodiments described herein rely on the use of one or more scFv domains, comprising the variable heavy and variable light chains, covalently linked using a linker, forming an antigen binding domain. Some embodiments herein use “standard” linkers, usually linkers of glycine and serine, as is well known in the art.

(141) The present invention further provides charged scFv linkers, to facilitate the separation in pI between a first and a second monomer. That is, by incorporating a charged scFv linker, either positive or negative (or both, in the case of scaffolds that use scFvs on different monomers), this allows the monomer comprising the charged linker to alter the pI without making further changes in the Fc domains. These charged linkers can be substituted into any scFv containing standard linkers. Again, as will be appreciated by those in the art, charged scFv linkers are used on the correct “strand” or monomer, according to the desired changes in pI. For example, as discussed herein, to make triple F format heterodimeric antibody, the original pI of the Fv region for each of the desired antigen binding domains are calculated, and one is chosen to make an scFv, and depending on the pI, either positive or negative linkers are chosen.

(142) In addition, in the case of anti-CD3 scFv regions, disulfide bonds can be engineered into the variable heavy and variable light chains to give additional stability. Suitable disulfide sequences in the context of anti-CD3 scFvs are shown in FIG. 8.

(143) Chimeric and Humanized Antibodies

(144) In some embodiments, the antibody can be a mixture from different species, e.g. a chimeric antibody and/or a humanized antibody. In general, both “chimeric antibodies” and “humanized antibodies” refer to antibodies that combine regions from more than one species. For example, “chimeric antibodies” traditionally comprise variable region(s) from a mouse (or rat, in some cases) and the constant region(s) from a human. “Humanized antibodies” generally refer to non-human antibodies that have had the variable-domain framework regions swapped for sequences found in human antibodies. Generally, in a humanized antibody, the entire antibody, except the CDRs, is encoded by a polynucleotide of human origin or is identical to such an antibody except within its CDRs. The CDRs, some or all of which are encoded by nucleic acids originating in a non-human organism, are grafted into the beta-sheet framework of a human antibody variable region to create an antibody, the specificity of which is determined by the engrafted CDRs. The creation of such antibodies is described in, e.g., WO 92/11018, Jones, 1986, Nature 321:522-525, Verhoeven et al., 1988, Science 239:1534-1536, all entirely incorporated by reference. “Backmutation” of selected acceptor framework residues to the corresponding donor residues is often required to regain affinity that is lost in the initial grafted construct (U.S. Pat. Nos. 5,530,101; 5,585,089; 5,693,761; 5,693,762; 6,180,370; 5,859,205; 5,821,337; 6,054,297; 6,407,213, all entirely incorporated by reference). The humanized antibody optimally also will comprise at least a portion of an immunoglobulin constant region, typically that of a human immunoglobulin, and thus will typically comprise a human Fc region. Humanized antibodies can also be generated using mice with a

genetically engineered immune system. Roque et al., 2004, *Biotechnol. Prog.* 20:639-654, entirely incorporated by reference. A variety of techniques and methods for humanizing and reshaping non-human antibodies are well known in the art (See Tsurushita & Vasquez, 2004, *Humanization of Monoclonal Antibodies*, *Molecular Biology of B Cells*, 533-545, Elsevier Science (USA), and references cited therein, all entirely incorporated by reference). Humanization methods include but are not limited to methods described in Jones et al., 1986, *Nature* 321:522-525; Riechmann et al., 1988; *Nature* 332:323-329; Verhoeyen et al., 1988, *Science*, 239:1534-1536; Queen et al., 1989, *Proc Natl Acad Sci, USA* 86:10029-33; He et al., 1998, *J. Immunol.* 160: 1029-1035; Carter et al., 1992, *Proc Natl Acad Sci USA* 89:4285-9, Presta et al., 1997, *Cancer Res.* 57(20):4593-9; Gorman et al., 1991, *Proc. Natl. Acad. Sci. USA* 88:4181-4185; O'Connor et al., 1998, *Protein Eng* 11:321-8, all entirely incorporated by reference. Humanization or other methods of reducing the immunogenicity of nonhuman antibody variable regions may include resurfacing methods, as described for example in Roguska et al., 1994, *Proc. Natl. Acad. Sci. USA* 91:969-973, entirely incorporated by reference. In one embodiment, the parent antibody has been affinity matured, as is known in the art. Structure-based methods may be employed for humanization and affinity maturation, for example as described in U.S. Ser. No. 11/004,590. Selection based methods may be employed to humanize and/or affinity mature antibody variable regions, including but not limited to methods described in Wu et al., 1999, *J. Mol. Biol.* 294:151-162; Baca et al., 1997, *J. Biol. Chem.* 272(16):10678-10684; Rosok et al., 1996, *J. Biol. Chem.* 271(37): 22611-22618; Rader et al., 1998, *Proc. Natl. Acad. Sci. USA* 95: 8910-8915; Krauss et al., 2003, *Protein Engineering* 16(10):753-759, all entirely incorporated by reference. Other humanization methods may involve the grafting of only parts of the CDRs, including but not limited to methods described in U.S. Ser. No. 09/810,510; Tan et al., 2002, *J. Immunol.* 169:1119-1125; De Pascalis et al., 2002, *J. Immunol.* 169:3076-3084, all entirely incorporated by reference.

(145) Heterodimeric Heavy Chain Constant Regions

(146) Accordingly, the present invention provides heterodimeric proteins based on the use of monomers containing variant heavy chain constant regions, and specifically the Fc domains, as a first domain. By “monomer” herein is meant one half of the heterodimeric protein. It should be noted that traditional antibodies are actually tetrameric (two heavy chains and two light chains). In the context of the present invention, one pair of heavy-light chains (if applicable, e.g. if the monomer comprises an Fab) is considered a “monomer”. Similarly, a heavy chain region comprising the scFv is considered a monomer. In the case where an Fv region is one fusion partner (e.g. heavy and light variable domains) and a non-antibody protein is another fusion partner, each “half” is considered a monomer. Essentially, each monomer comprises sufficient heavy chain constant region to allow heterodimerization engineering, whether that be all the constant region, e.g. CH1-hinge-CH2-CH3, the Fc region (CH2-CH3), or just the CH3 domain.

(147) The variant heavy chain constant regions can comprise all or part of the heavy chain constant region, including the full length construct, CH1-hinge-CH2-CH3, or portions thereof, including for example CH2-CH3 or CH3 alone. In addition, the heavy chain region of each monomer can be the same backbone (CH1-hinge-CH2-CH3 or CH2-CH3) or different. N- and C-terminal truncations and additions are also included within the definition; for example, some pI variants include the addition of charged amino acids to the C-terminus of the heavy chain domain.

(148) Thus, in general, one monomer of the present “triple F” construct is a scFv region-hinge-Fc domain) and the other is (VH-CH1-hinge-CH2-CH3 plus associated light chain), with heterodimerization variants, including steric, isotypic, charge steering, and pI variants, Fc and FcRn variants, ablation variants, and additional antigen binding domains (with optional linkers) included in these regions.

(149) In addition to the heterodimerization variants (e.g. steric and pI variants) outlined herein, the heavy chain regions may also contain additional amino acid substitutions, including changes for altering FcγR and FcRn binding as discussed below.

(150) In addition, some monomers can utilize linkers between the variant heavy chain constant region and the fusion partner. For the scFv portion of the “bottle-opener”, standard linkers as are known in the art can be used, or the charged scFv linkers described herein. In the case where additional fusion partners are made (e.g. FIGS. 1A-1M and 2A-2U), traditional peptide linkers can be used, including flexible linkers of glycine and serine, or the charged linkers of FIG. 9. In some cases, the linkers for use as components of the monomer are different from those defined below for the ADC constructs, and are in many embodiments not cleavable linkers (such as those susceptible to proteases), although cleavable linkers may find use in some embodiments.

(151) The heterodimerization variants include a number of different types of variants, including, but not limited to, steric variants (including charge variants) and pI variants, that can be optionally and independently combined with any other variants. In these embodiments, it is important to match “monomer A” with “monomer B”; that is, if a heterodimeric protein relies on both steric variants and pI variants, these need to be correctly matched to each monomer: e.g. the set of steric variants that work (1 set on monomer A, 1 set on monomer B) is combined with pI variant sets (1 set on monomer A, 1 set on monomer B), such that the variants on each monomer are designed to achieve the desired function, keeping in mind the pI “strandedness” such that steric variants that may alter pI are put on the appropriate monomer.

(152) It is important to note that the heterodimerization variants outlined herein (for example, including but not limited to those variants shown in FIGS. 3A-3C and 12A-12J), can be optionally and independently combined with any other variants, and on any other monomer. That is, what is important for the heterodimerization is that there are “sets” of variants, one set for one monomer and one set for the other. Whether these are combined from the FIGS. 1A to 1M (e.g. monomer 1 listings can go together) or switched (monomer 1 pI variants with monomer 2 steric variants) is irrelevant. However, as noted herein, “strandedness” should be preserved when combinations are made as outlined above. Furthermore, for the additional Fc variants (such as for FcγR binding, FcRn binding, etc.), either monomer, or both monomers, can include any of the listed variants, independently and optionally. In some cases, both monomers have the additional variants and in some only one monomer has the additional variants, or they can be combined.

(153) Heterodimerization Variants

(154) The present invention provides heterodimeric proteins, including heterodimeric antibodies in a variety of formats, which utilize heterodimeric variants to allow for heterodimeric formation and/or purification away from homodimers.

(155) Steric Variants

(156) In some embodiments, the formation of heterodimers can be facilitated by the addition of steric variants. That is, by changing amino acids in each heavy chain, different heavy chains are more likely to associate to form the heterodimeric structure than to form homodimers with the same Fc amino acid sequences. Suitable steric variants are included in FIG. 3A-3C, and in FIGS. 12A, 12B, 12C, 12D, 12F and 12G.

(157) One mechanism is generally referred to in the art as “knobs and holes”, referring to amino acid engineering that creates steric influences to favor heterodimeric formation and disfavor homodimeric formation can also optionally be used; this is sometimes referred to as “knobs and holes”, as described in U.S. Ser. No. 61/596,846, Ridgway et al., Protein Engineering 9(7):617 (1996); Atwell et al., J. Mol. Biol. 1997 270:26; U.S. Pat. No. 8,216,805, all of which are hereby incorporated by reference in their entirety. The Figures identify a number of “monomer A-monomer B” pairs that rely on “knobs and holes”. In addition, as described in Merchant et al., Nature Biotech. 16:677 (1998), these “knobs and hole” mutations can be combined with disulfide bonds to skew formation to heterodimerization.

(158) An additional mechanism that finds use in the generation of heterodimers is sometimes referred to as “electrostatic steering” as described in Gunasekaran et al., J. Biol. Chem. 285(25):19637 (2010), hereby incorporated by reference in its entirety. This is sometimes referred

to herein as “charge pairs”. In this embodiment, electrostatics are used to skew the formation towards heterodimerization. As those in the art will appreciate, these may also have an effect on pI, and thus on purification, and thus could in some cases also be considered pI variants. However, as these were generated to force heterodimerization and were not used as purification tools, they are classified as “steric variants”. These include, but are not limited to, D221E/P228E/L368E paired with D221R/P228R/K409R (e.g. these are “monomer corresponding sets) and C220E/P228E/368E paired with C220R/E224R/P228R/K409R.

(159) Additional monomer A and monomer B variants that can be combined with other variants, optionally and independently in any amount, such as pI variants outlined herein or other steric variants that are shown in FIG. 37 of US 2012/0149876, the figure and legend and SEQ ID NOs of which are incorporated expressly by reference herein.

(160) In some embodiments, the steric variants outlined herein can be optionally and independently incorporated with any pI variant (or other variants such as Fc variants, FcRn variants, etc.) into one or both monomers, and can be independently and optionally included or excluded from the proteins of the invention.

(161) pI (Isoelectric Point) Variants for Heterodimers

(162) In general, as will be appreciated by those in the art, there are two general categories of pI variants: those that increase the pI of the protein (basic changes) and those that decrease the pI of the protein (acidic changes). As described herein, all combinations of these variants can be done: one monomer may be wild type, or a variant that does not display a significantly different pI from wild-type, and the other can be either more basic or more acidic. Alternatively, each monomer is changed, one to more basic and one to more acidic.

(163) Preferred combinations of pI variants are shown in FIGS. 3A-3C and 12E.

(164) Heavy Chain pI Changes

(165) A number of pI variants are shown in FIGS. 54A-54C and 55. As outlined herein and shown in the figures, these changes are shown relative to IgG1, but all isotypes can be altered this way, as well as isotype hybrids. In the case where the heavy chain constant domain is from IgG2-4, R133E and R133Q can also be used.

(166) Antibody Heterodimers Light Chain Variants

(167) In the case of antibody based heterodimers, e.g. where at least one of the monomers comprises a light chain in addition to the heavy chain domain, pI variants can also be made in the light chain. Amino acid substitutions for lowering the pI of the light chain include, but are not limited to, K126E, K126Q, K145E, K145Q, N152D, S156E, K169E, S202E, K207E and adding peptide DEDE at the c-terminus of the light chain. Changes in this category based on the constant lambda light chain include one or more substitutions at R108Q, Q124E, K126Q, N138D, K145T and Q199E. In addition, increasing the pI of the light chains can also be done.

(168) Isotypic Variants

(169) In addition, many embodiments of the invention rely on the “importation” of pI amino acids at particular positions from one IgG isotype into another, thus reducing or eliminating the possibility of unwanted immunogenicity being introduced into the variants. A number of these are shown in FIGS. 10A and 10B. That is, IgG1 is a common isotype for therapeutic antibodies for a variety of reasons, including high effector function. However, the heavy constant region of IgG1 has a higher pI than that of IgG2 (8.10 versus 7.31). By introducing IgG2 residues at particular positions into the IgG1 backbone, the pI of the resulting monomer is lowered (or increased) and additionally exhibits longer serum half-life. For example, IgG1 has a glycine (pI 5.97) at position 137, and IgG2 has a glutamic acid (pI 3.22); importing the glutamic acid will affect the pI of the resulting protein. As is described below, a number of amino acid substitutions are generally required to significantly affect the pI of the variant antibody. However, it should be noted as discussed below that even changes in IgG2 molecules allow for increased serum half-life.

(170) In other embodiments, non-isotypic amino acid changes are made, either to reduce the overall

charge state of the resulting protein (e.g. by changing a higher pI amino acid to a lower pI amino acid), or to allow accommodations in structure for stability, etc. as is more further described below. (171) In addition, by pI engineering both the heavy and light constant domains, significant changes in each monomer of the heterodimer can be seen. As discussed herein, having the pIs of the two monomers differ by at least 0.5 can allow separation by ion exchange chromatography or isoelectric focusing, or other methods sensitive to isoelectric point.

(172) Calculating pI

(173) The pI of each monomer can depend on the pI of the variant heavy chain constant domain and the pI of the total monomer, including the variant heavy chain constant domain and the fusion partner. Thus, in some embodiments, the change in pI is calculated on the basis of the variant heavy chain constant domain, using the chart in FIG. 53. As discussed herein, which monomer to engineer is generally decided by the inherent pI of the Fv and scaffold regions. Alternatively, the pI of each monomer can be compared.

(174) Heterodimeric Fc Fusion Proteins

(175) In addition to heterodimeric antibodies, the invention provides heterodimeric proteins that comprise a first monomer comprising a variant Fc region and a first fusion partner and a second monomer, also comprising a variant Fc region and a second fusion partner. The variant Fc regions are engineered as herein for antibodies, and are thus different, and in general the first and second fusion partners are different as well. In some cases, where one monomer is antibody based (e.g. either comprising a standard heavy and light chain or a Fc domain with an scFv) and the other is an Fc fusion protein, the resulting heterodimeric protein is called a “fusionbody”.

(176) pI Variants that Also Confer Better FcRn In Vivo Binding

(177) In the case where the pI variant decreases the pI of the monomer, they can have the added benefit of improving serum retention in vivo.

(178) Although still under examination, Fc regions are believed to have longer half-lives in vivo, because binding to FcRn at pH 6 in an endosome sequesters the Fc (Ghetie and Ward, 1997 Immunol Today. 18(12): 592-598, entirely incorporated by reference). The endosomal compartment then recycles the Fc to the cell surface. Once the compartment opens to the extracellular space, the higher pH, ~7.4, induces the release of Fc back into the blood. In mice, Dall'Acqua et al. showed that Fc mutants with increased FcRn binding at pH 6 and pH 7.4 actually had reduced serum concentrations and the same half life as wild-type Fc (Dall'Acqua et al. 2002, J. Immunol. 169:5171-5180, entirely incorporated by reference). The increased affinity of Fc for FcRn at pH 7.4 is thought to forbid the release of the Fc back into the blood. Therefore, the Fc mutations that will increase Fc's half-life in vivo will ideally increase FcRn binding at the lower pH while still allowing release of Fc at higher pH. The amino acid histidine changes its charge state in the pH range of 6.0 to 7.4. Therefore, it is not surprising to find His residues at important positions in the Fc/FcRn complex.

(179) Recently it has been suggested that antibodies with variable regions that have lower isoelectric points may also have longer serum half-lives (Igawa et al., 2010 PEDS. 23(5): 385-392, entirely incorporated by reference). However, the mechanism of this is still poorly understood. Moreover, variable regions differ from antibody to antibody. Constant region variants with reduced pI and extended half-life would provide a more modular approach to improving the pharmacokinetic properties of antibodies, as described herein.

(180) pI variants that find use in this embodiment, as well as their use for purification optimization, are disclosed in FIG. 20A-20K.

(181) Combination of Heterodimeric Variants

(182) As will be appreciated by those in the art, all of the recited heterodimerization variants can be optionally and independently combined in any way, as long as they retain their “strandedness” or “monomer partition”. In addition, all of these variants can be combined into any of the heterodimerization formats.

(183) In the case of pI variants, while embodiments finding particular use are shown in the Figures, other combinations can be generated, following the basic rule of altering the pI difference between two monomers to facilitate purification.

(184) Nucleic Acids of the Invention

(185) The invention further provides nucleic acid compositions encoding the heterodimeric proteins of the invention. As will be appreciated by those in the art, the nucleic acid compositions will depend on the format and scaffold of the heterodimeric protein. Thus, for example, when the format requires three amino acid sequences, such as for the triple F format (e.g. a first amino acid monomer comprising an Fc domain and a scFv, a second amino acid monomer comprising a heavy chain and a light chain), three nucleic acid sequences can be incorporated into one or more expression vectors for expression. Similarly, some formats (e.g. dual scFv formats such as disclosed in FIG. 1M) only two nucleic acids are needed; again, they can be put into one or two expression vectors.

(186) Target Antigens

(187) The heterodimeric proteins of the invention may target virtually any antigens. The “triple F” format is particularly beneficial for targeting two (or more) distinct antigens. (As outlined herein, this targeting can be any combination of monovalent and divalent binding, depending on the format). Thus the immunoglobulins herein preferably co-engage two target antigens, although in some cases, three or four antigens can be monovalently engaged. Each monomer's specificity can be selected from the lists below. While the triple F immunoglobulins described herein are particularly beneficial for targeting distinct antigens, in some cases it may be beneficial to target only one antigen. That is, each monomer may have specificity for the same antigen.

(188) Particular suitable applications of the heterodimeric proteins herein are co-target pairs for which it is beneficial or critical to engage each target antigen monovalently. Such antigens may be, for example, immune receptors that are activated upon immune complexation. Cellular activation of many immune receptors occurs only by cross-linking, achieved typically by antibody/antigen immune complexes, or via effector cell to target cell engagement. For some immune receptors, for example the CD3 signaling receptor on T cells, activation only upon engagement with co-engaged target is critical, as nonspecific cross-linking in a clinical setting can elicit a cytokine storm and toxicity. Therapeutically, by engaging such antigens monovalently rather than multivalently, using the immunoglobulins herein, such activation occurs only in response to cross-linking only in the microenvironment of the primary target antigen. The ability to target two different antigens with different valencies is a novel and useful aspect of the present invention. Examples of target antigens for which it may be therapeutically beneficial or necessary to co-engage monovalently include but are not limited to immune activating receptors such as CD3, FcγRs, toll-like receptors (TLRs) such as TLR4 and TLR9, cytokine, chemokine, cytokine receptors, and chemokine receptors. In many embodiments, one of the antigen binding sites binds to CD3, and in some embodiments it is the scFv-containing monomer.

(189) Virtually any antigen may be targeted by the immunoglobulins herein, including but not limited to proteins, subunits, domains, motifs, and/or epitopes belonging to the following list of target antigens, which includes both soluble factors such as cytokines and membrane-bound factors, including transmembrane receptors: 17-IA, 4-1BB, 4Dc, 6-keto-PGF1a, 8-iso-PGF2a, 8-oxo-dG, A1 Adenosine Receptor, A33, ACE, ACE-2, Activin, Activin A, Activin AB, Activin B, Activin C, Activin RIA, Activin RIA ALK-2, Activin RIB ALK-4, Activin RIIA, Activin RIIB, ADAM, ADAM10, ADAM12, ADAM15, ADAM17/TACE, ADAMS, ADAMS, ADAMTS, ADAMTS4, ADAMTS5, Addressins, aFGF, ALCAM, ALK, ALK-1, ALK-7, alpha-1-antitrypsin, alpha-V/beta-1 antagonist, ANG, Ang, APAF-1, APE, APJ, APP, APRIL, AR, ARC, ART, Artemin, anti-Id, ASPARTIC, Atrial natriuretic factor, av/b3 integrin, Axl, b2M, B7-1, B7-2, B7-H, B-lymphocyte Stimulator (BlyS), BACE, BACE-1, Bad, BAFF, BAFF-R, Bag-1, BAK, Bax, BCA-1, BCAM, Bcl, BCMA, BDNF, b-ECGF, bFGF, BID, Bik, BIM, BLC, BL-CAM, BLK, BMP, BMP-2

BMP-2a, BMP-3 Osteogenin, BMP-4 BMP-2b, BMP-5, BMP-6 Vgr-1, BMP-7 (OP-1), BMP-8 (BMP-8a, OP-2), BMPR, BMPR-IA (ALK-3), BMPR-IB (ALK-6), BRK-2, RPK-1, BMPR-II (BRK-3), BMPs, b-NGF, BOK, Bombesin, Bone-derived neurotrophic factor, BPDE, BPDE-DNA, BTC, complement factor 3 (C3), C3a, C4, C5, C5a, C10, CA125, CAD-8, Calcitonin, cAMP, carcinoembryonic antigen (CEA), carcinoma-associated antigen, Cathepsin A, Cathepsin B, Cathepsin C/DPPI, Cathepsin D, Cathepsin E, Cathepsin H, Cathepsin L, Cathepsin O, Cathepsin S, Cathepsin V, Cathepsin X/Z/P, CBL, CCI, CCK2, CCL, CCL1, CCL11, CCL12, CCL13, CCL14, CCL15, CCL16, CCL17, CCL18, CCL19, CCL2, CCL20, CCL21, CCL22, CCL23, CCL24, CCL25, CCL26, CCL27, CCL28, CCL3, CCL4, CCL5, CCL6, CCL7, CCL8, CCL9/10, CCR, CCR1, CCR10, CCR10, CCR2, CCR3, CCR4, CCR5, CCR6, CCR7, CCR8, CCR9, CD1, CD2, CD3, CD3E, CD4, CD5, CD6, CD7, CD8, CD10, CD11a, CD11b, CD11c, CD13, CD14, CD15, CD16, CD18, CD19, CD20, CD21, CD22, CD23, CD25, CD27L, CD28, CD29, CD30, CD30L, CD32, CD33 (p67 proteins), CD34, CD38, CD40, CD40L, CD44, CD45, CD46, CD49a, CD52, CD54, CD55, CD56, CD61, CD64, CD66e, CD74, CD80 (B7-1), CD89, CD95, CD123, CD137, CD138, CD140a, CD146, CD147, CD148, CD152, CD164, CEACAM5, CFTR, cGMP, CINC, *Clostridium botulinum* toxin, *Clostridium perfringens* toxin, CKb8-1, CLC, CMV, CMV UL, CNTF, CNTN-1, COX, C-Ret, CRG-2, CT-1, CTACK, CTGF, CTLA-4, CX3CL1, CX3CR1, CXCL, CXCL1, CXCL2, CXCL3, CXCL4, CXCL5, CXCL6, CXCL7, CXCL8, CXCL9, CXCL10, CXCL11, CXCL12, CXCL13, CXCL14, CXCL15, CXCL16, CXCR, CXCR1, CXCR2, CXCR3, CXCR4, CXCR5, CXCR6, cytokeratin tumor-associated antigen, DAN, DCC, DcR3, DC-SIGN, Decay accelerating factor, des(1-3)-IGF-I (brain IGF-1), Dhh, digoxin, DNAM-1, Dnase, Dpp, DPPIV/CD26, Dtk, ECAD, EDA, EDA-A1, EDA-A2, EDAR, EGF, EGFR (ErbB-1), EMA, EMMPRIN, ENA, endothelin receptor, Enkephalinase, eNOS, Eot, eotaxin1, EpCAM, Ephrin B2/EphB4, EPO, ERCC, E-selectin, ET-1, Factor IIa, Factor VII, Factor VIII, Factor IX, fibroblast activation protein (FAP), Fas, FcR1, FEN-1, Ferritin, FGF, FGF-19, FGF-2, FGF3, FGF-8, FGFR, FGFR-3, Fibrin, FL, FLIP, Flt-3, Flt-4, Follicle stimulating hormone, Fractalkine, FZD1, FZD2, FZD3, FZD4, FZD5, FZD6, FZD7, FZD8, FZD9, FZD10, G250, Gas 6, GCP-2, GCSF, GD2, GD3, GDF, GDF-1, GDF-3 (Vgr-2), GDF-5 (BMP-14, CDMP-1), GDF-6 (BMP-13, CDMP-2), GDF-7 (BMP-12, CDMP-3), GDF-8 (Myostatin), GDF-9, GDF-15 (MIC-1), GDNF, GDNF, GFAP, GFRa-1, GFR-alpha1, GFR-alpha2, GFR-alpha3, GITR, Glucagon, Glut 4, glycoprotein IIb/IIIa (GP IIb/IIIa), GM-CSF, gp130, gp72, GRO, Growth hormone releasing factor, Hapten (NP-cap or NIP-cap), HB-EGF, HCC, HCMV gB envelope glycoprotein, HCMV) gH envelope glycoprotein, HCMV UL, Hemopoietic growth factor (HGF), Hep B gp120, heparanase, Her2, Her2/neu (ErbB-2), Her3 (ErbB-3), Her4 (ErbB-4), herpes simplex virus (HSV) gB glycoprotein, HSV gD glycoprotein, HGFA, High molecular weight melanoma-associated antigen (HMW-MAA), HIV gp120, HIV IIIB gp 120 V3 loop, HLA, HLA-DR, HM1.24, HMFG PEM, HRG, Hrk, human cardiac myosin, human cytomegalovirus (HCMV), human growth hormone (HGH), HVEM, 1-309, IAP, ICAM, ICAM-1, ICAM-3, ICE, ICOS, IFNg, Ig, IgA receptor, IgE, IGF, IGF binding proteins, IGF-1R, IGFBP, IGF-I, IGF-II, IL, IL-1, IL-1R, IL-2, IL-2R, IL-4, IL-4R, IL-5, IL-5R, IL-6, IL-6R, IL-8, IL-9, IL-10, IL-12, IL-13, IL-15, IL-18, IL-18R, IL-23, interferon (INF)-alpha, INF-beta, INF-gamma, Inhibin, iNOS, Insulin A-chain, Insulin B-chain, Insulin-like growth factor 1, integrin alpha2, integrin alpha3, integrin alpha4, integrin alpha4/beta1, integrin alpha4/beta7, integrin alpha5 (alphaV), integrin alpha5/beta1, integrin alpha5/beta3, integrin alpha6, integrin beta1, integrin beta2, interferon gamma, IP-10, I-TAC, JE, Kallikrein 2, Kallikrein 5, Kallikrein 6, Kallikrein 11, Kallikrein 12, Kallikrein 14, Kallikrein 15, Kallikrein L1, Kallikrein L2, Kallikrein L3, Kallikrein L4, KC, KDR, Keratinocyte Growth Factor (KGF), laminin 5, LAMP, LAP, LAP (TGF-1), Latent TGF-1, Latent TGF-1 bpl, LBP, LDGF, LECT2, Lefty, Lewis-Y antigen, Lewis-Y related antigen, LFA-1, LFA-3, Lfo, LIF, LIGHT, lipoproteins, LIX, LKN, Lptn, L-Selectin, LT-a, LT-b, LTB4, LTBP-1, Lung surfactant, Luteinizing hormone, Lymphotoxin Beta Receptor, Mac-1, MAdCAM, MAG, MAP2, MARC, MCAM, MCAM, MCK-2, MCP, M-CSF, MDC, Mer,

METALLOPROTEASES, MGDf receptor, MGMT, MHC (HLA-DR), MIF, MIG, MIP, MIP-1-alpha, MK, MMAC1, MMP, MMP-1, MMP-10, MMP-11, MMP-12, MMP-13, MMP-14, MMP-15, MMP-2, MMP-24, MMP-3, MMP-7, MMP-8, MMP-9, MPIF, Mpo, MSK, MSP, mucin (Muc1), MUC18, Muellerian-inhibitin substance, Mug, MuSK, NAIP, NAP, NCAD, N-Cadherin, NCA 90, NCAM, NCAM, Neprilysin, Neurotrophin-3, -4, or -6, Neurturin, Neuronal growth factor (NGF), NGFR, NGF-beta, nNOS, NO, NOS, Npn, NRG-3, NT, NTN, OB, OGG1, OPG, OPN, OSM, OX40L, OX40R, p150, p95, PADPr, Parathyroid hormone, PARC, PARP, PBR, PBSF, PCAD, P-Cadherin, PCNA, PDGF, PDGF, PDK-1, PECAM, PEM, PF4, PGE, PGF, PGI2, PGJ2, PIN, PLA2, placental alkaline phosphatase (PLAP), P1GF, PLP, PP14, Proinsulin, Prorelaxin, Protein C, PS, PSA, PSCA, prostate specific membrane antigen (PSMA), PTEN, PTHrp, Ptk, PTN, R51, RANK, RANKL, RANTES, RANTES, Relaxin A-chain, Relaxin B-chain, renin, respiratory syncytial virus (RSV) F, RSV Fgp, Ret, Rheumatoid factors, RLIP76, RPA2, RSK, 5100, SCF/KL, SDF-1, SERINE, Serum albumin, sFRP-3, Shh, SIGIRR, SK-1, SLAM, SLPI, SMAC, SMDF, SMOH, SOD, SPARC, Stat, STEAP, STEAP-II, TACE, TACI, TAG-72 (tumor-associated glycoprotein-72), TARC, TCA-3, T-cell receptors (e.g., T-cell receptor alpha/beta), TdT, TECK, TEM1, TEM5, TEM7, TEM8, TERT, testicular PLAP-like alkaline phosphatase, TfR, TGF, TGF-alpha, TGF-beta, TGF-beta Pan Specific, TGF-beta RI (ALK-5), TGF-beta RH, TGF-beta RHb, TGF-beta RIII, TGF-beta1, TGF-beta2, TGF-beta3, TGF-beta4, TGF-beta5, Thrombin, Thymus Ck-1, Thyroid stimulating hormone, Tie, TIMP, TIQ, Tissue Factor, TMEFF2, Tmpo, TMPRSS2, TNF, TNF-alpha, TNF-alpha beta, TNF-beta2, TNF-c, TNF-RI, TNF-RII, TNFRSF10A (TRAIL R1 Apo-2, DR4), TNFRSF10B (TRAIL R2 DR5, KILLER, TRICK-2A, TRICK-B), TNFRSF10C (TRAIL R3 DcR1, LIT, TRID), TNFRSF10D (TRAIL R4 DcR2, TRUND), TNFRSF11A (RANK ODF R, TRANCE R), TNFRSF11B (OPG OCIF, TR1), TNFRSF12 (TWEAK R FN14), TNFRSF13B (TACI), TNFRSF13C (BAFF R), TNFRSF14 (HVEM ATAR, HveA, LIGHT R, TR2), TNFRSF16 (NGFR p75NTR), TNFRSF17 (BCMA), TNFRSF18 (GITR ATTR), TNFRSF19 (TROY TAJ, TRADE), TNFRSF19L (RELT), TNFRSF1A (TNF RI CD120a, p55-60), TNFRSF1B (TNF RII CD120b, p75-80), TNFRSF26 (TNFRH3), TNFRSF3 (LTbR TNF RIII, TNFC R), TNFRSF4 (OX40 ACT35, TXGP1 R), TNFRSF5 (CD40 p50), TNFRSF6 (Fas Apo-1, APT1, CD95), TNFRSF6B (DcR3 M68, TR6), TNFRSF7 (CD27), TNFRSF8 (CD30), TNFRSF9 (4-1BB CD137, ILA), TNFRSF21 (DR6), TNFRSF22 (DcTRAIL R2 TNFRH2), TNFRST23 (DcTRAIL R1 TNFRH1), TNFRSF25 (DR3 Apo-3, LARD, TR-3, TRAMP, WSL-1), TNFSF10 (TRAIL Apo-2 Ligand, TL2), TNFSF11 (TRANCE/RANK Ligand ODF, OPG Ligand), TNFSF12 (TWEAK Apo-3 Ligand, DR3 Ligand), TNFSF13 (APRIL TALL2), TNFSF13B (BAFF BLYS, TALL1, THANK, TNFSF20), TNFSF14 (LIGHT HVEM Ligand, LTg), TNFSF15 (TL1A/VEGI), TNFSF18 (GITR Ligand AITR Ligand, TL6), TNFSF1A (TNF- α Conectin, DIF, TNFSF2), TNFSF1B (TNF-b LTa, TNFSF1), TNFSF3 (LTb TNFC, p33), TNFSF4 (OX40 Ligand gp34, TXGP1), TNFSF5 (CD40 Ligand CD154, gp39, HIGM1, IMD3, TRAP), TNFSF6 (Fas Ligand Apo-1 Ligand, APT1 Ligand), TNFSF7 (CD27 Ligand CD70), TNFSF8 (CD30 Ligand CD153), TNFSF9 (4-1BB Ligand CD137 Ligand), TP-1, t-PA, Tpo, TRAIL, TRAIL R, TRAIL-R1, TRAIL-R2, TRANCE, transferring receptor, TRF, Trk, TROP-2, TSG, TSLP, tumor-associated antigen CA 125, tumor-associated antigen expressing Lewis Y related carbohydrate, TWEAK, TXB2, Ung, uPAR, uPAR-1, Urokinase, VCAM, VCAM-1, VECAD, VE-Cadherin, VE-cadherin-2, VEGFR-1 (flt-1), VEGF, VEGFR, VEGFR-3 (flt-4), VEGI, VIM, Viral antigens, VLA, VLA-1, VLA-4, VNR integrin, von Willebrands factor, WIF-1, WNT1, WNT2, WNT2B/13, WNT3, WNT3A, WNT4, WNT5A, WNT5B, WNT6, WNT7A, WNT7B, WNT8A, WNT8B, WNT9A, WNT9A, WNT9B, WNT10A, WNT10B, WNT11, WNT16, XCL1, XCL2, XCR1, XCR1, XEDAR, XIAP, XPD, and receptors for hormones and growth factors. To form the bispecific or trispecific antibodies of the invention, antibodies to any combination of these antigens can be made; that is, each of these antigens can be optionally and independently included or excluded from a multispecific antibody according to the present invention.

(190) Exemplary antigens that may be targeted specifically by the immunoglobulins of the invention include but are not limited to: CD20, CD19, Her2, EGFR, EpCAM, CD3, FcγRIIIa (CD16), FcγRIIa (CD32a), FcγRIIb (CD32b), FcγRT (CD64), Toll-like receptors (TLRs) such as TLR4 and TLR9, cytokines such as IL-2, IL-5, IL-13, IL-12, IL-23, and TNFα, cytokine receptors such as IL-2R, chemokines, chemokine receptors, growth factors such as VEGF and HGF, and the like. To form the multispecific antibodies of the invention, antibodies to any combination of these antigens can be made; that is, each of these antigens can be optionally and independently included or excluded from a multispecific antibody according to the present invention.

(191) Particularly preferred combinations for bispecific antibodies are an antigen-binding domain to CD3 and an antigen binding domain to CD19; an antigen-binding domain to CD3 and an antigen binding domain to CD33; an antigen-binding domain to CD3 and an antigen binding domain to CD 38. Again, in many embodiments, the CD3 binding domain is the scFv, having an exemplary sequence as depicted in the Figures and/or CD3 CDRs as outlined.

(192) The choice of suitable target antigens and co-targets depends on the desired therapeutic application. Some targets that have proven especially amenable to antibody therapy are those with signaling functions. Other therapeutic antibodies exert their effects by blocking signaling of the receptor by inhibiting the binding between a receptor and its cognate ligand. Another mechanism of action of therapeutic antibodies is to cause receptor down regulation. Other antibodies do not work by signaling through their target antigen. The choice of co-targets will depend on the detailed biology underlying the pathology of the indication that is being treated.

(193) Monoclonal antibody therapy has emerged as an important therapeutic modality for cancer (Weiner et al., 2010, Nature Reviews Immunology 10:317-327; Reichert et al., 2005, Nature Biotechnology 23[9]:1073-1078; herein expressly incorporated by reference). For anti-cancer treatment it may be desirable to target one antigen (antigen-1) whose expression is restricted to the cancerous cells while co-targeting a second antigen (antigen-2) that mediates some immunological killing activity. For other treatments it may be beneficial to co-target two antigens, for example two angiogenic factors or two growth factors, that are each known to play some role in proliferation of the tumor. Exemplary co-targets for oncology include but are not limited to HGF and VEGF, IGF-1R and VEGF, Her2 and VEGF, CD19 and CD3, CD20 and CD3, Her2 and CD3, CD19 and FcγRIIIa, CD20 and FcγRIIIa, Her2 and FcγRIIIa. An immunoglobulin of the invention may be capable of binding VEGF and phosphatidylserine; VEGF and ErbB3; VEGF and PLGF; VEGF and ROBO4; VEGF and BSG2; VEGF and CDCP1; VEGF and ANPEP; VEGF and c-MET; HER-2 and ERB3; HER-2 and BSG2; HER-2 and CDCP1; HER-2 and ANPEP; EGFR and CD64; EGFR and BSG2; EGFR and CDCP1; EGFR and ANPEP; IGF1R and PDGFR; IGF1R and VEGF; IGF1R and CD20; CD20 and CD74; CD20 and CD30; CD20 and DR4; CD20 and VEGFR2; CD20 and CD52; CD20 and CD4; HGF and c-MET; HGF and NRP1; HGF and phosphatidylserine; ErbB3 and IGF1R; ErbB3 and IGF1,2; c-Met and Her-2; c-Met and NRP1; c-Met and IGF1R; IGF1,2 and PDGFR; IGF1,2 and CD20; IGF1,2 and IGF1R; IGF2 and EGFR; IGF2 and HER2; IGF2 and CD20; IGF2 and VEGF; IGF2 and IGF1R; IGF1 and IGF2; PDGFRa and VEGFR2; PDGFRa and PLGF; PDGFRa and VEGF; PDGFRa and c-Met; PDGFRa and EGFR; PDGFRb and VEGFR2; PDGFRb and c-Met; PDGFRb and EGFR; RON and c-Met; RON and MTSP1; RON and MSP; RON and CDCP1; VGFR1 and PLGF; VGFR1 and RON; VGFR1 and EGFR; VEGFR2 and PLGF; VEGFR2 and NRP1; VEGFR2 and RON; VEGFR2 and DLL4; VEGFR2 and EGFR; VEGFR2 and ROBO4; VEGFR2 and CD55; LPA and SIP; EPHB2 and RON; CTLA4 and VEGF; CD3 and EPCAM; CD40 and IL6; CD40 and IGF; CD40 and CD56; CD40 and CD70; CD40 and VEGFR1; CD40 and DR5; CD40 and DR4; CD40 and APRIL; CD40 and BCMA; CD40 and RANKL; CD28 and MAPG; CD80 and CD40; CD80 and CD30; CD80 and CD33; CD80 and CD74; CD80 and CD2; CD80 and CD3; CD80 and CD19; CD80 and CD4; CD80 and CD52; CD80 and VEGF; CD80 and DR5; CD80 and VEGFR2; CD22 and CD20; CD22 and CD80; CD22 and CD40; CD22 and CD23; CD22 and CD33; CD22 and CD74; CD22 and CD19;

CD22 and DR5; CD22 and DR4; CD22 and VEGF; CD22 and CD52; CD30 and CD20; CD30 and CD22; CD30 and CD23; CD30 and CD40; CD30 and VEGF; CD30 and CD74; CD30 and CD19; CD30 and DR5; CD30 and DR4; CD30 and VEGFR2; CD30 and CD52; CD30 and CD4; CD138 and RANKL; CD33 and FTL3; CD33 and VEGF; CD33 and VEGFR2; CD33 and CD44; CD33 and DR4; CD33 and DR5; DR4 and CD137; DR4 and IGF1,2; DR4 and IGF1R; DR4 and DR5; DR5 and CD40; DR5 and CD137; DR5 and CD20; DR5 and EGFR; DR5 and IGF1,2; DR5 and IGFR, DR5 and HER-2, and EGFR and DLL4. Other target combinations include one or more members of the EGF/erb-2/erb-3 family.

(194) Other targets (one or more) involved in oncological diseases that the immunoglobulins herein may bind include, but are not limited to those selected from the group consisting of: CD52, CD20, CD19, CD3, CD4, CD8, BMP6, IL12A, IL1A, IL1B, IL2, IL24, INHA, TNF, TNFSF10, BMP6, EGF, FGF1, FGF10, FGF11, FGF12, FGF13, FGF14, FGF16, FGF17, FGF18, FGF19, FGF2, FGF20, FGF21, FGF22, FGF23, FGF3, FGF4, FGF5, FGF6, FGF7, FGF8, FGF9, GRP, IGF1, IGF2, IL12A, IL1A, IL1B, IL2, INHA, TGFA, TGFB1, TGFB2, TGFB3, VEGF, CDK2, FGF10, FGF18, FGF2, FGF4, FGF7, IGF1R, IL2, BCL2, CD164, CDKN1A, CDKN1B, CDKN1C, CDKN2A, CDKN2B, CDKN2C, CDKN3, GNRH1, IGFBP6, IL1A, IL1B, ODZ1, PAWR, PLG, TGFB1I1, AR, BRCA1, CDK3, CDK4, CDK5, CDK6, CDK7, CDK9, E2F1, EGFR, ENO1, ERBB2, ESR1, ESR2, IGFBP3, IGFBP6, IL2, INSL4, MYC, NOX5, NR6A1, PAP, PCNA, PRKCQ, PRKD1, PRL, TP53, FGF22, FGF23, FGF9, IGFBP3, IL2, INHA, KLK6, TP53, CHGB, GNRH1, IGF1, IGF2, INHA, INSL3, INSL4, PRL, KLK6, SHBG, NR1D1, NR1H3, NR1I3, NR2F6, NR4A3, ESR1, ESR2, NROB1, NROB2, NR1D2, NR1H2, NR1H4, NR1I2, NR2C1, NR2C2, NR2E1, NR2E3, NR2F1, NR2F2, NR3C1, NR3C2, NR4A1, NR4A2, NR5A 1, NR5A2, NR6 PGR, RARB, FGF1, FGF2, FGF6, KLK3, KRT1, APOC1, BRCA1, CHGA, CHGB, CLU, COL1A1, COL6A1, EGF, ERBB2, ERK8, FGF1, FGF10, FGF11, FGF13, FGF14, FGF16, FGF17, FGF18, FGF2, FGF20, FGF21, FGF22, FGF23, FGF3, FGF4, FGF5, FGF6, FGF7, FGF8, FGF9, GNRH1, IGF1, IGF2, IGFBP3, IGFBP6, IL12A, IL1A, IL1B, IL2, IL24, INHA, INSL3, INSL4, KLK10, KLK12, KLK13, KLK14, KLK15, KLK3, KLK4, KLK5, KLK6, KLK9, MMP2, MMP9, MSMB, NTN4, ODZ1, PAP, PLAUI, PRL, PSAP, SERPINA3, SHBG, TGFA, TIMP3, CD44, CDH1, CDH10, CDH19, CDH20, CDH7, CDH9, CDH1, CDH10, CDH13, CDH18, CDH19, CDH20, CDH7, CDH8, CDH9, ROBO2, CD44, ILK, ITGA1, APC, CD164, COL6A1, MTSS1, PAP, TGFB1I1, AGR2, AIG1, AKAP1, AKAP2, CANT1, CAV1, CDH12, CLDN3, CLN3, CYB5, CYC1, DAB2IP, DES, DNCL1, ELAC2, ENO2, ENO3, FASN, FLJ12584, FLJ25530, GAGEB1, GAGEC1, GGT1, GSTP1, HIP1, HUMCYT2A, IL29, K6HF, KAI1, KRT2A, MD31, PART1, PATE, PCA3, PIAS2, PIK3CG, PPID, PR1, PSCA, SLC2A2, SLC33 SLC43 STEAP, STEAP2, TPM1, TPM2, TRPC6, ANGPT1, ANGPT2, ANPEP, ECGF1, EREG, FGF1, FGF2, FIGF, FLT1, JAG1, KDR, LAMAS, NRP1, NRP2, PGF, PLXDC1, STAB 1, VEGF, VEGFC, ANGPTL3, BAIL, COL4A3, IL8, LAMAS, NRP1, NRP2, STAB 1, ANGPTL4, PECAM1, PF4, PROK2, SERPINF1, TNFAIP2, CCL11, CCL2, CXCL1, CXCL10, CXCL3, CXCL5, CXCL6, CXCL9, IFNA1, IFNB1, IFNG, IL1B, IL6, MDK, EDG1, EFNA1, EFNA3, EFNB2, EGF, EPHB4, FGFR3, HGF, IGF1, ITGB3, PDGFA, TEK, TGFA, TGFB1, TGFB2, TGFB3, CCL2, CDH5, COL1A1, EDG1, ENG, ITGAV, ITGB3, THBS1, THBS2, BAD, BAG1, BCL2, CCNA1, CCNA2, CCND1, CCNE1, CCNE2, CDH1 (E-cadherin), CDKN1B (p27Kip1), CDKN2A (p16INK4a), COL6A1, CTNNB1 (b-catenin), CTSB (cathepsin B), ERBB2 (Her-2), ESR1, ESR2, F3 (TF), FOSL1 (FRA-1), GATA3, GSN (Gelsolin), IGFBP2, IL2RA, IL6, IL6R, IL6ST (glycoprotein 130), ITGA6 (a6 integrin), JUN, KLK5, KRT19, MAP2K7 (c-Jun), MKI67 (Ki-67), NGFB (GF), NGFR, NME1 (M23A), PGR, PLAUI (uPA), PTEN, SERPINB5 (maspin), SERPINE1 (PAI-1), TGFA, THBS1 (thrombospondin-1), TIE (Tie-1), TNFRSF6 (Fas), TNFSF6 (FasL), TOP2A (topoisomerase Ea), TP53, AZGP1 (zinc-a-glycoprotein), BPAG1 (plectin), CDKN1A (p21Wap1/Cip1), CLDN7 (claudin-7), CLU (clusterin), ERBB2 (Her-2), FGF1, FLRT1 (fibronectin), GABRP (GABAa), GNAS1, ID2, ITGA6 (a6 integrin), ITGB4 (b 4 integrin), KLF5 (GC Box BP), KRT19 (Keratin

19), KRTHB6 (hair-specific type II keratin), MACMARCKS, MT3 (metallothionein-III), MUC1 (mucin), PTGS2 (COX-2), RAC2 (p21Rac2), S100A2, SCGB1D2 (lipophilin B), SCGB2A1 (mammaglobin 2), SCGB2A2 (mammaglobin 1), SPRR1B (Spr1), THBS1, THBS2, THBS4, and TNFAIP2 (B94), RON, c-Met, CD64, DLL4, PLGF, CTLA4, phosphatidylserine, ROBO4, CD80, CD22, CD40, CD23, CD28, CD80, CD55, CD38, CD70, CD74, CD30, CD138, CD56, CD33, CD2, CD137, DR4, DR5, RANKL, VEGFR2, PDGFR, VEGFR1, MTSP1, MSP, EPHB2, EPHA1, EPHA2, EpCAM, PGE2, NKG2D, LPA, SIP, APRIL, BCMA, MAPG, FLT3, PDGFR alpha, PDGFR beta, ROR1, PSMA, PSCA, SCD1, and CD59. To form the bispecific or trispecific antibodies of the invention, antibodies to any combination of these antigens can be made; that is, each of these antigens can be optionally and independently included or excluded from a multispecific antibody according to the present invention.

(195) Monoclonal antibody therapy has become an important therapeutic modality for treating autoimmune and inflammatory disorders (Chan & Carter, 2010, Nature Reviews Immunology 10:301-316; Reichert et al., 2005, Nature Biotechnology 23[9]:1073-1078; herein expressly incorporated by reference). Many proteins have been implicated in general autoimmune and inflammatory responses, and thus may be targeted by the immunoglobulins of the invention. Autoimmune and inflammatory targets include but are not limited to C5, CCL1 (1-309), CCL11 (eotaxin), CCL13 (mcp-4), CCL15 (MIP-1d), CCL16 (HCC-4), CCL17 (TARC), CCL18 (PARC), CCL19, CCL2 (mcp-1), CCL20 (MIP-3a), CCL21 (MIP-2), CCL23 (MPIF-1), CCL24 (MPIF-2/eotaxin-2), CCL25 (TECK), CCL26, CCL3 (MIP-1a), CCL4 (MIP-1b), CCL5 (RANTES), CCL7 (mcp-3), CCL8 (mcp-2), CXCL1, CXCL10 (IP-10), CXCL11 (1-TAC/IP-9), CXCL12 (SDF1), CXCL13, CXCL14, CXCL2, CXCL3, CXCL5 (ENA-78/LIX), CXCL6 (GCP-2), CXCL5, IL13, IL8, CCL13 (mcp-4), CCR1, CCR2, CCR3, CCR4, CCR5, CCR6, CCR7, CCR8, CCR9, CX3CR1, IL8RA, XCR1 (CCXCR1), IFNA2, IL10, IL13, IL17C, IL1A, IL1B, IL1F10, IL1F5, IL1F6, IL1F7, IL1F8, IL1F9, IL22, IL5, IL8, IL9, LTA, LTB, MIF, SCYE1 (endothelial Monocyte-activating cytokine), SPP1, TNF, TNFSF5, IFNA2, IL10RA, IL10RB, IL13, IL13RA1, IL5RA, IL9, IL9R, ABCF1, BCL6, C3, C4A, CEBPB, CRP, ICEBERG, IL1R1, IL1RN, IL8RB, LTB4R, TOLLIP, FADD, IRAK1, IRAK2, MYD88, NCK2, TNFAIP3, TRADD, TRAF1, TRAF2, TRAF3, TRAF4, TRAF5, TRAF6, ACVR1, ACVR1B, ACVR2, ACVR2B, ACVRL1, CD28, CD3E, CD3G, CD3Z, CD69, CD80, CD86, CNR1, CTLA4, CYSLTR1, FCER1A, FCER2, FCGR3A, GPR44, HAVCR2, OPRD1, P2RX7, TLR2, TLR3, TLR4, TLR5, TLR6, TLR7, TLR8, TLR9, TLR10, BLR1, CCL1, CCL2, CCL3, CCL4, CCL5, CCL7, CCL8, CCL11, CCL13, CCL15, CCL16, CCL17, CCL18, CCL19, CCL20, CCL21, CCL22, CCL23, CCL24, CCL25, CCR1, CCR2, CCR3, CCR4, CCR5, CCR6, CCR7, CCR8, CCR9, CX3CL1, CX3CR1, CXCL1, CXCL2, CXCL3, CXCL5, CXCL6, CXCL10, CXCL11, CXCL12, CXCL13, CXCR4, GPR2, SCYE1, SDF2, XCL1, XCL2, XCR1, AMH, AMHR2, BMPR1A, BMPR1B, BMPR2, C19orf10 (IL27w), CER1, CSF1, CSF2, CSF3, DKFZp451J0118, FGF2, GFI1, IFNA1, IFNB1, IFNG, IGF1, IL1A, IL1B, IL1R1, IL1R2, IL2, IL2RA, IL2RB, IL2RG, IL3, IL4, IL4R, IL5, IL5RA, IL6, IL6R, IL6ST, IL7, IL8, IL8RA, IL8RB, IL9, IL9R, IL10, IL10RA, IL10RB, IL11, IL12RA, IL12A, IL12B, IL12RB1, IL12RB2, IL13, IL13RA1, IL13RA2, IL15, IL15RA, IL16, IL17, IL17R, IL18, IL18R1, IL19, IL20, KITLG, LEP, LTA, LTB, LTB4R, LTB4R2, LTBR, MIF, NPPB, PDGFB, TBX21, TDGF1, TGFA, TGFB1, TGFB1I1, TGFB2, TGFB3, TGFB1, TGFB1R1, TGFB1R2, TGFB1R3, TH1L, TNF, TNFRSF1A, TNFRSF1B, TNFRSF7, TNFRSF8, TNFRSF9, TNFRSF11A, TNFRSF21, TNFSF4, TNFSF5, TNFSF6, TNFSF11, VEGF, ZFPM2, and RNF110 (ZNF144). To form the bispecific or trispecific antibodies of the invention, antibodies to any combination of these antigens can be made; that is, each of these antigens can be optionally and independently included or excluded from a multispecific antibody according to the present invention.

(196) Exemplary co-targets for autoimmune and inflammatory disorders include but are not limited to IL-1 and TNFalpha, IL-6 and TNFalpha, IL-6 and IL-1, IgE and IL-13, IL-1 and IL-13, IL-4 and IL-13, IL-5 and IL-13, IL-9 and IL-13, CD19 and FcγRIIb, and CD79 and FcγRIIb.

(197) Immunoglobulins of the invention with specificity for the following pairs of targets to treat inflammatory disease are contemplated: TNF and IL-17A; TNF and RANKL; TNF and VEGF; TNF and SOST; TNF and DKK; TNF and α V β 3; TNF and NGF; TNF and IL-23p19; TNF and IL-6; TNF and SOST; TNF and IL-6R; TNF and CD-20; IgE and IL-13; IL-13 and IL23p19; IgE and IL-4; IgE and IL-9; IgE and IL-9; IgE and IL-13; IL-13 and IL-9; IL-13 and IL-4; IL-13 and IL-9; IL-13 and IL-9; IL-13 and IL-4; IL-13 and IL-23p19; IL-13 and IL-9; IL-6R and VEGF; IL-6R and IL-17A; IL-6R and RANKL; IL-17A and IL-1 β ; IL-1 β and RANKL; IL-1 β and VEGF; RANKL and CD-20; IL-1 α and IL-1 β ; IL-1 α and IL-1 β .

(198) Pairs of targets that the immunoglobulins described herein can bind and be useful to treat asthma may be determined. In an embodiment, such targets include, but are not limited to, IL-13 and IL-1 β , since IL-1 β is also implicated in inflammatory response in asthma; IL-13 and cytokines and chemokines that are involved in inflammation, such as IL-13 and IL-9; IL-13 and IL-4; IL-13 and IL-5; IL-13 and IL-25; IL-13 and TARC; IL-13 and MDC; IL-13 and MIF; IL-13 and TGF- β ; IL-13 and LHR agonist; IL-13 and CL25; IL-13 and SPRR2a; IL-13 and SPRR2b; and IL-13 and ADAMS. The immunoglobulins herein may have specificity for one or more targets involved in asthma selected from the group consisting of CSF1 (MCSF), CSF2 (GM-CSF), CSF3 (GCSF), FGF2, IFNA1, IFNB1, IFNG, histamine and histamine receptors, IL1A, IL1B, IL2, IL3, IL4, IL5, IL6, IL7, IL8, IL9, IL10, IL11, IL12A, IL12B, IL13, IL14, IL15, IL16, IL17, IL18, IL19, KITLG, PDGFB, IL2RA, IL4R, IL5RA, IL8RA, IL8RB, IL12RB1, IL12RB2, IL13RA1, IL13RA2, IL18R1, TSLP, CCL1, CCL2, CCL3, CCL4, CCL5, CCL7, CCL8, CCL13, CCL17, CCL18, CCL19, CCL20, CCL22, CCL24, CX3CL1, CXCL1, CXCL2, CXCL3, XCL1, CCR2, CCR3, CCR4, CCR5, CCR6, CCR7, CCR8, CX3CR1, GPR2, XCR1, FOS, GATA3, JAK1, JAK3, STAT3, TBX21, TGFB1, TNF, TNFSF6, YY1, CYSLTR1, FCER1A, FCER2, LTB4R, TB4R2, LTBR, and Chitinase. To form the bispecific or trispecific antibodies of the invention, antibodies to any combination of these antigens can be made; that is, each of these antigens can be optionally and independently included or excluded from a multispecific antibody according to the present invention.

(199) Pairs of targets involved in rheumatoid arthritis (RA) may be co-targeted by the invention, including but not limited to TNF and IL-18; TNF and IL-12; TNF and IL-23; TNF and IL-1 β ; TNF and MIF; TNF and IL-17; and TNF and IL-15.

(200) Antigens that may be targeted in order to treat systemic lupus erythematosus (SLE) by the immunoglobulins herein include but are not limited to CD-20, CD-22, CD-19, CD28, CD4, CD80, HLA-DRA, IL10, IL2, IL4, TNFRSF5, TNFRSF6, TNFSF5, TNFSF6, BLR1, HDAC4, HDAC5, HDAC7A, HDAC9, ICOSL, IGBP1, MS4A1, RGS1, SLA2, CD81, IFNB1, IL10, TNFRSF5, TNFRSF7, TNFSF5, AICDA, BLNK, GALNAC4S-6ST, HDAC4, HDAC5, HDAC7A, HDAC9, IL10, IL11, IL4, INHA, INHBA, KLF6, TNFRSF7, CD28, CD38, CD69, CD80, CD83, CD86, DPP4, FCER2, IL2RA, TNFRSF8, TNFSF7, CD24, CD37, CD40, CD72, CD74, CD79A, CD79B, CR2, ILIR2, ITGA2, ITGA3, MS4A1, ST6GAL1, CDIC, CHST10, HLA-A, HLA-DRA, and NT5E.; CTLA4, B7.1, B7.2, BlyS, BAFF, C5, IL-4, IL-6, IL-10, IFN- α , and TNF- α . To form the bispecific or trispecific antibodies of the invention, antibodies to any combination of these antigens can be made; that is, each of these antigens can be optionally and independently included or excluded from a multispecific antibody according to the present invention.

(201) The immunoglobulins herein may target antigens for the treatment of multiple sclerosis (MS), including but not limited to IL-12, TWEAK, IL-23, CXCL13, CD40, CD40L, IL-18, VEGF, VLA-4, TNF, CD45RB, CD200, IFN γ , GM-CSF, FGF, C5, CD52, and CCR2. An embodiment includes co-engagement of anti-IL-12 and TWEAK for the treatment of MS.

(202) One aspect of the invention pertains to immunoglobulins capable of binding one or more targets involved in sepsis, in an embodiment two targets, selected from the group consisting TNF, IL-1, MIF, IL-6, IL-8, IL-18, IL-12, IL-23, FasL, LPS, Toll-like receptors, TLR-4, tissue factor, MIP-2, ADORA2A, CASP1, CASP4, IL-10, IL-1B, NF κ B1, PROC, TNFRSF1A, CSF3, CCR3,

ILIRN, MIF, NFκB1, PTAFR, TLR2, TLR4, GPR44, HMOX1, midkine, IRAK1, NFκB2, SERPINA1, SERPINE1, and TREM1. To form the bispecific or trispecific antibodies of the invention, antibodies to any combination of these antigens can be made; that is, each of these antigens can be optionally and independently included or excluded from a multispecific antibody according to the present invention.

(203) In some cases, immunoglobulins herein may be directed against antigens for the treatment of infectious diseases.

(204) Antigen Binding Domains

(205) As will be appreciated by those in the art, there are two basic types of antigen binding domains, those that resemble antibody antigen binding domains (e.g. comprising a set of 6 CDRs) and those that can be ligands or receptors, for example, that bind to targets without the use of CDRs.

(206) Modified Antibodies

(207) In addition to the modifications outlined above, other modifications can be made. For example, the molecules may be stabilized by the incorporation of disulphide bridges linking the VH and VL domains (Reiter et al., 1996, Nature Biotech. 14:1239-1245, entirely incorporated by reference). In addition, there are a variety of covalent modifications of antibodies that can be made as outlined below.

(208) Covalent modifications of antibodies are included within the scope of this invention, and are generally, but not always, done post-translationally. For example, several types of covalent modifications of the antibody are introduced into the molecule by reacting specific amino acid residues of the antibody with an organic derivatizing agent that is capable of reacting with selected side chains or the N- or C-terminal residues.

(209) Cysteinyl residues most commonly are reacted with α-haloacetates (and corresponding amines), such as chloroacetic acid or chloroacetamide, to give carboxymethyl or carboxyamidomethyl derivatives. Cysteinyl residues may also be derivatized by reaction with bromotrifluoroacetone, α-bromo-β-(5-imidozoyl)propionic acid, chloroacetyl phosphate, N-alkylmaleimides, 3-nitro-2-pyridyl disulfide, methyl 2-pyridyl disulfide, p-chloromercuribenzoate, 2-chloromercuri-4-nitrophenol, or chloro-7-nitrobenzo-2-oxa-1,3-diazole and the like.

(210) In addition, modifications at cysteines are particularly useful in antibody-drug conjugate (ADC) applications, further described below. In some embodiments, the constant region of the antibodies can be engineered to contain one or more cysteines that are particularly “thiol reactive”, so as to allow more specific and controlled placement of the drug moiety. See for example U.S. Pat. No. 7,521,541, incorporated by reference in its entirety herein.

(211) Histidyl residues are derivatized by reaction with diethylpyrocarbonate at pH 5.5-7.0 because this agent is relatively specific for the histidyl side chain. Para-bromophenacyl bromide also is useful; the reaction is preferably performed in 0.1M sodium cacodylate at pH 6.0.

(212) Lysinyl and amino terminal residues are reacted with succinic or other carboxylic acid anhydrides. Derivatization with these agents has the effect of reversing the charge of the lysinyl residues. Other suitable reagents for derivatizing alpha-amino-containing residues include imidoesters such as methyl picolinimidate; pyridoxal phosphate; pyridoxal; chloroborohydride; trinitrobenzenesulfonic acid; O-methylisourea; 2,4-pentanedione; and transaminase-catalyzed reaction with glyoxylate.

(213) Arginyl residues are modified by reaction with one or several conventional reagents, among them phenylglyoxal, 2,3-butanedione, 1,2-cyclohexanedione, and ninhydrin. Derivatization of arginine residues requires that the reaction be performed in alkaline conditions because of the high pKa of the guanidine functional group. Furthermore, these reagents may react with the groups of lysine as well as the arginine epsilon-amino group.

(214) The specific modification of tyrosyl residues may be made, with particular interest in introducing spectral labels into tyrosyl residues by reaction with aromatic diazonium compounds or

tetranitromethane. Most commonly, N-acetylimidazole and tetranitromethane are used to form O-acetyl tyrosyl species and 3-nitro derivatives, respectively. Tyrosyl residues are iodinated using 1251 or 1311 to prepare labeled proteins for use in radioimmunoassay, the chloramine T method described above being suitable.

(215) Carboxyl side groups (aspartyl or glutamyl) are selectively modified by reaction with carbodiimides ($R'-N=C=N-R'$), where R and R' are optionally different alkyl groups, such as 1-cyclohexyl-3-(2-morpholinyl-4-ethyl) carbodiimide or 1-ethyl-3-(4-azonia-4,4-dimethylpentyl) carbodiimide. Furthermore, aspartyl and glutamyl residues are converted to asparaginyl and glutaminyl residues by reaction with ammonium ions.

(216) Derivatization with bifunctional agents is useful for crosslinking antibodies to a water-insoluble support matrix or surface for use in a variety of methods, in addition to methods described below. Commonly used crosslinking agents include, e.g., 1,1-bis(diazoacetyl)-2-phenylethane, glutaraldehyde, N-hydroxysuccinimide esters, for example, esters with 4-azidosalicylic acid, homobifunctional imidoesters, including disuccinimidyl esters such as 3,3'-dithiobis (succinimidylpropionate), and bifunctional maleimides such as bis-N-maleimido-1,8-octane. Derivatizing agents such as methyl-3-[(p-azidophenyl)dithio]propioimide yield photoactivatable intermediates that are capable of forming crosslinks in the presence of light. Alternatively, reactive water-insoluble matrices such as cynomolgusogen bromide-activated carbohydrates and the reactive substrates described in U.S. Pat. Nos. 3,969,287; 3,691,016; 4,195,128; 4,247,642; 4,229,537; and 4,330,440, all entirely incorporated by reference, are employed for protein immobilization.

(217) Glutaminyl and asparaginyl residues are frequently deamidated to the corresponding glutamyl and aspartyl residues, respectively. Alternatively, these residues are deamidated under mildly acidic conditions. Either form of these residues falls within the scope of this invention.

(218) Other modifications include hydroxylation of proline and lysine, phosphorylation of hydroxyl groups of seryl or threonyl residues, methylation of the α -amino groups of lysine, arginine, and histidine side chains (T. E. Creighton, *Proteins: Structure and Molecular Properties*, W. H. Freeman & Co., San Francisco, pp. 79-86 [1983], entirely incorporated by reference), acetylation of the N-terminal amine, and amidation of any C-terminal carboxyl group.

(219) In addition, as will be appreciated by those in the art, labels (including fluorescent, enzymatic, magnetic, radioactive, etc. can all be added to the antibodies (as well as the other compositions of the invention).

(220) Glycosylation

(221) Another type of covalent modification is alterations in glycosylation. In another embodiment, the antibodies disclosed herein can be modified to include one or more engineered glycoforms. By "engineered glycoform" as used herein is meant a carbohydrate composition that is covalently attached to the antibody, wherein said carbohydrate composition differs chemically from that of a parent antibody. Engineered glycoforms may be useful for a variety of purposes, including but not limited to enhancing or reducing effector function. A preferred form of engineered glycoform is afucosylation, which has been shown to be correlated to an increase in ADCC function, presumably through tighter binding to the Fc γ RIIIa receptor. In this context, "afucosylation" means that the majority of the antibody produced in the host cells is substantially devoid of fucose, e.g. 90-95-98% of the generated antibodies do not have appreciable fucose as a component of the carbohydrate moiety of the antibody (generally attached at N297 in the Fc region). Defined functionally, afucosylated antibodies generally exhibit at least a 50% or higher affinity to the Fc γ RIIIa receptor.

(222) Engineered glycoforms may be generated by a variety of methods known in the art (Umali et al., 1999, *Nat Biotechnol* 17:176-180; Davies et al., 2001, *Biotechnol Bioeng* 74:288-294; Shields et al., 2002, *J Biol Chem* 277:26733-26740; Shinkawa et al., 2003, *J Biol Chem* 278:3466-3473; U.S. Pat. No. 6,602,684; U.S. Ser. No. 10/277,370; U.S. Ser. No. 10/113,929; PCT WO

00/61739A1; PCT WO 01/29246A1; PCT WO 02/31140A1; PCT WO 02/30954A1, all entirely incorporated by reference; (Potelligent® technology [Biowa, Inc., Princeton, NJ]; GlycoMAb® glycosylation engineering technology [Glycart Biotechnology AG, Zurich, Switzerland]). Many of these techniques are based on controlling the level of fucosylated and/or bisecting oligosaccharides that are covalently attached to the Fc region, for example by expressing an IgG in various organisms or cell lines, engineered or otherwise (for example Lec-13 CHO cells or rat hybridoma YB2/0 cells, by regulating enzymes involved in the glycosylation pathway (for example FUT8 [α 1,6-fucosyltransferase] and/or β 1-4-N-acetylglucosaminyltransferase III [GnTIII]), or by modifying carbohydrate(s) after the IgG has been expressed. For example, the “sugar engineered antibody” or “SEA technology” of Seattle Genetics functions by adding modified saccharides that inhibit fucosylation during production; see for example 20090317869, hereby incorporated by reference in its entirety. Engineered glycoform typically refers to the different carbohydrate or oligosaccharide; thus an antibody can include an engineered glycoform.

(223) Alternatively, engineered glycoform may refer to the IgG variant that comprises the different carbohydrate or oligosaccharide. As is known in the art, glycosylation patterns can depend on both the sequence of the protein (e.g., the presence or absence of particular glycosylation amino acid residues, discussed below), or the host cell or organism in which the protein is produced. Particular expression systems are discussed below.

(224) Glycosylation of polypeptides is typically either N-linked or O-linked. N-linked refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. The tri-peptide sequences asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline, are the recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain. Thus, the presence of either of these tri-peptide sequences in a polypeptide creates a potential glycosylation site. O-linked glycosylation refers to the attachment of one of the sugars N-acetylgalactosamine, galactose, or xylose, to a hydroxyamino acid, most commonly serine or threonine, although 5-hydroxyproline or 5-hydroxylysine may also be used.

(225) Addition of glycosylation sites to the antibody is conveniently accomplished by altering the amino acid sequence such that it contains one or more of the above-described tri-peptide sequences (for N-linked glycosylation sites). The alteration may also be made by the addition of, or substitution by, one or more serine or threonine residues to the starting sequence (for O-linked glycosylation sites). For ease, the antibody amino acid sequence is preferably altered through changes at the DNA level, particularly by mutating the DNA encoding the target polypeptide at preselected bases such that codons are generated that will translate into the desired amino acids.

(226) Another means of increasing the number of carbohydrate moieties on the antibody is by chemical or enzymatic coupling of glycosides to the protein. These procedures are advantageous in that they do not require production of the protein in a host cell that has glycosylation capabilities for N- and O-linked glycosylation. Depending on the coupling mode used, the sugar(s) may be attached to (a) arginine and histidine, (b) free carboxyl groups, (c) free sulfhydryl groups such as those of cysteine, (d) free hydroxyl groups such as those of serine, threonine, or hydroxyproline, (e) aromatic residues such as those of phenylalanine, tyrosine, or tryptophan, or (f) the amide group of glutamine. These methods are described in WO 87/05330 and in Aplin and Wriston, 1981, CRC Crit. Rev. Biochem., pp. 259-306, both entirely incorporated by reference.

(227) Removal of carbohydrate moieties present on the starting antibody (e.g. post-translationally) may be accomplished chemically or enzymatically. Chemical deglycosylation requires exposure of the protein to the compound trifluoromethanesulfonic acid, or an equivalent compound. This treatment results in the cleavage of most or all sugars except the linking sugar (N-acetylglucosamine or N-acetylgalactosamine), while leaving the polypeptide intact. Chemical deglycosylation is described by Hakimuddin et al., 1987, Arch. Biochem. Biophys. 259:52 and by Edge et al., 1981, Anal. Biochem. 118:131, both entirely incorporated by reference. Enzymatic cleavage of carbohydrate moieties on polypeptides can be achieved by the use of a variety of endo-

and exo-glycosidases as described by Thotakura et al., 1987, Meth. Enzymol. 138:350, entirely incorporated by reference. Glycosylation at potential glycosylation sites may be prevented by the use of the compound tunicamycin as described by Duskin et al., 1982, J. Biol. Chem. 257:3105, entirely incorporated by reference. Tunicamycin blocks the formation of protein-N-glycoside linkages.

(228) Another type of covalent modification of the antibody comprises linking the antibody to various nonproteinaceous polymers, including, but not limited to, various polyols such as polyethylene glycol, polypropylene glycol or polyoxyalkylenes, in the manner set forth in, for example, 2005-2006 PEG Catalog from Nektar Therapeutics (available at the Nektar website) U.S. Pat. Nos. 4,640,835; 4,496,689; 4,301,144; 4,670,417; 4,791,192 or 4,179,337, all entirely incorporated by reference. In addition, as is known in the art, amino acid substitutions may be made in various positions within the antibody to facilitate the addition of polymers such as PEG. See for example, U.S. Publication No. 2005/0114037A1, entirely incorporated by reference.

(229) Additional Fc Variants for Additional Functionality

(230) In addition to pI amino acid variants, there are a number of useful Fc amino acid modification that can be made for a variety of reasons, including, but not limited to, altering binding to one or more FcγR receptors, altered binding to FcRn receptors, etc.

(231) Accordingly, the proteins of the invention can include amino acid modifications, including the heterodimerization variants outlined herein, which includes the pI variants and steric variants. Each set of variants can be independently and optionally included or excluded from any particular heterodimeric protein.

(232) FcγR Variants

(233) Accordingly, there are a number of useful Fc substitutions that can be made to alter binding to one or more of the FcγR receptors. Substitutions that result in increased binding as well as decreased binding can be useful. For example, it is known that increased binding to FcγRIIIa generally results in increased ADCC (antibody dependent cell-mediated cytotoxicity; the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell). Similarly, decreased binding to FcγRIIb (an inhibitory receptor) can be beneficial as well in some circumstances. Amino acid substitutions that find use in the present invention include those listed in U.S. Ser. No. 11/124,620 (particularly FIG. 41), Ser. Nos. 11/174,287, 11/396,495, 11/538,406, all of which are expressly incorporated herein by reference in their entirety and specifically for the variants disclosed therein. Particular variants that find use include, but are not limited to, 236A, 239D, 239E, 332E, 332D, 239D/332E, 267D, 267E, 328F, 267E/328F, 236A/332E, 239D/332E/330Y, 239D, 332E/330L, 243A, 243L, 264A, 264V and 299T.

(234) In addition, there are additional Fc substitutions that find use in increased binding to the FcRn receptor and increased serum half life, as specifically disclosed in U.S. Ser. No. 12/341,769, hereby incorporated by reference in its entirety, including, but not limited to, 434S, 434A, 428L, 308F, 259I, 428L/434S, 259I/308F, 436I/428L, 436I or V/434S, 436V/428L and 259I/308F/428L.

(235) Linkers

(236) The present invention optionally provides linkers as needed, for example in the addition of additional antigen binding sites, as depicted for example in FIG. 2A-2U, where “the other end” of the molecule contains additional antigen binding components. In addition, as outlined below, linkers are optionally also used in antibody drug conjugate (ADC) systems. When used to join the components of the central mAb-Fv constructs, the linker is generally a polypeptide comprising two or more amino acid residues joined by peptide bonds and are used to link one or more of the components of the present invention. Such linker polypeptides are well known in the art (see e.g., Holliger, P., et al. (1993) Proc. Natl. Acad. Sci. USA 90:6444-6448; Poljak, R. J., et al. (1994) Structure 2:1121-1123). A variety of linkers may find use in some embodiments described herein. As will be appreciated by those in the art, there are at least three different linker types used in the

present invention.

(237) “Linker” herein is also referred to as “linker sequence”, “spacer”, “tethering sequence” or grammatical equivalents thereof. Homo- or hetero-bifunctional linkers as are well known (see, 1994 Pierce Chemical Company catalog, technical section on cross-linkers, pages 155-200, incorporated entirely by reference). A number of strategies may be used to covalently link molecules together. These include, but are not limited to polypeptide linkages between N- and C-termini of proteins or protein domains, linkage via disulfide bonds, and linkage via chemical cross-linking reagents. In one aspect of this embodiment, the linker is a peptide bond, generated by recombinant techniques or peptide synthesis. The linker peptide may predominantly include the following amino acid residues: Gly, Ser, Ala, or Thr. The linker peptide should have a length that is adequate to link two molecules in such a way that they assume the correct conformation relative to one another so that they retain the desired activity. In one embodiment, the linker is from about 1 to 50 amino acids in length, preferably about 1 to 30 amino acids in length. In one embodiment, linkers of 1 to 20 amino acids in length may be used. Useful linkers include glycine-serine polymers, including for example (GS)_n, (GSGGS)_n, (GGGGS)_n, and (GGGS)_n, where n is an integer of at least one, glycine-alanine polymers, alanine-serine polymers, and other flexible linkers. Alternatively, a variety of nonproteinaceous polymers, including but not limited to polyethylene glycol (PEG), polypropylene glycol, polyoxyalkylenes, or copolymers of polyethylene glycol and polypropylene glycol, may find use as linkers, that is may find use as linkers.

(238) Other linker sequences may include any sequence of any length of CL/CH1 domain but not all residues of CL/CH1 domain; for example the first 5-12 amino acid residues of the CL/CH1 domains. Linkers can be derived from immunoglobulin light chain, for example C_κ or C_λ. Linkers can be derived from immunoglobulin heavy chains of any isotype, including for example C_γ1, C_γ2, C_γ3, C_γ4, C_α1, C_α2, C_δ, C_ε, and C_μ. Linker sequences may also be derived from other proteins such as Ig-like proteins (e.g. TCR, FcR, KIR), hinge region-derived sequences, and other natural sequences from other proteins.

(239) Antibody-Drug Conjugates

(240) In some embodiments, the multispecific antibodies of the invention are conjugated with drugs to form antibody-drug conjugates (ADCs). In general, ADCs are used in oncology applications, where the use of antibody-drug conjugates for the local delivery of cytotoxic or cytostatic agents allows for the targeted delivery of the drug moiety to tumors, which can allow higher efficacy, lower toxicity, etc. An overview of this technology is provided in Ducry et al., *Bioconjugate Chem.*, 21:5-13 (2010), Carter et al., *Cancer J.* 14(3):154 (2008) and Senter, *Current Opin. Chem. Biol.* 13:235-244 (2009), all of which are hereby incorporated by reference in their entirety.

(241) Thus the invention provides multispecific antibodies conjugated to drugs. Generally, conjugation is done by covalent attachment to the antibody, as further described below, and generally relies on a linker, often a peptide linkage (which, as described below, may be designed to be sensitive to cleavage by proteases at the target site or not). In addition, as described above, linkage of the linker-drug unit (LU-D) can be done by attachment to cysteines within the antibody. As will be appreciated by those in the art, the number of drug moieties per antibody can change, depending on the conditions of the reaction, and can vary from 1:1 to 10:1 drug:antibody. As will be appreciated by those in the art, the actual number is an average.

(242) Thus the invention provides multispecific antibodies conjugated to drugs. As described below, the drug of the ADC can be any number of agents, including but not limited to cytotoxic agents such as chemotherapeutic agents, growth inhibitory agents, toxins (for example, an enzymatically active toxin of bacterial, fungal, plant, or animal origin, or fragments thereof), or a radioactive isotope (that is, a radioconjugate) are provided. In other embodiments, the invention further provides methods of using the ADCs.

(243) Drugs for use in the present invention include cytotoxic drugs, particularly those which are used for cancer therapy. Such drugs include, in general, DNA damaging agents, anti-metabolites, natural products and their analogs. Exemplary classes of cytotoxic agents include the enzyme inhibitors such as dihydrofolate reductase inhibitors, and thymidylate synthase inhibitors, DNA intercalators, DNA cleavers, topoisomerase inhibitors, the anthracycline family of drugs, the *vinca* drugs, the mitomycins, the bleomycins, the cytotoxic nucleosides, the pteridine family of drugs, diynenes, the podophyllotoxins, dolastatins, maytansinoids, differentiation inducers, and taxols.

(244) Members of these classes include, for example, methotrexate, methopterin, dichloromethotrexate, 5-fluorouracil, 6-mercaptopurine, cytosine arabinoside, melphalan, leurosine, leurosideine, actinomycin, daunorubicin, doxorubicin, mitomycin C, mitomycin A, caminomycin, aminopterin, tallysomyacin, podophyllotoxin and podophyllotoxin derivatives such as etoposide or etoposide phosphate, vinblastine, vincristine, vindesine, taxanes including taxol, taxotere retinoic acid, butyric acid, N8-acetyl spermidine, camptothecin, calicheamicin, esperamicin, ene-diynes, duocarmycin A, duocarmycin SA, calicheamicin, camptothecin, maytansinoids (including DM1), monomethylauristatin E (MMAE), monomethylauristatin F (MMAF), and maytansinoids (DM4) and their analogues.

(245) Toxins may be used as antibody-toxin conjugates and include bacterial toxins such as diphtheria toxin, plant toxins such as ricin, small molecule toxins such as geldanamycin (Mandler et al (2000) J. Nat. Cancer Inst. 92(19):1573-1581; Mandler et al (2000) Bioorganic & Med. Chem. Letters 10:1025-1028; Mandler et al (2002) Bioconjugate Chem. 13:786-791), maytansinoids (EP 1391213; Liu et al., (1996) Proc. Natl. Acad. Sci. USA 93:8618-8623), and calicheamicin (Lode et al (1998) Cancer Res. 58:2928; Hinman et al (1993) Cancer Res. 53:3336-3342). Toxins may exert their cytotoxic and cytostatic effects by mechanisms including tubulin binding, DNA binding, or topoisomerase inhibition.

(246) Conjugates of a multispecific antibody and one or more small molecule toxins, such as a maytansinoids, dolastatins, auristatins, a trichothecene, calicheamicin, and CC1065, and the derivatives of these toxins that have toxin activity, are contemplated.

(247) Maytansinoids

(248) Maytansine compounds suitable for use as maytansinoid drug moieties are well known in the art, and can be isolated from natural sources according to known methods, produced using genetic engineering techniques (see Yu et al (2002) PNAS 99:7968-7973), or maytansinol and maytansinol analogues prepared synthetically according to known methods. As described below, drugs may be modified by the incorporation of a functionally active group such as a thiol or amine group for conjugation to the antibody.

(249) Exemplary maytansinoid drug moieties include those having a modified aromatic ring, such as: C-19-dechloro (U.S. Pat. No. 4,256,746) (prepared by lithium aluminum hydride reduction of ansamycin P2); C-20-hydroxy (or C-20-demethyl)+/-C-19-dechloro (U.S. Pat. Nos. 4,361,650 and 4,307,016) (prepared by demethylation using *Streptomyces* or *Actinomyces* or dechlorination using LAH); and C-20-demethoxy, C-20-acyloxy (—OCOR), +/-dechloro (U.S. Pat. No. 4,294,757) (prepared by acylation using acyl chlorides) and those having modifications at other positions

(250) Exemplary maytansinoid drug moieties also include those having modifications such as: C-9-SH (U.S. Pat. No. 4,424,219) (prepared by the reaction of maytansinol with H₂S or P₂S₅); C-14-alkoxymethyl(demethoxy/CH₂OR) (U.S. Pat. No. 4,331,598); C-14-hydroxymethyl or acyloxymethyl (CH₂OH or CH₂OAc) (U.S. Pat. No. 4,450,254) (prepared from *Nocardia*); C-15-hydroxy/acyloxy (U.S. Pat. No. 4,364,866) (prepared by the conversion of maytansinol by *Streptomyces*); C-15-methoxy (U.S. Pat. Nos. 4,313,946 and 4,315,929) (isolated from *Trewia nudiflora*); C-18-N-demethyl (U.S. Pat. Nos. 4,362,663 and 4,322,348) (prepared by the demethylation of maytansinol by *Streptomyces*); and 4,5-deoxy (U.S. Pat. No. 4,371,533) (prepared by the titanium trichloride/LAH reduction of maytansinol).

(251) Of particular use are DM1 (disclosed in U.S. Pat. No. 5,208,020, incorporated by reference) and DM4 (disclosed in U.S. Pat. No. 7,276,497, incorporated by reference). See also a number of additional maytansinoid derivatives and methods in 5,416,064, WO/01/24763, 7,303,749, 7,601,354, U.S. Ser. No. 12/631,508, WO02/098883, 6,441,163, 7,368,565, WO02/16368 and WO04/1033272, all of which are expressly incorporated by reference in their entirety.

(252) ADCs containing maytansinoids, methods of making same, and their therapeutic use are disclosed, for example, in U.S. Pat. Nos. 5,208,020; 5,416,064; 6,441,163 and European Patent EP 0 425 235 B1, the disclosures of which are hereby expressly incorporated by reference. Liu et al., Proc. Natl. Acad. Sci. USA 93:8618-8623 (1996) described ADCs comprising a maytansinoid designated DM1 linked to the monoclonal antibody C242 directed against human colorectal cancer. The conjugate was found to be highly cytotoxic towards cultured colon cancer cells, and showed antitumor activity in an in vivo tumor growth assay.

(253) Chari et al., Cancer Research 52:127-131 (1992) describe ADCs in which a maytansinoid was conjugated via a disulfide linker to the murine antibody A7 binding to an antigen on human colon cancer cell lines, or to another murine monoclonal antibody TA.1 that binds the HER-2/neu oncogene. The cytotoxicity of the TA.1-maytansinoid conjugate was tested in vitro on the human breast cancer cell line SK-BR-3, which expresses 3×10^5 HER-2 surface antigens per cell. The drug conjugate achieved a degree of cytotoxicity similar to the free maytansinoid drug, which could be increased by increasing the number of maytansinoid molecules per antibody molecule. The A7-maytansinoid conjugate showed low systemic cytotoxicity in mice.

(254) Auristatins and Dolastatins

(255) In some embodiments, the ADC comprises a multispecific antibody conjugated to dolastatins or dolastatin peptidic analogs and derivatives, the auristatins (U.S. Pat. Nos. 5,635,483; 5,780,588). Dolastatins and auristatins have been shown to interfere with microtubule dynamics, GTP hydrolysis, and nuclear and cellular division (Woyke et al (2001) Antimicrob. Agents and Chemother. 45(12):3580-3584) and have anticancer (U.S. Pat. No. 5,663,149) and antifungal activity (Pettit et al (1998) Antimicrob. Agents Chemother. 42:2961-2965). The dolastatin or auristatin drug moiety may be attached to the antibody through the N (amino) terminus or the C (carboxyl) terminus of the peptidic drug moiety (WO 02/088172).

(256) Exemplary auristatin embodiments include the N-terminus linked monomethylauristatin drug moieties DE and DF, disclosed in "Senter et al, Proceedings of the American Association for Cancer Research, Volume 45, Abstract Number 623, presented Mar. 28, 2004 and described in United States Patent Publication No. 2005/0238648, the disclosure of which is expressly incorporated by reference in its entirety.

(257) An exemplary auristatin embodiment is MMAE (see U.S. Pat. No. 6,884,869 expressly incorporated by reference in its entirety).

(258) Another exemplary auristatin embodiment is MMAF (see US 2005/0238649, 5,767,237 and 6,124,431, expressly incorporated by reference in their entirety).

(259) Additional exemplary embodiments comprising MMAE or MMAF and various linker components (described further herein) have the following structures and abbreviations (wherein Ab means antibody and p is 1 to about 8):

(260) Typically, peptide-based drug moieties can be prepared by forming a peptide bond between two or more amino acids and/or peptide fragments. Such peptide bonds can be prepared, for example, according to the liquid phase synthesis method (see E. Schroder and K. Lubke, "The Peptides", volume 1, pp 76-136, 1965, Academic Press) that is well known in the field of peptide chemistry. The auristatin/dolastatin drug moieties may be prepared according to the methods of: U.S. Pat. Nos. 5,635,483; 5,780,588; Pettit et al (1989) J. Am. Chem. Soc. 111:5463-5465; Pettit et al (1998) Anti-Cancer Drug Design 13:243-277; Pettit, G. R., et al. Synthesis, 1996, 719-725; Pettit et al (1996) J. Chem. Soc. Perkin Trans. 1 5:859-863; and Doronina (2003) Nat Biotechnol 21(7):778-784.

(261) Calicheamicin

(262) In other embodiments, the ADC comprises an antibody of the invention conjugated to one or more calicheamicin molecules. For example, Mylotarg is the first commercial ADC drug and utilizes calicheamicin yl as the payload (see U.S. Pat. No. 4,970,198, incorporated by reference in its entirety). Additional calicheamicin derivatives are described in U.S. Pat. Nos. 5,264,586, 5,384,412, 5,550,246, 5,739,116, 5,773,001, 5,767,285 and 5,877,296, all expressly incorporated by reference. The calicheamicin family of antibiotics are capable of producing double-stranded DNA breaks at sub-picomolar concentrations. For the preparation of conjugates of the calicheamicin family, see U.S. Pat. Nos. 5,712,374, 5,714,586, 5,739,116, 5,767,285, 5,770,701, 5,770,710, 5,773,001, 5,877,296 (all to American Cyanamid Company). Structural analogues of calicheamicin which may be used include, but are not limited to, γ 1I, α 2I, α 2I, N-acetyl- γ 1I, PSAG and Oil (Hinman et al., Cancer Research 53:3336-3342 (1993), Lode et al., Cancer Research 58:2925-2928 (1998) and the aforementioned U.S. patents to American Cyanamid). Another anti-tumor drug that the antibody can be conjugated is QFA which is an antifolate. Both calicheamicin and QFA have intracellular sites of action and do not readily cross the plasma membrane. Therefore, cellular uptake of these agents through antibody mediated internalization greatly enhances their cytotoxic effects.

(263) Duocarmycins

(264) CC-1065 (see 4,169,888, incorporated by reference) and duocarmycins are members of a family of antitumor antibiotics utilized in ADCs. These antibiotics appear to work through sequence-selectively alkylating DNA at the N3 of adenine in the minor groove, which initiates a cascade of events that result in apoptosis.

(265) Important members of the duocarmycins include duocarmycin A (U.S. Pat. No. 4,923,990, incorporated by reference) and duocarmycin SA (U.S. Pat. No. 5,101,038, incorporated by reference), and a large number of analogues as described in U.S. Pat. Nos. 7,517,903, 7,691,962, 5,101,038; 5,641,780; 5,187,186; 5,070,092; 5,070,092; 5,641,780; 5,101,038; 5,084,468, 5,475,092, 5,585,499, 5,846,545, WO2007/089149, WO2009/017394A1, 5,703,080, 6,989,452, 7,087,600, 7,129,261, 7,498,302, and 7,507,420, all of which are expressly incorporated by reference.

(266) Other Cytotoxic Agents

(267) Other antitumor agents that can be conjugated to the antibodies of the invention include BCNU, streptozocin, vincristine and 5-fluorouracil, the family of agents known collectively LL-E33288 complex described in U.S. Pat. Nos. 5,053,394, 5,770,710, as well as esperamicins (U.S. Pat. No. 5,877,296).

(268) Enzymatically active toxins and fragments thereof which can be used include diphtheria A chain, nonbinding active fragments of diphtheria toxin, exotoxin A chain (from *Pseudomonas aeruginosa*), ricin A chain, abrin A chain, modeccin A chain, alpha-sarcin, *Aleurites fordii* proteins, dianthin proteins, *Phytolaca americana* proteins (PAPI, PAPII, and PAP-S), *Momordica charantia* inhibitor, curcin, crotin, *Sapaonaria officinalis* inhibitor, gelonin, mitogellin, restrictocin, phenomycin, enomycin and the tricothecenes. See, for example, WO 93/21232 published Oct. 28, 1993.

(269) The present invention further contemplates an ADC formed between an antibody and a compound with nucleolytic activity (e.g., a ribonuclease or a DNA endonuclease such as a deoxyribonuclease; DNase).

(270) For selective destruction of the tumor, the antibody may comprise a highly radioactive atom. A variety of radioactive isotopes are available for the production of radioconjugated antibodies. Examples include At211, I131, I125, Y90, Re186, Re188, Sm153, Bi212, P32, Pb212 and radioactive isotopes of Lu.

(271) The radio- or other labels may be incorporated in the conjugate in known ways. For example, the peptide may be biosynthesized or may be synthesized by chemical amino acid synthesis using

suitable amino acid precursors involving, for example, fluorine-19 in place of hydrogen. Labels such as Tc99m or I123, Re186, Re188 and In111 can be attached via a cysteine residue in the peptide. Yttrium-90 can be attached via a lysine residue. The IODOGEN method (Fraker et al (1978) Biochem. Biophys. Res. Commun. 80: 49-57 can be used to incorporate Iodine-123. "Monoclonal Antibodies in Immunoscintigraphy" (Chatal, CRC Press 1989) describes other methods in detail.

(272) For compositions comprising a plurality of antibodies, the drug loading is represented by p , the average number of drug molecules per Antibody. Drug loading may range from 1 to 20 drugs (D) per Antibody. The average number of drugs per antibody in preparation of conjugation reactions may be characterized by conventional means such as mass spectroscopy, ELISA assay, and HPLC. The quantitative distribution of Antibody-Drug-Conjugates in terms of p may also be determined.

(273) In some instances, separation, purification, and characterization of homogeneous Antibody-Drug-conjugates where p is a certain value from Antibody-Drug-Conjugates with other drug loadings may be achieved by means such as reverse phase HPLC or electrophoresis. In exemplary embodiments, p is 2, 3, 4, 5, 6, 7, or 8 or a fraction thereof.

(274) The generation of Antibody-drug conjugate compounds can be accomplished by any technique known to the skilled artisan. Briefly, the Antibody-drug conjugate compounds can include a multispecific antibody as the Antibody unit, a drug, and optionally a linker that joins the drug and the binding agent.

(275) A number of different reactions are available for covalent attachment of drugs and/or linkers to binding agents. This can be accomplished by reaction of the amino acid residues of the binding agent, for example, antibody molecule, including the amine groups of lysine, the free carboxylic acid groups of glutamic and aspartic acid, the sulfhydryl groups of cysteine and the various moieties of the aromatic amino acids. A commonly used non-specific method of covalent attachment is the carbodiimide reaction to link a carboxy (or amino) group of a compound to amino (or carboxy) groups of the antibody. Additionally, bifunctional agents such as dialdehydes or imidoesters have been used to link the amino group of a compound to amino groups of an antibody molecule.

(276) Also available for attachment of drugs to binding agents is the Schiff base reaction. This method involves the periodate oxidation of a drug that contains glycol or hydroxy groups, thus forming an aldehyde which is then reacted with the binding agent. Attachment occurs via formation of a Schiff base with amino groups of the binding agent. Isothiocyanates can also be used as coupling agents for covalently attaching drugs to binding agents. Other techniques are known to the skilled artisan and within the scope of the present invention.

(277) In some embodiments, an intermediate, which is the precursor of the linker, is reacted with the drug under appropriate conditions. In other embodiments, reactive groups are used on the drug and/or the intermediate. The product of the reaction between the drug and the intermediate, or the derivatized drug, is subsequently reacted with an multispecific antibody of the invention under appropriate conditions.

(278) It will be understood that chemical modifications may also be made to the desired compound in order to make reactions of that compound more convenient for purposes of preparing conjugates of the invention. For example a functional group e.g. amine, hydroxyl, or sulfhydryl, may be appended to the drug at a position which has minimal or an acceptable effect on the activity or other properties of the drug

(279) ADC Linker Units

(280) Typically, the antibody-drug conjugate compounds comprise a Linker unit between the drug unit and the antibody unit. In some embodiments, the linker is cleavable under intracellular or extracellular conditions, such that cleavage of the linker releases the drug unit from the antibody in the appropriate environment. For example, solid tumors that secrete certain proteases may serve as

the target of the cleavable linker; in other embodiments, it is the intracellular proteases that are utilized. In yet other embodiments, the linker unit is not cleavable and the drug is released, for example, by antibody degradation in lysosomes.

(281) In some embodiments, the linker is cleavable by a cleaving agent that is present in the intracellular environment (for example, within a lysosome or endosome or caveolea). The linker can be, for example, a peptidyl linker that is cleaved by an intracellular peptidase or protease enzyme, including, but not limited to, a lysosomal or endosomal protease. In some embodiments, the peptidyl linker is at least two amino acids long or at least three amino acids long or more.

(282) Cleaving agents can include, without limitation, cathepsins B and D and plasmin, all of which are known to hydrolyze dipeptide drug derivatives resulting in the release of active drug inside target cells (see, e.g., Dubowchik and Walker, 1999, *Pharm. Therapeutics* 83:67-123). Peptidyl linkers that are cleavable by enzymes that are present in CD38-expressing cells. For example, a peptidyl linker that is cleavable by the thiol-dependent protease cathepsin-B, which is highly expressed in cancerous tissue, can be used (e.g., a Phe-Leu or a Gly-Phe-Leu-Gly linker (SEQ ID NO: 460)). Other examples of such linkers are described, e.g., in U.S. Pat. No. 6,214,345, incorporated herein by reference in its entirety and for all purposes.

(283) In some embodiments, the peptidyl linker cleavable by an intracellular protease is a Val-Cit linker or a Phe-Lys linker (see, e.g., U.S. Pat. No. 6,214,345, which describes the synthesis of doxorubicin with the val-cit linker).

(284) In other embodiments, the cleavable linker is pH-sensitive, that is, sensitive to hydrolysis at certain pH values. Typically, the pH-sensitive linker hydrolyzable under acidic conditions. For example, an acid-labile linker that is hydrolyzable in the lysosome (for example, a hydrazone, semicarbazone, thiosemicarbazone, cis-aconitic amide, orthoester, acetal, ketal, or the like) may be used. (See, e.g., U.S. Pat. Nos. 5,122,368; 5,824,805; 5,622,929; Dubowchik and Walker, 1999, *Pharm. Therapeutics* 83:67-123; Neville et al., 1989, *Biol. Chem.* 264:14653-14661.) Such linkers are relatively stable under neutral pH conditions, such as those in the blood, but are unstable at below pH 5.5 or 5.0, the approximate pH of the lysosome. In certain embodiments, the hydrolyzable linker is a thioether linker (such as, e.g., a thioether attached to the therapeutic agent via an acylhydrazone bond (see, e.g., U.S. Pat. No. 5,622,929).

(285) In yet other embodiments, the linker is cleavable under reducing conditions (for example, a disulfide linker). A variety of disulfide linkers are known in the art, including, for example, those that can be formed using SATA (N-succinimidyl-5-acetylthioacetate), SPDP (N-succinimidyl-3-(2-pyridyldithio)propionate), SPDB (N-succinimidyl-3-(2-pyridyldithio)butyrate) and SMPT (N-succinimidyl-oxycarbonyl-alpha-methyl-alpha-(2-pyridyl-dithio)toluene)-, SPDB and SMPT. (See, e.g., Thorpe et al., 1987, *Cancer Res.* 47:5924-5931; Wawrzynczak et al., In *Immunoconjugates: Antibody Conjugates in Radioimagery and Therapy of Cancer* (C. W. Vogel ed., Oxford U. Press, 1987. See also U.S. Pat. No. 4,880,935.)

(286) In other embodiments, the linker is a malonate linker (Johnson et al., 1995, *Anticancer Res.* 15:1387-93), a maleimidobenzoyl linker (Lau et al., 1995, *Bioorg-Med-Chem.* 3(10):1299-1304), or a 3'-N-amide analog (Lau et al., 1995, *Bioorg-Med-Chem.* 3(10):1305-12).

(287) In yet other embodiments, the linker unit is not cleavable and the drug is released by antibody degradation. (See U.S. Publication No. 2005/0238649 incorporated by reference herein in its entirety and for all purposes).

(288) In many embodiments, the linker is self-immolative. As used herein, the term "self-immolative Spacer" refers to a bifunctional chemical moiety that is capable of covalently linking together two spaced chemical moieties into a stable tripartite molecule. It will spontaneously separate from the second chemical moiety if its bond to the first moiety is cleaved. See for example, WO 2007059404A2, WO06110476A2, WO05112919A2, WO2010/062171, WO09/017394, WO07/089149, WO 07/018431, WO04/043493 and WO02/083180, which are directed to drug-cleavable substrate conjugates where the drug and cleavable substrate are

optionally linked through a self-immolative linker and which are all expressly incorporated by reference.

(289) Often the linker is not substantially sensitive to the extracellular environment. As used herein, “not substantially sensitive to the extracellular environment,” in the context of a linker, means that no more than about 20%, 15%, 10%, 5%, 3%, or no more than about 1% of the linkers, in a sample of antibody-drug conjugate compound, are cleaved when the antibody-drug conjugate compound presents in an extracellular environment (for example, in plasma).

(290) Whether a linker is not substantially sensitive to the extracellular environment can be determined, for example, by incubating with plasma the antibody-drug conjugate compound for a predetermined time period (for example, 2, 4, 8, 16, or 24 hours) and then quantitating the amount of free drug present in the plasma.

(291) In other, non-mutually exclusive embodiments, the linker promotes cellular internalization. In certain embodiments, the linker promotes cellular internalization when conjugated to the therapeutic agent (that is, in the milieu of the linker-therapeutic agent moiety of the antibody-drug conjugate compound as described herein). In yet other embodiments, the linker promotes cellular internalization when conjugated to both the auristatin compound and the multispecific antibodies of the invention.

(292) A variety of exemplary linkers that can be used with the present compositions and methods are described in WO 2004-010957, U.S. Publication No. 2006/0074008, U.S. Publication No. 20050238649, and U.S. Publication No. 2006/0024317 (each of which is incorporated by reference herein in its entirety and for all purposes).

(293) Drug Loading

(294) Drug loading is represented by p and is the average number of Drug moieties per antibody in a molecule. Drug loading (“ p ”) may be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more moieties (D) per antibody, although frequently the average number is a fraction or a decimal. Generally, drug loading of from 1 to 4 is frequently useful, and from 1 to 2 is also useful. ADCs of the invention include collections of antibodies conjugated with a range of drug moieties, from 1 to 20. The average number of drug moieties per antibody in preparations of ADC from conjugation reactions may be characterized by conventional means such as mass spectroscopy and, ELISA assay.

(295) The quantitative distribution of ADC in terms of p may also be determined. In some instances, separation, purification, and characterization of homogeneous ADC where p is a certain value from ADC with other drug loadings may be achieved by means such as electrophoresis.

(296) For some antibody-drug conjugates, p may be limited by the number of attachment sites on the antibody. For example, where the attachment is a cysteine thiol, as in the exemplary embodiments above, an antibody may have only one or several cysteine thiol groups, or may have only one or several sufficiently reactive thiol groups through which a linker may be attached. In certain embodiments, higher drug loading, e.g. $p > 5$, may cause aggregation, insolubility, toxicity, or loss of cellular permeability of certain antibody-drug conjugates. In certain embodiments, the drug loading for an ADC of the invention ranges from 1 to about 8; from about 2 to about 6; from about 3 to about 5; from about 3 to about 4; from about 3.1 to about 3.9; from about 3.2 to about 3.8; from about 3.2 to about 3.7; from about 3.2 to about 3.6; from about 3.3 to about 3.8; or from about 3.3 to about 3.7. Indeed, it has been shown that for certain ADCs, the optimal ratio of drug moieties per antibody may be less than 8, and may be about 2 to about 5. See US 2005-0238649 A1 (herein incorporated by reference in its entirety).

(297) In certain embodiments, fewer than the theoretical maximum of drug moieties are conjugated to an antibody during a conjugation reaction. An antibody may contain, for example, lysine residues that do not react with the drug-linker intermediate or linker reagent, as discussed below. Generally, antibodies do not contain many free and reactive cysteine thiol groups which may be linked to a drug moiety; indeed most cysteine thiol residues in antibodies exist as disulfide bridges.

In certain embodiments, an antibody may be reduced with a reducing agent such as dithiothreitol (DTT) or tricarboylethylphosphine (TCEP), under partial or total reducing conditions, to generate reactive cysteine thiol groups. In certain embodiments, an antibody is subjected to denaturing conditions to reveal reactive nucleophilic groups such as lysine or cysteine.

(298) The loading (drug/antibody ratio) of an ADC may be controlled in different ways, e.g., by: (i) limiting the molar excess of drug-linker intermediate or linker reagent relative to antibody, (ii) limiting the conjugation reaction time or temperature, (iii) partial or limiting reductive conditions for cysteine thiol modification, (iv) engineering by recombinant techniques the amino acid sequence of the antibody such that the number and position of cysteine residues is modified for control of the number and/or position of linker-drug attachments (such as thioMab or thioFab prepared as disclosed herein and in WO2006/034488 (herein incorporated by reference in its entirety)).

(299) It is to be understood that where more than one nucleophilic group reacts with a drug-linker intermediate or linker reagent followed by drug moiety reagent, then the resulting product is a mixture of ADC compounds with a distribution of one or more drug moieties attached to an antibody. The average number of drugs per antibody may be calculated from the mixture by a dual ELISA antibody assay, which is specific for antibody and specific for the drug. Individual ADC molecules may be identified in the mixture by mass spectroscopy and separated by HPLC, e.g. hydrophobic interaction chromatography.

(300) In some embodiments, a homogeneous ADC with a single loading value may be isolated from the conjugation mixture by electrophoresis or chromatography.

(301) Methods of Determining Cytotoxic Effect of ADCs

(302) Methods of determining whether a Drug or Antibody-Drug conjugate exerts a cytostatic and/or cytotoxic effect on a cell are known. Generally, the cytotoxic or cytostatic activity of an Antibody Drug conjugate can be measured by: exposing mammalian cells expressing a target protein of the Antibody Drug conjugate in a cell culture medium; culturing the cells for a period from about 6 hours to about 5 days; and measuring cell viability. Cell-based in vitro assays can be used to measure viability (proliferation), cytotoxicity, and induction of apoptosis (caspase activation) of the Antibody Drug conjugate.

(303) For determining whether an Antibody Drug conjugate exerts a cytostatic effect, a thymidine incorporation assay may be used. For example, cancer cells expressing a target antigen at a density of 5,000 cells/well of a 96-well plated can be cultured for a 72-hour period and exposed to 0.5 μ Ci of 3H-thymidine during the final 8 hours of the 72-hour period. The incorporation of 3H-thymidine into cells of the culture is measured in the presence and absence of the Antibody Drug conjugate.

(304) For determining cytotoxicity, necrosis or apoptosis (programmed cell death) can be measured. Necrosis is typically accompanied by increased permeability of the plasma membrane; swelling of the cell, and rupture of the plasma membrane. Apoptosis is typically characterized by membrane blebbing, condensation of cytoplasm, and the activation of endogenous endonucleases. Determination of any of these effects on cancer cells indicates that an Antibody Drug conjugate is useful in the treatment of cancers.

(305) Cell viability can be measured by determining in a cell the uptake of a dye such as neutral red, trypan blue, or ALAMARTM blue (see, e.g., Page et al., 1993, Intl. J. Oncology 3:473-476). In such an assay, the cells are incubated in media containing the dye, the cells are washed, and the remaining dye, reflecting cellular uptake of the dye, is measured spectrophotometrically. The protein-binding dye sulforhodamine B (SRB) can also be used to measure cytotoxicity (Skehan et al., 1990, J. Natl. Cancer Inst. 82:1107-12).

(306) Alternatively, a tetrazolium salt, such as MTT, is used in a quantitative colorimetric assay for mammalian cell survival and proliferation by detecting living, but not dead, cells (see, e.g., Mosmann, 1983, J. Immunol. Methods 65:55-63).

(307) Apoptosis can be quantitated by measuring, for example, DNA fragmentation. Commercial

photometric methods for the quantitative in vitro determination of DNA fragmentation are available. Examples of such assays, including TUNEL (which detects incorporation of labeled nucleotides in fragmented DNA) and ELISA-based assays, are described in Biochemica, 1999, no. 2, pp. 34-37 (Roche Molecular Biochemicals).

(308) Apoptosis can also be determined by measuring morphological changes in a cell. For example, as with necrosis, loss of plasma membrane integrity can be determined by measuring uptake of certain dyes (e.g., a fluorescent dye such as, for example, acridine orange or ethidium bromide). A method for measuring apoptotic cell number has been described by Duke and Cohen, Current Protocols in Immunology (Coligan et al. eds., 1992, pp. 3.17.1-3.17.16). Cells also can be labeled with a DNA dye (e.g., acridine orange, ethidium bromide, or propidium iodide) and the cells observed for chromatin condensation and margination along the inner nuclear membrane. Other morphological changes that can be measured to determine apoptosis include, e.g., cytoplasmic condensation, increased membrane blebbing, and cellular shrinkage.

(309) The presence of apoptotic cells can be measured in both the attached and “floating” compartments of the cultures. For example, both compartments can be collected by removing the supernatant, trypsinizing the attached cells, combining the preparations following a centrifugation wash step (e.g., 10 minutes at 2000 rpm), and detecting apoptosis (e.g., by measuring DNA fragmentation). (See, e.g., Piazza et al., 1995, Cancer Research 55:3110-16).

(310) In vivo, the effect of a therapeutic composition of the multispecific antibody of the invention can be evaluated in a suitable animal model. For example, xenogenic cancer models can be used, wherein cancer explants or passaged xenograft tissues are introduced into immune compromised animals, such as nude or SCID mice (Klein et al., 1997, Nature Medicine 3: 402-408). Efficacy can be measured using assays that measure inhibition of tumor formation, tumor regression or metastasis, and the like.

(311) The therapeutic compositions used in the practice of the foregoing methods can be formulated into pharmaceutical compositions comprising a carrier suitable for the desired delivery method. Suitable carriers include any material that when combined with the therapeutic composition retains the anti-tumor function of the therapeutic composition and is generally non-reactive with the patient's immune system. Examples include, but are not limited to, any of a number of standard pharmaceutical carriers such as sterile phosphate buffered saline solutions, bacteriostatic water, and the like (see, generally, Remington's Pharmaceutical Sciences 16th Edition, A. Osal., Ed., 1980).

(312) Antibody Compositions for In Vivo Administration

(313) Formulations of the antibodies used in accordance with the present invention are prepared for storage by mixing an antibody having the desired degree of purity with optional pharmaceutically acceptable carriers, excipients or stabilizers (Remington's Pharmaceutical Sciences 16th edition, Osol, A. Ed. [1980]), in the form of lyophilized formulations or aqueous solutions. Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations employed, and include buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives (such as octadecyldimethylbenzyl ammonium chloride; hexamethonium chloride; benzalkonium chloride, benzethonium chloride; phenol, butyl or benzyl alcohol; alkyl parabens such as methyl or propyl paraben; catechol; resorcinol; cyclohexanol; 3-pentanol; and m-cresol); low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, histidine, arginine, or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g. Zn-protein complexes); and/or non-ionic surfactants such as TWEEN™, PLURONICS™ or polyethylene glycol (PEG).

(314) The formulation herein may also contain more than one active compound as necessary for the particular indication being treated, preferably those with complementary activities that do not adversely affect each other. For example, it may be desirable to provide antibodies with other specificities. Alternatively, or in addition, the composition may comprise a cytotoxic agent, cytokine, growth inhibitory agent and/or small molecule antagonist. Such molecules are suitably present in combination in amounts that are effective for the purpose intended.

(315) The active ingredients may also be entrapped in microcapsules prepared, for example, by coacervation techniques or by interfacial polymerization, for example, hydroxymethylcellulose or gelatin-microcapsules and poly-(methylmethacrylate) microcapsules, respectively, in colloidal drug delivery systems (for example, liposomes, albumin microspheres, microemulsions, nano-particles and nanocapsules) or in macroemulsions. Such techniques are disclosed in Remington's Pharmaceutical Sciences 16th edition, Osol, A. Ed. (1980).

(316) The formulations to be used for in vivo administration should be sterile, or nearly so. This is readily accomplished by filtration through sterile filtration membranes.

(317) Sustained-release preparations may be prepared. Suitable examples of sustained-release preparations include semipermeable matrices of solid hydrophobic polymers containing the antibody, which matrices are in the form of shaped articles, e.g. films, or microcapsules. Examples of sustained-release matrices include polyesters, hydrogels (for example, poly(2-hydroxyethyl-methacrylate), or poly(vinylalcohol)), polylactides (U.S. Pat. No. 3,773,919), copolymers of L-glutamic acid and .gamma. ethyl-L-glutamate, non-degradable ethylene-vinyl acetate, degradable lactic acid-glycolic acid copolymers such as the LUPRON DEPOT™ (injectable microspheres composed of lactic acid-glycolic acid copolymer and leuprolide acetate), and poly-D-(-)-3-hydroxybutyric acid. While polymers such as ethylene-vinyl acetate and lactic acid-glycolic acid enable release of molecules for over 100 days, certain hydrogels release proteins for shorter time periods.

(318) When encapsulated antibodies remain in the body for a long time, they may denature or aggregate as a result of exposure to moisture at 37° C., resulting in a loss of biological activity and possible changes in immunogenicity. Rational strategies can be devised for stabilization depending on the mechanism involved. For example, if the aggregation mechanism is discovered to be intermolecular S—S bond formation through thio-disulfide interchange, stabilization may be achieved by modifying sulfhydryl residues, lyophilizing from acidic solutions, controlling moisture content, using appropriate additives, and developing specific polymer matrix compositions.

(319) Administrative Modalities

(320) The antibodies and chemotherapeutic agents of the invention are administered to a subject, in accord with known methods, such as intravenous administration as a bolus or by continuous infusion over a period of time, by intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, intrathecal, oral, topical, or inhalation routes. Intravenous or subcutaneous administration of the antibody is preferred.

(321) Treatment Modalities

(322) In the methods of the invention, therapy is used to provide a positive therapeutic response with respect to a disease or condition. By "positive therapeutic response" is intended an improvement in the disease or condition, and/or an improvement in the symptoms associated with the disease or condition. For example, a positive therapeutic response would refer to one or more of the following improvements in the disease: (1) a reduction in the number of neoplastic cells; (2) an increase in neoplastic cell death; (3) inhibition of neoplastic cell survival; (5) inhibition (i.e., slowing to some extent, preferably halting) of tumor growth; (6) an increased patient survival rate; and (7) some relief from one or more symptoms associated with the disease or condition.

(323) Positive therapeutic responses in any given disease or condition can be determined by standardized response criteria specific to that disease or condition. Tumor response can be assessed for changes in tumor morphology (i.e., overall tumor burden, tumor size, and the like) using

screening techniques such as magnetic resonance imaging (MM) scan, x-radiographic imaging, computed tomographic (CT) scan, bone scan imaging, endoscopy, and tumor biopsy sampling including bone marrow aspiration (BMA) and counting of tumor cells in the circulation.

(324) In addition to these positive therapeutic responses, the subject undergoing therapy may experience the beneficial effect of an improvement in the symptoms associated with the disease.

(325) Thus for B cell tumors, the subject may experience a decrease in the so-called B symptoms, i.e., night sweats, fever, weight loss, and/or urticaria. For pre-malignant conditions, therapy with an multispecific therapeutic agent may block and/or prolong the time before development of a related malignant condition, for example, development of multiple myeloma in subjects suffering from monoclonal gammopathy of undetermined significance (MGUS).

(326) An improvement in the disease may be characterized as a complete response. By “complete response” is intended an absence of clinically detectable disease with normalization of any previously abnormal radiographic studies, bone marrow, and cerebrospinal fluid (CSF) or abnormal monoclonal protein in the case of myeloma.

(327) Such a response may persist for at least 4 to 8 weeks, or sometimes 6 to 8 weeks, following treatment according to the methods of the invention. Alternatively, an improvement in the disease may be categorized as being a partial response. By “partial response” is intended at least about a 50% decrease in all measurable tumor burden (i.e., the number of malignant cells present in the subject, or the measured bulk of tumor masses or the quantity of abnormal monoclonal protein) in the absence of new lesions, which may persist for 4 to 8 weeks, or 6 to 8 weeks.

(328) Treatment according to the present invention includes a “therapeutically effective amount” of the medicaments used. A “therapeutically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve a desired therapeutic result.

(329) A therapeutically effective amount may vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of the medicaments to elicit a desired response in the individual. A therapeutically effective amount is also one in which any toxic or detrimental effects of the antibody or antibody portion are outweighed by the therapeutically beneficial effects.

(330) A “therapeutically effective amount” for tumor therapy may also be measured by its ability to stabilize the progression of disease. The ability of a compound to inhibit cancer may be evaluated in an animal model system predictive of efficacy in human tumors.

(331) Alternatively, this property of a composition may be evaluated by examining the ability of the compound to inhibit cell growth or to induce apoptosis by in vitro assays known to the skilled practitioner. A therapeutically effective amount of a therapeutic compound may decrease tumor size, or otherwise ameliorate symptoms in a subject. One of ordinary skill in the art would be able to determine such amounts based on such factors as the subject's size, the severity of the subject's symptoms, and the particular composition or route of administration selected.

(332) Dosage regimens are adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus may be administered, several divided doses may be administered over time or the dose may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

(333) The specification for the dosage unit forms of the present invention are dictated by and directly dependent on (a) the unique characteristics of the active compound and the particular therapeutic effect to be achieved, and (b) the limitations inherent in the art of compounding such an active compound for the treatment of sensitivity in individuals.

(334) The efficient dosages and the dosage regimens for the multispecific antibodies used in the

present invention depend on the disease or condition to be treated and may be determined by the persons skilled in the art.

(335) An exemplary, non-limiting range for a therapeutically effective amount of an multispecific antibody used in the present invention is about 0.1-100 mg/kg, such as about 0.1-50 mg/kg, for example about 0.1-20 mg/kg, such as about 0.1-10 mg/kg, for instance about 0.5, about such as 0.3, about 1, or about 3 mg/kg. In another embodiment, the antibody is administered in a dose of 1 mg/kg or more, such as a dose of from 1 to 20 mg/kg, e.g. a dose of from 5 to 20 mg/kg, e.g. a dose of 8 mg/kg.

(336) A medical professional having ordinary skill in the art may readily determine and prescribe the effective amount of the pharmaceutical composition required. For example, a physician or a veterinarian could start doses of the medicament employed in the pharmaceutical composition at levels lower than that required in order to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved.

(337) In one embodiment, the multispecific antibody is administered by infusion in a weekly dosage of from 10 to 500 mg/kg such as of from 200 to 400 mg/kg. Such administration may be repeated, e.g., 1 to 8 times, such as 3 to 5 times. The administration may be performed by continuous infusion over a period of from 2 to 24 hours, such as of from 2 to 12 hours.

(338) In one embodiment, the multispecific antibody is administered by slow continuous infusion over a long period, such as more than 24 hours, if required to reduce side effects including toxicity.

(339) In one embodiment the multispecific antibody is administered in a weekly dosage of from 250 mg to 2000 mg, such as for example 300 mg, 500 mg, 700 mg, 1000 mg, 1500 mg or 2000 mg, for up to 8 times, such as from 4 to 6 times. The administration may be performed by continuous infusion over a period of from 2 to 24 hours, such as of from 2 to 12 hours. Such regimen may be repeated one or more times as necessary, for example, after 6 months or 12 months. The dosage may be determined or adjusted by measuring the amount of compound of the present invention in the blood upon administration by for instance taking out a biological sample and using anti-idiotypic antibodies which target the antigen binding region of the multispecific antibody.

(340) In a further embodiment, the multispecific antibody is administered once weekly for 2 to 12 weeks, such as for 3 to 10 weeks, such as for 4 to 8 weeks.

(341) In one embodiment, the multispecific antibody is administered by maintenance therapy, such as, e.g., once a week for a period of 6 months or more.

(342) In one embodiment, the multispecific antibody is administered by a regimen including one infusion of an multispecific antibody followed by an infusion of an multispecific antibody conjugated to a radioisotope. The regimen may be repeated, e.g., 7 to 9 days later.

(343) As non-limiting examples, treatment according to the present invention may be provided as a daily dosage of an antibody in an amount of about 0.1-100 mg/kg, such as 0.5, 0.9, 1.0, 1.1, 1.5, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 40, 45, 50, 60, 70, 80, 90 or 100 mg/kg, per day, on at least one of day 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40, or alternatively, at least one of week 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 after initiation of treatment, or any combination thereof, using single or divided doses of every 24, 12, 8, 6, 4, or 2 hours, or any combination thereof.

(344) In some embodiments the multispecific antibody molecule thereof is used in combination with one or more additional therapeutic agents, e.g. a chemotherapeutic agent. Non-limiting examples of DNA damaging chemotherapeutic agents include topoisomerase I inhibitors (e.g., irinotecan, topotecan, camptothecin and analogs or metabolites thereof, and doxorubicin); topoisomerase II inhibitors (e.g., etoposide, teniposide, and daunorubicin); alkylating agents (e.g., melphalan, chlorambucil, busulfan, thiotepa, ifosfamide, carmustine, lomustine, semustine, streptozocin, decarbazine, methotrexate, mitomycin C, and cyclophosphamide); DNA intercalators (e.g., cisplatin, oxaliplatin, and carboplatin); DNA intercalators and free radical generators such as

bleomycin; and nucleoside mimetics (e.g., 5-fluorouracil, capecitabine, gemcitabine, fludarabine, cytarabine, mercaptopurine, thioguanine, pentostatin, and hydroxyurea).

(345) Chemotherapeutic agents that disrupt cell replication include: paclitaxel, docetaxel, and related analogs; vincristine, vinblastin, and related analogs; thalidomide, lenalidomide, and related analogs (e.g., CC-5013 and CC-4047); protein tyrosine kinase inhibitors (e.g., imatinib mesylate and gefitinib); proteasome inhibitors (e.g., bortezomib); NF- κ B inhibitors, including inhibitors of I- κ B kinase; antibodies which bind to proteins overexpressed in cancers and thereby downregulate cell replication (e.g., trastuzumab, rituximab, cetuximab, and bevacizumab); and other inhibitors of proteins or enzymes known to be upregulated, over-expressed or activated in cancers, the inhibition of which downregulates cell replication.

(346) In some embodiments, the antibodies of the invention can be used prior to, concurrent with, or after treatment with Velcade® (bortezomib).

(347) All cited references are herein expressly incorporated by reference in their entirety.

(348) Whereas particular embodiments of the invention have been described above for purposes of illustration, it will be appreciated by those skilled in the art that numerous variations of the details may be made without departing from the invention as described in the appended claims.

EXAMPLES

(349) Examples are provided below to illustrate the present invention. These examples are not meant to constrain the present invention to any particular application or theory of operation. For all constant region positions discussed in the present invention, numbering is according to the EU index as in Kabat (Kabat et al., 1991, Sequences of Proteins of Immunological Interest, 5th Ed., United States Public Health Service, National Institutes of Health, Bethesda, entirely incorporated by reference). Those skilled in the art of antibodies will appreciate that this convention consists of nonsequential numbering in specific regions of an immunoglobulin sequence, enabling a normalized reference to conserved positions in immunoglobulin families. Accordingly, the positions of any given immunoglobulin as defined by the EU index will not necessarily correspond to its sequential sequence.

Example 1. Design of Non-Native Charge Substitutions to Reduce pI

(350) Antibody constant chains were modified with lower pI by engineering substitutions in the constant domains. Reduced pI can be engineered by making substitutions of basic amino acids (K or R) to acidic amino acids (D or E), which result in the largest decrease in pI. Mutations of basic amino acids to neutral amino acids and neutral amino acids to acidic amino acids will also result in a decrease in pI. A list of amino acid pK values can be found in Table 1 of Bjellqvist et al., 1994, Electrophoresis 15:529-539.

(351) We chose to explore substitutions in the antibody CH1 (Cy1) and CL (Ckappa or CK) regions (sequences are shown in FIG. 13) because, unlike the Fc region, they do not interact with native ligands that impact the antibody's pharmacological properties. In deciding which positions to mutate, the surrounding environment and number of contacts the WT amino acid makes with its neighbors was taken into account such as to minimize the impact of a substitution or set of substitutions on structure and/or function. The solvent accessibility or fraction exposed of each CH1 and CK position was calculated using relevant crystal structures of antibody Fab domains. The results are shown in FIGS. 2 and 3 of U.S. Ser. No. 13/648,951 for the Cy1 and CK respectively (Figures and accompanying legends are expressly incorporated herein by reference). Design was guided further by examining the CH1 and CL domains for positions that are isotypic between the immunoglobulin isotypes (IgG1, IgG2, IgG3, and IgG4). Because such variations occur naturally, such positions are expected to be amenable to substitution. Based on this analysis, a number of substitutions were identified that reduce pI but are predicted to have minimal impact on the biophysical properties of the domains.

(352) As for all the heterodimeric proteins herein, genes encoding the heavy and light chains of the antibodies were constructed in the mammalian expression vector pTT5. The human IgG1 constant

chain gene was obtained from IMAGE clones and subcloned into the pTT5 vector. VH and VL genes encoding the anti-VEGF antibodies were synthesized commercially (Blue Heron Biotechnologies, Bothell WA), and subcloned into the vectors encoding the appropriate CL and IgG1 constant chains. Amino acid modifications were constructed using site-directed mutagenesis using the QuikChange® site-directed mutagenesis methods (Stratagene, La Jolla CA). All DNA was sequenced to confirm the fidelity of the sequences.

(353) Plasmids containing heavy chain gene (VH-Cy1-Cy2-Cy3) were co-transfected with plasmid containing light chain gene (VL-C κ) into 293E cells using lipofectamine (Invitrogen, Carlsbad CA) and grown in FreeStyle 293 media (Invitrogen, Carlsbad CA). After 5 days of growth, the antibodies were purified from the culture supernatant by protein A affinity using the Mab Select resin (GE Healthcare). Antibody concentrations were determined by bicinchoninic acid (BCA) assay (Pierce).

(354) The pI engineered mAbs were generally characterized by SDS PAGE on an Agilent Bioanalyzer, by size exclusion chromatography (SEC), isoelectric focusing (IEF) gel electrophoresis, binding to antigen by Biacore, and differential scanning calorimetry (DSC). All mAbs showed high purity on SDS-PAGE and SEC. IEF gels indicated that each variant had the designed isoelectric point. Generally the binding analysis on Biacore showed that pI engineered variants bound to antigen with similar affinity as the parent antibodies, indicating that the designed substitutions did not perturb the function of the mAb. DSC in the Figures show which variants generally had high thermostability.

(355) Pharmacokinetic experiments for serum half life as appropriate were performed in B6 mice that are homozygous knock-outs for murine FcRn and heterozygous knock-ins of human FcRn (mFcRn $-/-$, hFcRn $+$) (Petkova et al., 2006, *Int Immunol* 18(12):1759-69, entirely incorporated by reference), herein referred to as hFcRn or hFcRn $+$ mice.

(356) A single, intravenous tail vein injection of antibody (2 mg/kg) was given to groups of 4-7 female mice randomized by body weight (20-30 g range). Blood (~50 μ l) was drawn from the orbital plexus at each time point, processed to serum, and stored at -80° C. until analysis. Antibody concentrations were determined using an ELISA assay. Serum concentration of antibody was measured using recombinant antigen as capture reagent, and detection was carried out with biotinylated anti-human kappa antibody and europium-labeled streptavidin. The time resolved fluorescence signal was collected. PK parameters were determined for individual mice with a non-compartmental model using WinNonLin (Pharsight Inc, Mountain View CA). Nominal times and dose were used with uniform weighing of points.

Example 2. Engineering Approaches to Constant Region pI Engineering

(357) Reduction in the pI of a protein or antibody can be carried out using a variety of approaches. At the most basic level, residues with high pKa's (lysine, arginine, and to some extent histidine) are replaced with neutral or negative residues, and/or neutral residues are replaced with low pKa residues (aspartic acid and glutamic acid). The particular replacements may depend on a variety of factors, including location in the structure, role in function, and immunogenicity.

(358) Because immunogenicity is a concern, efforts can be made to minimize the risk that a substitution that lowers the pI will elicit immunogenicity. One way to minimize risk is to minimize the mutational load of the variants, i.e. to reduce the pI with the fewest number of mutations. Charge swapping mutations, where a K, R, or H is replaced with a D or E, have the greatest impact on reducing pI, and so these substitutions are preferred. Another approach to minimizing the risk of immunogenicity while reducing pI is to utilize substitutions from homologous human proteins. Thus for antibody constant chains, the isotypic differences between the IgG subclasses (IgG1, IgG2, IgG3, and IgG4) provide low-risk substitutions. Because immune recognition occurs at a local sequence level, i.e. MHC II and T-cell receptors recognize epitopes typically 9 residues in length, pI-altering substitutions may be accompanied by isotypic substitutions proximal in sequence. In this way, epitopes can be extended to match a natural isotype. Such substitutions

would thus make up epitopes that are present in other human IgG isotypes, and thus would be expected to be tolerized.

(359) One approach for engineering changes in pI is to use isotype switching, as described herein. (360) Another approach to engineering lower pI into proteins and antibodies is to fuse negatively charged residues to the N- or C-termini. Thus for example, peptides consisting principally of aspartic acids and glutamic acid may be fused to the N-terminus or C-terminus to the antibody heavy chain, light chain or both. Because the N-termini are structurally close to the antigen binding site, the C-termini are preferred.

(361) Based on the described engineering approaches, a number of variants were designed to alter the isoelectric point of the antibody heavy chain (Fc region generally) and in some cases the light chain.

Example 3. Isotypic Light Chain Constant Region Variants

(362) Homology between CK and C λ , is not as high as between the IgG subclasses, however the sequence and structural homology that exists was still used to guide substitutions to create an isotypic low-pI light chain constant region. In FIG. 56, positions with residues contributing to a higher pI (K, R, and H) or lower pI (D and E) are highlighted in bold. Gray indicates lysine, arginines, and histidines that may be substituted, preferably with aspartic or glutamic acids, to lower the isoelectric point. These variants, alone or in any combination, can independently and optionally be combined with all other heavy chain variants in scaffolds that have at least one light chain.

Example 4. Purifying Mixtures of Antibody Variants with Modified Isoelectric Points

(363) Substitutions that modify the antibody isoelectric point may be introduced into one or more chains of an antibody variant to facilitate analysis and purification. For instance, heterodimeric antibodies such as those disclosed in US2011/0054151A1 can be purified by modifying the isoelectric point of one chain, so that the multiple species present after expression and Protein A purification can be purified by methods that separate proteins based on differences in charge, such as ion exchange chromatography.

(364) As an example, the heavy chain of bevacizumab was modified by introducing substitutions to lower its isoelectric point such that the difference in charges between the three species produced when WT-IgG1-HC, low-pI-HC, and WT-LC are transfected in 293E cells is large enough to facilitate purification by anion exchange chromatography. Clones were created as described above, and transfection and initial purification by Protein A chromatography is also as described above. Sequences of the three chains “Heavy chain 1 of XENP10653”, “Heavy chain 2 of XENP10653”, and “Light chain of XENP10653” in the Figures. After Protein A purification, three species with nearly identical molecular weights, but different charges are obtained. These are the WT-IgG1-HC/WT-IgG1-HC homodimer (pI=8.12), WT-IgG1-HC/low-pI-HC heterodimer (pI=6.89), and low-pI-HC/low-pI-HC homodimer (pI=6.20). The mixture was loaded onto a GE HiTrap Q HP column in 20 mM Tris, pH 7.6 and eluted with a step-wise gradient of NaCl consisting of 50 mM, 100 mM, and finally 200 mM NaCl in the same Tris buffer. Elution was monitored by A280, and each fraction analyzed on Invitrogen pH 3-10 IEF gels with Novex running buffer and these results are shown in FIG. 40. WT-IgG1-HC/WT-IgG1-HC homodimer does not bind to the anion exchange column at pH 7.6 and is thus present in the flowthrough and wash (lanes 1-2). The desired heterodimer elutes with 50 mM NaCl (lane 3), while the low-pI-HC/low-pI-HC homodimer binds tightest to the column and elutes at 100 (lane 4) and 200 mM (lane 5) NaCl. Thus the desired heterodimer variant, which is difficult to purify by other means because of its similar molecular weight to the other two species, is easily purified by the introduction of low pI substitutions into one chain. This method of purifying antibodies by engineering the isoelectric point of each chain can be applied to methods of purifying various bispecific antibody constructs. The method is particularly useful when the desired species in the mixture has similar molecular weight and other properties such that normal purification techniques are not capable of separating the desired species

in high yield.

Example 5. Design of Non-Native Charge Substitutions to Alter pI

(365) The pI of antibody constant chains were altered by engineering substitutions in the constant domains. Reduced pI can be engineered by making substitutions of basic amino acids (K or R) to acidic amino acids (D or E), which result in the largest decrease in pI. Mutations of basic amino acids to neutral amino acids and neutral amino acids to acidic amino acids will also result in a decrease in pI. Conversely, increased pI can be engineered by making substitutions of acidic amino acids (D or E) to basic amino acids (K or R), which result in the largest increase in pI. Mutations of acidic amino acids to neutral amino acids and neutral amino acids to basic amino acids will also result in an increase in pI. A list of amino acid pK values can be found in Table 1 of Bjellqvist et al., 1994, Electrophoresis 15:529-539.

(366) In deciding which positions to mutate, the surrounding environment and number of contacts the WT amino acid makes with its neighbors was taken into account such as to minimize the impact of a substitution or set of substitutions on structure and/or function. The solvent accessibility or fraction exposed of each constant region position was calculated using relevant crystal structures. Based on this analysis, a number of substitutions were identified that reduce or increase pI but are predicted to have minimal impact on the biophysical properties of the domains.

(367) Calculation of protein pI was performed as follows. First, a count was taken of the number of D, E, C, H, K, R, and Y amino acids as well as the number of N- and C-termini present in the protein. Then, the pI was calculated by identifying the pH for which the protein has an overall charge of zero. This was done by calculating the net charge of the protein at a number of test pH values. Test pH values were set in an iterative manner, stepping up from a low pH of 0 to a high pH of 14 by increments of 0.001 until the charge of the protein reached or surpassed zero. Net charge of a protein at a given pH was calculated by the following formula:

$$(368) q_{\text{protein}}(\text{pH}) = \sum_{i=H,K,R,\text{Ntermini}} \frac{N_i}{1 + 10^{\text{pH} - \text{pK}_i}} - \sum_{i=D,E,C,Y,\text{Ctermini}} \frac{N_i}{1 + 10^{\text{pK}_i - \text{pH}}}$$

where $q_{\text{sub.protein}}(\text{pH})$ is the net charge on the protein at the given pH, N_i is the number of amino acid i (or N- or C-termini) present in the protein, and pK_i is the pK of amino acid i (or N- or C-termini).

Example 6. Purifying Mixtures of Antibody Variants with Modified Isoelectric Points

(369) Variants were first purified by Protein A, and then loaded onto a GE Healthcare HiTrap SP HP cation exchange column in 50 mM IVIES (pH 6.0) and eluted with an NaCl gradient.

Following elution, fractions from each peak were loaded onto a Lonza IsoGel IEF plate (pH range 7-11) for analysis. Separation of the middle pI heterodimer is achieved in each case, with separation improved when the heterodimer has a larger difference in pI from the homodimers.

Example 7. Stability of pI Isosteric Variants

(370) Differential scanning fluorimetry (DSF) was used to evaluate the stability of antibodies containing isosteric pI substitutions. DSF experiments were performed using a Bio-Rad CFX Connect Real-Time PCR Detection System. Proteins were mixed with SYPRO Orange fluorescent dye and diluted to 0.25 or 0.50 mg/mL in PBS. The final concentration of SYPRO Orange was 10 $\times$. After an initial 10 minute incubation period at 25 $^{\circ}$ C., proteins were heated from 25 to 95 $^{\circ}$ C. using a heating rate of 1 $^{\circ}$ C./min. A fluorescence measurement was taken every 30 sec. Melting temperatures were calculated using the instrument software. The results are shown in FIG. 44. The results indicated that isosteric(+) pI variants had lower stability. We therefore made further variants to reduce the number of substitutions on the increased pI side, but results showed that only E269Q had a small effect on stability, while E272Q and E283Q had large negative impacts on stability.

Example 8. Design of Charged scFv Linkers to Enable IEX Purification of scFv Containing Heterodimeric Bispecific Antibodies

(371) We have previously engineered the antibody constant regions of heterodimeric antibodies to have higher or lower pI using both isotypic and isosteric charge substitutions. These methods enable efficient IEX purification of heterodimeric species, but may impact stability or

immunogenicity of the antibodies due to the unnatural substitutions introduced. For scFv containing heterodimeric bispecific antibodies (Examples are shown in FIG. 1), another region to introduce charged substitutions is the scFv linker that connects the VH and VL of scFv constructs. The most common linker used is (GGGGS)₃ or (GGGGS)₄, which has been shown to be flexible enough to allow stable scFv formation without diabody formation. These sequences are already unnatural, and contain little sequence specificity for likely immunogenic epitopes. Therefore we thought that introducing charged substitutions into scFv linkers may be a good strategy to enable IEX purification of heterodimeric bispecific species containing scFvs. Various positively and negatively charged scFv linkers were designed and are shown in 9. All linkers are novel constructs except for the “Whitlow” linker which was reported by Whitlow et al., (Whitlow M, Protein Eng. 1993 (8), 989-995.). Linkers designated as 6paxA_1 (+A) and 3hsc_2 (-A) were taken from a database of unstructured regions in human proteins obtained from PDB files and these linkers are approximately the same length as (GGGGS)₃ and contain positive or negative charges. Other linkers are based on introducing repetitive Lys or Glu residues, as well as Lys-Pro motifs designed to reduce the chance of proteolytic degradation in the positively charged linkers.

(372) Charged linkers were first evaluated for biophysical behavior in the scFv-His format and then were later constructed in anti-CD19×CD3 Fab-scFv-Fc bispecific format. Genes encoding the scFv of engineered forms of the anti-CD3 antibody SP34 or the anti-CD19 4G7 antibody were constructed in the mammalian expression vector pTT5. For full-length constructs, the human IgG1 constant chain gene was obtained from IMAGE clones and subcloned into the pTT5 vector. scFv genes were synthesized commercially (Blue Heron Biotechnologies, Bothell WA. Amino acid modifications were constructed using site-directed mutagenesis using the QuikChange® site-directed mutagenesis methods (Stratagene, La Jolla CA). All DNA was sequenced to confirm the fidelity of the sequences.

(373) Plasmids containing scFv or heavy chain and light chain genes were transfected (or co-transfected for full-length formats) into 293E cells using lipofectamine (Invitrogen, Carlsbad CA) and grown in FreeStyle 293 media (Invitrogen, Carlsbad CA). After 5 days of growth, the antibodies were purified from the culture supernatant by protein A (full-length) using the Mab Select resin (GE Healthcare) or using Ni-NTA resins for His-tagged scFvs. Heterodimers were further purified by ion exchange chromatography (IEX) to assess the ability of the altered pI heavy chains to enable efficient purification. Examples of IEX purifications for an anti-CD19×CD3 bispecific containing a positively charged linker in the CD3 scFv is shown in FIG. 49. Antibody concentrations were determined by bicinchoninic acid (BCA) assay (Pierce).

(374) The pI engineered scFvs or antibodies were characterized by SDS-PAGE, size exclusion chromatography (SEC), isoelectric focusing (IEF) gel electrophoresis, and/or differential scanning fluorimetry (DSF).

Example 9. Stability and Behavior of scFvs Containing Charged Linkers

(375) Anti-CD3 scFv's and anti-CD19 scFv's containing positively or negatively charged linkers, respectively, were evaluated for SEC behavior as well as for stability using DSF. Differential scanning fluorimetry (DSF) was used to evaluate the stability of scFvs containing charged linkers. DSF experiments were performed using a Bio-Rad CFX Connect Real-Time PCR Detection System. Proteins were mixed with SYPRO Orange fluorescent dye and diluted to 0.25 or 0.50 mg/mL in PBS. The final concentration of SYPRO Orange was 10×. After an initial 10 minute incubation period at 25° C., proteins were heated from 25 to 95° C. using a heating rate of 1° C./min. A fluorescence measurement was taken every 30 sec. Melting temperatures were calculated using the instrument software. T_m values for scFvs are shown in FIG. 45. Charged linkers had only marginal impacts on overall scFv stability as indicated by their T_m values. SEC chromatograms obtained from purified scFvs are shown in FIG. 46. Highly charged linkers have a longer elution time and noticeable peak tails indicating that too much charge causes the scFvs to stick to the SEC resin longer than expected. Binding results for positively charged anti-CD3 scFvs binding to CD4+

T cells (FIG. 47) indicated that binding of most scFvs was similar, with the exception of the very highly charged (GKGKS)₄ scFv, which showed weaker binding. No off-target binding was detected when gating for CD20⁺ cells in PBMCs. However, when off-target binding was tested using SP34 cells, some amount of off-target binding was seen with the highest charged linkers at high concentrations (FIG. 48).

(376) Positively charged scFv linkers on the anti-CD3 scFv in an anti-CD19×CD3 Fab-scFv-Fc construct had the unexpected property of reducing the amount of high molecular weight aggregation SEC chromatograms of two bispecific constructs (13121—with standard (GGGGS)₄ linker) and (13124—with charged linker (GKPGS)₄) incubated at various concentrations confirmed this phenomenon.

(377) Activity of anti-CD19×CD3 constructs containing charged scFv linkers in the anti-CD3 scFv was evaluated using an RTCC assay with PBMCs and Fab-scFv-Fc format bispecific anti-CD19×CD3 antibodies containing different scFv linkers (FIG. 51). Linkers have little impact on RTCC activity, except for the highly charged linker (GKGKS)₃ which has lower activity.

(378) Sequences for all constructs of the invention are shown in the Figures.

Claims

1. A heterodimeric protein comprising a heterodimeric Fc domain, the heterodimeric Fc domain comprising: a) a first monomer comprising a first variant Fc domain comprising amino acid substitutions L368D, K370S, M428L and N434S; and b) a second monomer comprising a second variant Fc domain comprising amino acid substitutions S364K, E357Q, M428L and N434S, wherein numbering is according to EU numbering, and wherein the first variant Fc domain and the second variant Fc domain are variant human IgG Fc domains.
 2. A heterodimeric protein according to claim 1, wherein the first monomer further comprises a first fusion partner, and the second monomer further comprises a second fusion partner.
 3. A heterodimeric protein according to claim 2, wherein the first monomer comprises a third fusion partner.
 4. A heterodimeric protein according to claim 3, wherein the second monomer comprises a fourth fusion partner.
 5. A heterodimeric protein according to claim 2, wherein the fusion partners are independently selected from the group consisting of an immunoglobulin component, a peptide, a cytokine, a chemokine, an immune receptor and a blood factor.
 6. A heterodimeric protein according to claim 5, wherein the immunoglobulin component is selected from the group consisting of Fab, VH, VL, scFv, scFv2, and dAb.
 7. A heterodimeric protein according to claim 5, wherein both fusion partners are immunoglobulin components.
 8. A nucleic acid composition encoding the heterodimeric protein of claim 1 comprising: a) a first nucleic acid encoding the first monomer; and b) a second nucleic acid encoding the second monomer.
 9. An expression vector composition encoding the heterodimeric protein of claim 1 comprising: a) a first expression vector comprising a first nucleic acid encoding the first monomer; and b) a second expression vector comprising a second nucleic acid encoding the second monomer.
 10. A host cell comprising the expression vector composition of claim 9.
 11. A method of making a heterodimeric protein comprising culturing a host cell according to claim 10 under conditions whereby a heterodimeric protein comprising a heterodimeric Fc domain is produced, and recovering the heterodimeric protein.
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